

The mathcomp-eulerian formalization

A Stanley-style reading of permutation statistics in Rocq/MathComp

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About this document

This document is a Stanley-style reading of the [mathcomp-eulerian](#) formalization. Each definition and statement is given twice: once in ordinary mathematical English (in the spirit of Stanley *Enumerative Combinatorics I*, Chapter 1) and once as the verbatim Rocq text, with a clickable link to the line in the repository on GitHub. The intention is that a mathematician can read top to bottom like a textbook and at any point click through to verify the formal statement.

Status. This edition covers all of Stanley §1.3 (cycles, Stirling numbers, inversions, major index, Foata), §1.4 (descents, Eulerian numbers, $\beta_n(S)$, q -analogues), §1.6.2 (longest alternating subsequence), §1.6.3 (toggle action, ω -map, and the headline Corollary 1.6.5), and §1.6.4 (André’s reflection method, Chapter 9): the André recurrence $2E_{n+2} = \sum_k \binom{n+1}{k} E_k E_{n+1-k}$ is now fully proved, with no admits, in the active build chain.

First-time readers. Read Chapter 1 once, then Chapters 2–9 top to bottom. Use Appendix B as a quick reference on second reading. The primer and glossary together contain everything a Stanley reader needs to know about mathcomp to verify the formal statements; you do not need to install Rocq.

Auxiliary lemmas. The narrative chapters cover the ~ 70 most-cited results in the formalization. The full library contains ~ 830 named results (Definition/Lemma/Theorem/Corollary), all of which are listed in Appendix A with their `coqdoc` docstrings and clickable links to GitHub. Auxiliary lemmas (those not narrated in a chapter) are flagged `[aux]`; narrated entries carry a `[Ch X §X.Y]` badge pointing back to where they appear in the body. The catalog is auto-generated from the `.v` files; adding a docstring to a result automatically enriches both the catalog and the deployed `coqdoc` HTML.

Conventions. The single recurring shift is 0-indexing. Stanley writes $w \in \mathfrak{S}_n$ for permutations of $\{1, \dots, n\}$ and $D(w) \subseteq \{1, \dots, n-1\}$ for the descent set; we use `{perm 'In}` for permutations of $\{0, \dots, n-1\}$ and `{set 'In-1}` for the descent set. Concretely:

$$\text{our } \{\text{perm 'I}_{n+1}\} = \text{Stanley's } \mathfrak{S}_{n+1}.$$

A complete translation table is in [docs/READING_GUIDE.md](#); a worked example through Stanley’s $w = [3, 1, 4, 2]$ is in the same file.

Reading the formal text. Each Rocq block looks like this:

[\[Rocq, descent.v:25\]](#)

```
Definition descent_set s : {set 'I_n} := [set i | is_descent s i].
```

Decoded. $D(s) = \{i \in \{0, \dots, n-1\} : s(i) > s(i+1)\}$.

The bracketed link at the top is clickable and opens the exact line on GitHub. The Rocq body is verbatim; identifiers and keywords are highlighted but no mathematical substitution has

been performed. The shaded *Decoded* line below the block restates the formal text in plain mathematics, mechanically substituting mathcomp casts for their meaning. Side by side, the two should say the same thing; if they do not, that is exactly the kind of discrepancy worth flagging. Some definitions also carry a **Design choice** sidebar that explains why a particular formalization was chosen when a more natural-looking alternative exists. These boxes are for first reading and can be skipped on a re-read.

What is *not* included here. Full proofs are not included; only the formal *statements* are shown alongside the informal mathematics, with a one-paragraph proof sketch for each non-trivial result. The full proof is always one click away in the linked source. The cd-index development (`mmtree.v`, `psi_*.v`) is project-internal scaffolding not in Stanley and is not covered.

Verifying axiom-freeness. Every theorem in this document is kernel-validated by Rocq with no axioms or admits. To check this yourself for, e.g., the headline Corollary 8.7:

```
echo 'From mathcomp_eulerian Require Import beta_swap.
      Print Assumptions beta_alt_max.' | rocq top -R . mathcomp_eulerian
```

prints `Closed under the global context`, meaning no axiom is used anywhere in its transitive dependencies.

Chapter 1

A combinatorialist's mathcomp primer

This chapter is a focused reference for the few mathcomp primitives that appear in Chapter 5. It is not a tutorial; it gives just enough vocabulary to read the formal blocks and recognise that they say what the informal text claims. Skim it once on first read; come back to it as a glossary when re-reading.

1.1 Ordinals: the type `'I_n`

The type `'I_n` (read “I-sub-n”) is mathcomp’s *finite type* of natural numbers strictly less than n . An inhabitant is a pair: a natural number `i : nat` and a *proof* `i < n`. The two together are written `Ordinal i (proof : i < n)` but a reader normally never constructs ordinals by hand — the type appears as the index of a sum or a permutation domain, and inhabitants come from the typing context.

`ord0 : 'I_n.+1` the value 0, with its proof $0 < n + 1$. Always available when the bound is at least 1.

`ord_max : 'I_n.+1` the value n , with its proof. The largest element of `'I_n.+1`.

`val i : nat` strips an ordinal back to its underlying natural number, forgetting the proof.

The *cardinality* of `'I_n` is exactly n (lemma `card_ord` in `fintype.v`). So `#|'I_4| = 4`.

1.2 Casts: `widen_ord`, `lift`, `unlift`

There are exactly two natural maps $'I_n \rightarrow 'I_{n+1}$:

`widen_ord (h : n ≤ n+1) i` sends `i : 'I_n` to the same numerical value, viewed inside the bigger type. Diagrammatically:

$$i : 'I_n \xrightarrow{\text{widen_ord}} i : 'I_{n+1}.$$

The h argument is a proof that $n \leq n + 1$, supplied by `leqnSn n`. Mathcomp users write `widen_ord (leqnSn n) i` reflexively.

`lift (p : 'I_n.+1) i` sends `i : 'I_n` to a value in `'I_n.+1` *skipping* the value p . With $p = \text{ord0}$, this shifts every value up by one:

$$i : 'I_n \xrightarrow{\text{lift ord0}} i + 1 : 'I_{n+1}.$$

The general `lift p i` skips p , which is exactly what one needs when removing one position from a permutation.

The two casts together name two specific elements of `'I_n.+1` from the same source $i : 'I_n$: the value i itself, and the value $i + 1$. This is the idiom that lets us write “the comparison between consecutive positions i and $i + 1$ ” without ever leaving the well-typed world of finite ordinals.

The partial inverse of `lift p` is `unlift p`, of type `'I_n.+1 → option 'I_n`. Concretely: `unlift p i = None` if $i = p$, and `unlift p i = Some j` otherwise, with j the unique witness of $i = \text{lift } p \ j$. This is the canonical Rocq idiom for the case-split “is the input the special index p , or one of the others?” — it appears in every inductive bijection on ordinals.

1.3 Finite sets: `{set T}`

For T a finite type (such as `'I_n`), `{set T}` is the type of *finite subsets of T* , with *extensional* equality: two sets are equal iff they have the same elements.

`{set i : T | P i}` set comprehension — the set of $i : T$ satisfying the boolean predicate P . The vertical bar reads as “such that”, as in mathematical notation.

`#|S|` the cardinality of S as a natural number.

`i \in S` membership test, returning a `bool`.

`~: S` complement, `S :| : T` union, `S :& : T` intersection, `S :\ : T` difference.

`{set T}` is itself a finite type, which means big operators (sums, products) can be indexed by sets. This matters for the Eulerian recurrence and for the partition-by-descent-set identity `beta_eulerian`.

1.4 Permutations: `{perm T}`

`{perm T}` is the type of *bijections $T \rightarrow T$* for T a finite type. An inhabitant is a function together with a proof of bijectivity, packaged so that:

`s i` applies the permutation s to $i : T$ as a function. (No special syntax is needed: `{perm T}` *coerces* to a function.)

`1%g` is the identity permutation. The `%g` selects mathcomp’s group scope.

`(s * t)%g i` is composition. Mathcomp’s convention is that $(s * t) i = t (s i)$ (right-to-left for the *action*; the lemma is `permM`).

`perm (h : injective f)` constructs a permutation from an injective function $f : T \rightarrow T$ (on a finite type, injective implies bijective).

The cardinality of `{perm 'I_n}` is $n!$ (Stanley’s $|\mathfrak{S}_n|$); this is the lemma `card_perm` or `card_Sn` in mathcomp.

1.5 Big operators: `\sum`, `\prod`

Mathcomp’s `bigop.v` provides indexed sums and products with a uniform syntax:

`\sum_(i : T) f i` sums `f i` over every `i : T` (`T` a finite type).

`\sum_(i < n) f i` same, with `i : 'I_n`.

`\sum_(i \in S) f i` sums over an explicit set `S`.

`\sum_(i | P i) f i` sums over indices satisfying the boolean filter `P`.

The vertical bar in `\sum_(i | P i)` is a *filter*, not the disjunction symbol. Replace `\sum` by `\prod` for products. Several recurring *moves* appear in the proofs:

`partition_big f xpredT` rewrite `\sum_x g x` as `\sum_y \sum_(x | f x = y) g x` — partition by the value of `f`.

`bigID p` split a sum into the part where the predicate `p` holds and the part where it does not.

`reindex ...` change of summation variable along a bijection.

These names appear in the proofs we will not reprint in this document, but they are worth recognising if you click through to a `.v` file.

1.6 Sections and section variables

Every formalization file in this project opens with

```
Section X.
Variable n : nat.
Definition foo := ...      (* implicitly takes n *)
Lemma bar : ...           (* implicitly takes n *)
End X.
```

Inside the section, definitions and lemmas may use `n` as if it were globally fixed. At `End X`, every definition and lemma is automatically generalised to take `n` as an explicit argument. This is why `descent.v` reads `Variable n : nat` once at the top and then `Definition is_descent s i := ...` without an explicit `n`, but outside the file `is_descent` is invoked as `is_descent n s i`. (For the mathematician: think of `Variable` as “Let `n` be a natural number” at the start of a chapter.)

1.7 The takeaway

Three points are enough to read Chapter 5:

1. `{perm 'I_n.+1}` is mathcomp’s $\mathfrak{S}_{n+1}$. Apply a permutation to a position by writing `s i`; the descent set is a `{set 'I_n}` (positions, not values).
2. The two casts `widen_ord (leqnSn n) i` and `lift_ord0 i` pick out positions `i` and `i + 1` inside `'I_n.+1` from a single `i : 'I_n`. The reader does not need to verify cast arithmetic by hand — the kernel rejects the file unless every cast is the unique one of its type.
3. Big operators `\sum_(...)` are just sums; the syntax variants control the index range. The vertical bar is a filter.

With these in hand, the formal blocks in Chapter 5 read directly. Each formal block is followed by a one-line *Decoded* reading that mechanically substitutes the mathcomp casts for their mathematical meaning, so the reader has both forms side by side.

Chapter 2

Cycles and Stirling numbers

This chapter formalizes Stanley §1.3.1 (cycle representation of permutations) and §1.3.2 (Stirling numbers of the first kind). The two main objects are the cycle count $c(w)$ and the (signless) Stirling cycle number $c(n, k)$ that counts permutations by $c(w)$. They are the cycle-side analogues of $\text{des}(w)$ and $A(n, k)$ from Chapter 5.

2.1 Cycle count

Definition 2.1 (Cycle decomposition, cycle count). Every $w \in \mathfrak{S}_n$ admits a unique decomposition into a set of *disjoint cycles*; let $c(w)$ denote the number of such cycles.

[Rocq, cycles.v:24]

```
Definition cycle_count {T : finType} (s : {perm T}) : nat :=  
  #|porbits s|.
```

Decoded. $c(s)$ = number of orbits of the action of the cyclic group $\langle s \rangle$ on the underlying set — that is, the number of cycles in the disjoint-cycle decomposition.

Design choice: why porbits, not a manual cycle decomposition?

A naive formalization would build the cycle decomposition by hand: pick an element, follow the orbit, mark seen, repeat. Mathcomp’s `porbits` (in `perm.v`) does this once and exposes the result as a `{set {set T}}` — a set of orbits — together with all the lemmas one would want (`porbits_partition`, `cover_porbits`, etc.). Reusing it costs nothing and gives proofs like `cycle_count_id` in three lines instead of thirty. The cardinality of the orbit set is exactly the cycle count.

Lemma 2.2 (Cycle count bound). $c(w) \leq |T|$ for every $w \in \mathfrak{S}_T$, with equality iff w is the identity.

[Rocq, cycles.v:44]

```
Lemma cycle_count_le {T : finType} (s : {perm T}) :  
  cycle_count s ≤ #|T|.
```

Decoded. The number of cycles is at most the number of elements (each cycle has at least one element).

[Rocq, cycles.v:64]

```
Lemma cycle_count_id (T : finType) :  
  cycle_count (1 : {perm T}) = #|T|.
```

Decoded. The identity has every element as a fixed point, i.e. a 1-cycle; the cycle count equals $|T|$.

2.2 Stirling cycle numbers

Definition 2.3 (Signless Stirling cycle numbers). The number $c(n, k)$ is the count of permutations of $[n] = \{1, \dots, n\}$ with exactly k cycles.

[Rocq, cycles.v:33]

```
Definition stirling_c (n k : nat) : nat :=
  #| [set s : {perm 'I_n} | cycle_count s == k] |.
```

Decoded. $c(n, k) = |\{s \in \mathfrak{S}_n : c(s) = k\}|$, the number of perms of length n with k cycles. *Direct match to Stanley* (no off-by-one): we use `{perm 'I_n}`, of size n , exactly mirroring Stanley's S_n .

Lemma 2.4 (Row sum recovers $n!$). $\sum_{k=0}^n c(n, k) = n!$

[Rocq, cycles.v:98]

```
Lemma stirling_c_row_sum_fact n :
  \sum_ (k < n.+1) stirling_c n k = n'!.
```

Decoded. Sum over $k \in \{0, \dots, n\}$ of $c(n, k)$ equals $n!$, since the sets partition $\mathfrak{S}_n$ by cycle count.

Proof sketch. The sets $\{s : c(s) = k\}$ for k ranging over all possible cycle counts partition $\mathfrak{S}_n$. Total cardinality is $|\mathfrak{S}_n| = n!$. $\square$

Theorem 2.5 (Stirling cycle recurrence). For $n, k \geq 0$,

$$c(n+1, k+1) = n \cdot c(n, k+1) + c(n, k).$$

[Rocq, stirling_fiber.v:341]

```
Lemma stirling_c_rec n k :
  stirling_c n.+1 k.+1 = n * stirling_c n k.+1 + stirling_c n k.
```

Decoded. Stanley's recurrence for the signless Stirling cycle numbers: a permutation of $[n+1]$ with $k+1$ cycles either inserts the element $n+1$ into one of the existing n cycle-positions of a perm of $[n]$ with $k+1$ cycles (n choices), or sits as a new fixed point in a perm of $[n]$ with k cycles.

Proof sketch. Partition perms of $[n+1]$ with $k+1$ cycles by whether $n+1$ is a fixed point. If yes, the rest is a perm of $[n]$ with k cycles (giving $c(n, k)$). If no, $n+1$ sits inside an existing cycle; there are n positions to insert it, giving $n \cdot c(n, k+1)$.

The formal proof uses an auxiliary construction `insert_after` (in `stirling_fiber.v`) that realises the bijection $[n] \times \{s \in \mathfrak{S}_n : c(s) = k+1\} \rightarrow \{\sigma \in \mathfrak{S}_{n+1} : \sigma(n+1) \neq n+1, c(\sigma) = k+1\}$. $\square$

Chapter 3

Inversions and the major index

This chapter formalizes Stanley §1.3.3 (the inversion count inv and the major index maj , the two classical Mahonian statistics) and the basic identities each satisfies. The headline result — their equidistribution over $\mathfrak{S}_n$ — requires the Foata bijection and is the subject of Chapter 4.

3.1 Inversions

Definition 3.1 (Inversion). A pair (i, j) with $1 \leq i < j \leq n$ is an *inversion* of $w \in \mathfrak{S}_n$ if $w_i > w_j$.

[Rocq, inversions.v:26]

```
Definition is_inv s (i j : 'I_n.+1) : bool :=
  (i < j) && (s j < s i).
```

Decoded. (i, j) is an inversion of s iff $i < j$ and $s(j) < s(i)$ (i.e. $s(i) > s(j)$).

Definition 3.2 (Inversion set, inversion count). $\text{Inv}(w) = \{(i, j) : i < j, w_i > w_j\}$, and $\text{inv}(w) = |\text{Inv}(w)|$.

[Rocq, inversions.v:30]

```
Definition inv_set s : {set 'I_n.+1 * 'I_n.+1} :=
  [set ij | is_inv s ij.1 ij.2].
```

Decoded. The set of inversion pairs.

[Rocq, inversions.v:34]

```
Definition inv s : nat := #|inv_set s|.
```

Decoded. $\text{inv}(s)$ is the cardinality of the inversion set.

Lemma 3.3 (Identity has no inversions). $\text{inv}(\text{id}) = 0$.

[Rocq, inversions.v:41]

```
Lemma inv_id : inv (1 : {perm 'I_n.+1}) = 0.
```

Decoded. The identity permutation has no (i, j) with $s(i) > s(j)$, hence no inversions.

Lemma 3.4 (Inversion bound). $\text{inv}(w) \leq \binom{n}{2}$ for every $w \in \mathfrak{S}_n$.

[Rocq, inversions.v:136]

Lemma `inv_le s : inv s ≤ 'C(n.+1, 2).`

Decoded. $\text{inv}(s) \leq \binom{n+1}{2}$, the number of pairs (i, j) with $i < j$ in $\mathfrak{S}_{n+1}$.

Lemma 3.5 (Reversal identity for inv). $\text{inv}(w^{\text{rev}}) = \binom{n}{2} - \text{inv}(w)$.

[Rocq, `inversions.v:219`]

Lemma `inv_rev_perm s :`
`inv (rev_perm s) = 'C(n.+1, 2) - inv s.`

Decoded. The reverse permutation has the “opposite” inversion count: each pair (i, j) with $i < j$ is either an inversion of s or of its reverse, but not both. Hence the two counts sum to $\binom{n+1}{2}$.

3.2 The major index

Definition 3.6 (Major index). The *major index* of $w \in \mathfrak{S}_n$ is the sum of (1-indexed) descent positions:

$$\text{maj}(w) = \sum_{i \in D(w)} i.$$

[Rocq, `inversions.v:81`]

Definition `maj s : nat := \sum_(i in descent_set s) (val i).+1.`

Decoded. $\text{maj}(s) = \sum_{i \in D(s)} (i + 1)$. The `(val i).+1` adds 1 to recover Stanley’s 1-indexing from our 0-indexed descent positions.

Design choice: why `(val i).+1` inside the sum, not in the descent set?

We could “correct” the indexing once for all by storing 1-indexed positions in the descent set. We don’t, because the descent set type `{set 'I_n}` would no longer fit the ambient type system: 1-indexed positions in $\{1, \dots, n-1\}$ would need `{set 'I_n.-1}` with values shifted by 1, and every set-level lemma (complement, partition, big-op) would carry a translation. Doing the index shift inside `maj` keeps the descent-set machinery clean and isolates the translation to the one definition that needs it. The price is a single `(val i).+1` in the formula; the benefit is that `descent_set`, `beta`, and `set_is_alt` all live in the same type without conversion casts.

Lemma 3.7 (Major-index bound). $\text{maj}(w) \leq \binom{n}{2}$ for every $w \in \mathfrak{S}_n$.

[Rocq, `inversions.v:158`]

Lemma `maj_le s : maj s ≤ 'C(n.+1, 2).`

Decoded. $\text{maj}(s) \leq \binom{n+1}{2}$, attained by the reverse identity.

Example 3.8 (Stanley’s $w = [3, 1, 4, 2]$, statistics). Take $w = [3, 1, 4, 2] \in \mathfrak{S}_4$ as in Example 5.3. Pairs (i, j) with $w_i > w_j$: $(1, 2)$ since $3 > 1$, $(1, 4)$ since $3 > 2$, $(3, 4)$ since $4 > 2$. Hence $\text{inv}(w) = 3$. Descent positions: $D(w) = \{1, 3\}$, so $\text{maj}(w) = 1 + 3 = 4$. In our types, $\text{inv}(s) = 3$ and $\text{maj}(s) = (0 + 1) + (2 + 1) = 4$, matching.

Chapter 4

Foata's bijection and MacMahon's equidistribution

This chapter formalizes Stanley §1.3.4: Foata's first fundamental bijection on $\mathfrak{S}_n$, which sends permutations of major index k bijectively to permutations of inversion count k . The corollary is MacMahon's equidistribution theorem.

4.1 The Foata transformation on words

The bijection is built from a recursive construction on words. Given a word u and a new letter a , classify the previous letters of u according to whether they are $< a$ or $> a$ (depending on the last letter of u), split u into blocks at the classifying letters, and cyclically rotate each block (last $\rightarrow$ front). Then append a .

[Rocq, foata.v:70-77]

```
Definition foata_step (a : nat) (u : seq nat) : seq nat :=
  match u with
  | [::]  $\Rightarrow$  [:: a]
  | _ :: _  $\Rightarrow$ 
    let x := last 0 u in
    let P := if x < a then (fun y : nat  $\Rightarrow$  y < a) else (fun y  $\Rightarrow$  a < y) in
    flatten (map cyc_last_to_front (split_blocks P u)) ++ [:: a]
  end.
```

Decoded. Empty word: just $[a]$. Otherwise: pick the predicate P (" $< a$ " or " $> a$ " depending on the sign of $\text{last}(u)$ vs a), `split_blocks P u` cuts u into blocks ending at P -letters, `cyc_last_to_front` rotates each block (last letter to front), `flatten` concatenates, then append a .

[Rocq, foata.v:80]

```
Definition foata (w : seq nat) : seq nat :=
  foldl (fun u a  $\Rightarrow$  foata_step a u) [::] w.
```

Decoded. Process w left-to-right, applying `foata_step` at each new letter starting from the empty word.

Example 4.1 (Stanley's running example). $w = [3, 1, 4, 5, 9, 2, 6]$. Then:

$$\text{foata}(w) = [3, 4, 1, 5, 2, 9, 6].$$

Verified inside `foata.v` (lemma `sanity_inv_eq_maj` at line 125 confirms by `Compute`). Stanley's hand computation on p. 24 gives the same value.

4.2 Lifting to permutations

The seq-level bijection is wrapped into a permutation via `seq_to_perm`, which packages a uniqueness-and-bound proof obligation into a `{perm 'I_n.+1}`.

[Rocq, foata.v:1529]

```
Definition foata_perm (s : {perm 'I_n.+1}) : {perm 'I_n.+1} :=
  seq_to_perm (foata_perm_to_seq_size s) (foata_perm_to_seq_uniq s)
  (foata_perm_to_seq_bnd s).
```

Decoded. `foata_perm s` is the permutation whose one-line sequence is `foata` applied to the one-line sequence of `s`. The three named lemmas discharge the side-conditions (size, distinctness, bound) needed to package a `seq nat` back into a `{perm}`.

Design choice: why a seq-level bijection lifted to perms?

Foata's bijection is defined inductively on the word (seq), not on the permutation as a function. Trying to define it directly on `{perm 'I_n.+1}` would force every step to re-establish “the result is a permutation,” an obligation that the seq-level construction discharges once at the end. We pay for the wrapping once (in `seq_to_perm`); in exchange the bijection itself has the natural, recursive form Stanley describes.

4.3 The equidistribution theorem

Theorem 4.2 ($\text{inv}(\text{foata } w) = \text{maj}(w)$). *For every $w \in \mathfrak{S}_n$,*

$$\text{inv}(\text{foata}(w)) = \text{maj}(w).$$

[Rocq, foata.v:1576]

```
Lemma foata_perm_inv_maj n (s : {perm 'I_n.+1}) :
  inv (foata_perm s) = maj s.
```

Decoded. The Foata transformation converts maj-statistics to inv-statistics: for every `s`, the inversion count of `foata(s)` equals the major index of `s`.

Lemma 4.3 (`foata_perm` is injective). *The map $s \mapsto \text{foata_perm } s$ is injective on $\mathfrak{S}_{n+1}$.*

[Rocq, foata.v:1586]

```
Lemma foata_perm_inj n : injective (@foata_perm n).
```

Decoded. Different inputs give different outputs (proven by constructing an explicit inverse `foata_inv` at the seq level).

Theorem 4.4 (MacMahon's equidistribution). *For every n, k , the number of permutations of $\mathfrak{S}_n$ with $\text{inv} = k$ equals the number with $\text{maj} = k$:*

$$\#\{w \in \mathfrak{S}_n : \text{inv}(w) = k\} = \#\{w \in \mathfrak{S}_n : \text{maj}(w) = k\}.$$

[Rocq, foata.v:1598]

```
Theorem inv_maj_equidistr n k :
  #| [set s : {perm 'I_n.+1} | inv s == k] |
  = #| [set s : {perm 'I_n.+1} | maj s == k] |.
```

Decoded. The two count functions `inv` and `maj` have the same distribution over $\mathfrak{S}_{n+1}$.

Proof sketch. The map $s \mapsto \text{foata_perm } s$ sends $\{s : \text{maj}(s) = k\}$ into $\{t : \text{inv}(t) = k\}$ (Theorem 4.2); it is injective; on a finite set, injective implies surjective. So the two sets are in bijection, hence have equal cardinality. $\square$

Chapter 5

Descents and Eulerian numbers

This chapter formalizes Stanley §1.4 (Stanley *Enumerative Combinatorics I*, 2nd ed., pp. 30–38). The central objects are the *descent set* of a permutation, the *Eulerian numbers* $A(n, k)$ that count permutations by descent count, and the set-refined count $\beta_n(S)$ that counts permutations by their full descent set.

5.1 The descent set

Definition 5.1 (Descent). Let $w = (w_1, w_2, \dots, w_n) \in \mathfrak{S}_n$. A position $i \in \{1, \dots, n-1\}$ is a *descent* of w if $w_i > w_{i+1}$.

[Rocq, descent.v:22]

```
Definition is_descent s i : bool :=
  s (widen_ord (leqnSn n) i) > s (lift_ord0 i).
```

Decoded. $s(i) > s(i+1)$ for $i \in \{0, \dots, n-1\}$ and $s : \{0, \dots, n\} \rightarrow \{0, \dots, n\}$ a bijection.

! Design choice: why `widen_ord` and `lift`, not `s i > s i.+1`?

The naive form `s i > s i.+1` does not type-check. The permutation `s : {perm 'I_n.+1}` expects an argument of type `'I_n.+1`, but a descent slot i lives in `'I_n` (there are n adjacencies for $n+1$ positions). The two casts `widen_ord (leqnSn n) i` and `lift_ord0 i` are the canonical maps that ship i into the larger type `'I_n.+1` as the values i and $i+1$ respectively (Primer 1.2). There is no way to “just cast” — the two casts encode *which* element of `'I_n.+1` we mean. The verbosity is what makes the formalization correct by typing.

The result type `bool` (rather than `Prop`) makes `is_descent` computable: `Compute is_descent s i` returns `true` or `false` for any concrete `s` and `i`. Mathcomp’s small-scale reflection bridges `is_descent s i = true` with the corresponding proposition on demand.

Definition 5.2 (Descent set). The *descent set* of $w \in \mathfrak{S}_n$ is

$$D(w) = \{i \in \{1, \dots, n-1\} : w_i > w_{i+1}\}.$$

[Rocq, descent.v:25]

```
Definition descent_set s : {set 'I_n} := [set i | is_descent s i].
```

Decoded. $D(s) = \{i \in \{0, \dots, n-1\} : s(i) > s(i+1)\}.$

Design choice: why $\{\text{set } 'I_n\}$, not $\text{pred } 'I_n$ or $\text{seq } 'I_n$?

Three options exist for representing a finite collection of positions: a boolean predicate ($\text{pred } T$), a list ($\text{seq } T$), or a set ($\{\text{set } T\}$). $\{\text{set } 'I_n\}$ is the right choice here for two reasons. First, it has *extensional equality*: two sets are equal iff they have the same elements, so $\text{descent_set } s = D$ is the condition we want, with no axiom of function extensionality required. Second, $\{\text{set } 'I_n\}$ is itself a finite type, which means we can sum over it with mathcomp’s big operators — the lemma `beta_eulerian` below uses $\sum_{(D : \{\text{set } 'I_n\} \mid \#|D| == k)}$ exactly because $\{\text{set}\}$ supports this. $\text{pred } 'I_n$ fails the first test (extensional equality of predicates needs `funext`); $\text{seq } 'I_n$ carries spurious order and multiplicity.

$D(w)$ is a subset of $\{1, \dots, n-1\}$ in Stanley; in our types it is a subset of $\{0, \dots, n-1\} = 'I_n$. The correspondence is Stanley’s $i \leftrightarrow$ our $(i-1)$.

Example 5.3 (Stanley’s $w = [3, 1, 4, 2]$, both ways). Take $w = [3, 1, 4, 2] \in \mathfrak{S}_4$. Then $D(w) = \{1, 3\}$ since $w_1 = 3 > 1 = w_2$ and $w_3 = 4 > 2 = w_4$. In our types, the corresponding $s : \{\text{perm } 'I_4\}$ has $s\ 0 = 2, s\ 1 = 0, s\ 2 = 3, s\ 3 = 1$ (values shifted to $\{0, 1, 2, 3\}$). Then:

```

Compute is_descent s 0.      (* = true, since s 0 = 2 > 0 = s 1 *)
Compute is_descent s 1.      (* = false, since s 1 = 0 < 3 = s 2 *)
Compute is_descent s 2.      (* = true, since s 2 = 3 > 1 = s 3 *)
Compute descent_set s.       (* = [set 0; 2] : {set 'I_3} *)
Compute des s.               (* = 2 *)

```

After the index shift our $\{0, 2\}$ corresponds to Stanley’s $\{1, 3\}$, and $\text{des}(s) = 2 = \text{des}(w)$. The computation is what the kernel does on the elaborated term — the formal definition matches the informal one not by editorial promise but by the test running here on the page.

Definition 5.4 (Descent number, ascent number). $\text{des}(w) = |D(w)|$ and $\text{asc}(w) = (n-1) - \text{des}(w)$.

[Rocq, descent.v:27]

```

Definition des s : nat := #|descent_set s|.

```

Decoded. $\text{des}(s) = |D(s)|$, the number of descent positions.

[Rocq, descent.v:29]

```

Definition asc s : nat := n - des s.

```

Decoded. $\text{asc}(s) = n - \text{des}(s)$, the number of ascent positions (here n is the number of descent slots, i.e. Stanley’s $n-1$).

Design choice: why is the parameter n “perm size minus one”?

A permutation of $n+1$ elements has n adjacencies, so the descent set lives in $\{0, \dots, n-1\}$ regardless of how we parameterise. Two choices: name the parameter after the perm size (forcing every descent-set type to be $\{\text{set } 'I_{n-1}\}$ with a `.-1`), or name it after the descent slot count (forcing the perm type to be $\{\text{perm } 'I_{n+1}\}$ with a `+.1`). We picked the latter because it removes a `.-1` from every theorem about descent sets and from every `eulerian n k` statement. The cost is a single conversion: **our parameter n is Stanley’s $n-1$** , and our `eulerian n k` is Stanley’s $A(n+1, k)$. This is a project convention, not a mathcomp requirement; mathcomp itself uses both styles in different files. We flag the off-by-one explicitly at every theorem statement that crosses it.

Lemma 5.5 (Descent count is bounded). $\text{des}(w) \leq n-1$ for every $w \in \mathfrak{S}_n$.

[Rocq, descent.v:38]

```

Lemma des_le s : des s ≤ n.

```

Decoded. $\text{des}(s) \leq n$ for every $s : \{\text{perm } 'I_{n+1}\}$ (equivalently $\text{des}(w) \leq |w| - 1$).

Proof sketch. $D(w) \subseteq \{0, \dots, n-1\}$ which has n elements; the cardinality is at most n . $\square$

Lemma 5.6 (Identity has no descents). $\text{des}(\text{id}) = 0$.

[Rocq, descent.v:44]

```
Lemma des_id : des (1 : {perm 'I_n.+1}) = 0.
```

Decoded. $\text{des}(\text{id}) = 0$, where $1\mathbf{g}$ is the group identity, i.e. the identity permutation.

Proof sketch. For the identity, $si = i$ at every position, so $s(i) = i < i + 1 = s(i + 1)$ — no descent slot. $\square$

5.2 Reversal symmetry

A useful symmetry: reversing a permutation flips ascents and descents at corresponding positions.

Definition 5.7 (Reverse permutation). Let $w \in \mathfrak{S}_n$ and define w^{rev} by $w_i^{\text{rev}} = w_{n+1-i}$.

[Rocq, descent.v:80]

```
Definition rev_perm_ord : {perm 'I_n.+1} := perm (@rev_ord_inj
  n.+1).
```

Decoded. The position-reversal involution $i \mapsto n - i$, packaged as a permutation. `perm (h : injective f)` is the constructor that turns an injective function into a `{perm T}`.

[Rocq, descent.v:85]

```
Definition rev_perm s : {perm 'I_n.+1} := rev_perm_ord * s.
```

Decoded. $w^{\text{rev}}(i) = s(n - i)$, the reversal of s viewed at the level of positions.

Design choice: why `rev_perm_ord * s`, not `s * rev_perm_ord`?

Mathcomp's group convention is $(s * t) i = t (s i)$ (see the lemma `permM`). With `rev_perm := rev_perm_ord * s`, applying gives $(\text{rev_perm_ord} * s) i = s (\text{rev_perm_ord } i) = s (n - i)$. This reverses *positions* (Stanley's $w_i^{\text{rev}} = w_{n+1-i}$). Post-composing instead would reverse *values*, a different operation. Picking the wrong side here would silently change the lemma's content; the choice matches Stanley by direct check on a small example.

Lemma 5.8 (Descents of the reverse). *Position i is a descent of w^{rev} iff position $n - i$ is not a descent of w :*

$$i \in D(w^{\text{rev}}) \iff (n - i) \notin D(w).$$

[Rocq, descent.v:90]

```
Lemma is_descent_rev s (i : 'I_n) :
  is_descent (rev_perm s) i = ~~ is_descent s (rev_ord i).
```

Decoded. Position i is a descent of w^{rev} iff position $n - i$ is *not* a descent of w . The `~~` is boolean negation.

Proof sketch. Direct calculation: $(\text{rev_perms}) j = s(n - j)$, and the comparison $s(n - i) > s(n - i - 1)$ is the negation of $s(n - i - 1) > s(n - i)$. $\square$

5.3 Eulerian numbers

Definition 5.9 (Eulerian number). The *Eulerian number* $A(n, k)$ is the number of permutations of $\{1, \dots, n\}$ with exactly k descents:

$$A(n, k) = \#\{w \in \mathfrak{S}_n : \text{des}(w) = k\}.$$

[Rocq, eulerian.v:14]

```
Definition eulerian (n k : nat) : nat :=
  #| [set s : {perm 'I_n.+1} | des s == k] |.
```

Decoded. $\text{eulerian } n k = |\{s \in \mathfrak{S}_{n+1} : \text{des}(s) = k\}|$. The $==$ is boolean equality on nat , and $\#|\dots|$ is set cardinality.

Off-by-one alert. The Rocq parameter n is “permutation size minus one” (see the design-choice sidebar above). Concretely, $\text{eulerian } n k$ counts perms of $\{\text{perm 'I}_{n+1}\}$, which are Stanley’s $\mathfrak{S}_{n+1}$. Hence

$$\text{eulerian } n k = A(n+1, k).$$

For instance, $\text{eulerian } 3 1$ corresponds to Stanley’s $A(4, 1) = 11$ (Stanley table p. 36).

```
Compute eulerian 0 0.      (* = 1      -- Stanley A(1,0) = 1 *)
Compute eulerian 1 0.      (* = 1      -- Stanley A(2,0) = 1 *)
Compute eulerian 1 1.      (* = 1      -- Stanley A(2,1) = 1 *)
Compute eulerian 2 0.      (* = 1      -- Stanley A(3,0) = 1 *)
Compute eulerian 2 1.      (* = 4      -- Stanley A(3,1) = 4 *)
Compute eulerian 2 2.      (* = 1      -- Stanley A(3,2) = 1 *)
```

The numbers match Stanley’s table on p. 36. Computing eulerian on $\{\text{perm 'I}_{n+1}\}$ requires enumerating the set; for $n \geq 4$ the brute-force enumeration is slow, but for the small cases above it returns immediately and the values agree.

Lemma 5.10 (Row sum). $\sum_{k=0}^{n-1} A(n, k) = n!$

[Rocq, eulerian.v:31]

```
Lemma eulerian_row_sum_fact n :
  \sum_ (k < n.+1) eulerian n k = n.+1'!.
```

Decoded. $\sum_{k=0}^n A(n+1, k) = (n+1)!$. The $'!$ is mathcomp’s factorial notation; $\backslash\text{sum_}(k < n.+1)$ sums over $k \in \{0, \dots, n\}$.

Proof sketch. The summands partition $\mathfrak{S}_{n+1}$ by descent count; the total is $|\mathfrak{S}_{n+1}| = (n+1)!$. $\square$

Lemma 5.11 (Out-of-range vanishing). $A(n, k) = 0$ for $k \geq n$.

[Rocq, eulerian.v:34]

```
Lemma eulerian_out_of_range n k :
  n < k  $\rightarrow$  eulerian n k = 0.
```

Decoded. If $k > n$ then $A(n+1, k) = 0$, since perms of length $n+1$ have at most n descents.

Proof sketch. A permutation has at most n descent positions (Lemma 5.5); none can have $k > n$ descents. $\square$

Lemma 5.12 (The unique perm with no descents is the identity). If $\text{des}(w) = 0$ then $w = \text{id}$.

[Rocq, eulerian.v:41]

```
Lemma des0_id n (s : {perm 'I_n.+1}) : des s = 0  $\rightarrow$  s = 1%g.
```

Decoded. If $\text{des}(s) = 0$ then s is the identity permutation.

Proof sketch. $\text{des}(s) = 0$ means $s(i) < s(i+1)$ for every adjacent pair i , so s is strictly increasing. The only strictly increasing function $'\text{I}_{n+1} \rightarrow '\text{I}_{n+1}$ is the identity. $\square$

5.4 The insert-max bijection

A central tool for the Eulerian recurrence is the bijection that inserts the maximum value $n + 1$ at a chosen position into a permutation of $\{1, \dots, n\}$.

Definition 5.13 (Insert-max). For $t \in \mathfrak{S}_n$ and $p \in \{1, \dots, n + 1\}$, let $\text{insert_max}(t, p) \in \mathfrak{S}_{n+1}$ be the permutation obtained by inserting the value $n + 1$ at position p in the one-line form of t .

[Rocq, eulerian.v:134-157]

```

Definition insert_max_fun (i : 'I_n.+2) : 'I_n.+2 :=
  match unlift p i with
  | Some j  $\Rightarrow$  widen_ord (leqnSn _) (t j)
  | None  $\Rightarrow$  ord_max
  end.

Definition insert_max_perm : {perm 'I_n.+2} := perm
  insert_max_fun_inj.

```

Decoded. For $i \in \{0, \dots, n + 1\}$: if $i = p$, output $n + 1$ (the new max); otherwise, write $i = \text{lift } p \ j$ for the unique $j \in \{0, \dots, n\}$ that skips p , and output $t(j)$ cast into the larger type.

Design choice: why match unlift p i with?

The fundamental case-split for inserting at position p is “is this position p or not?”. Mathcomp encodes this as the option-valued partial inverse $\text{unlift } p$: it returns None exactly when $i = p$, and $\text{Some } j$ otherwise — where $j : 'I_{n+1}$ is the unique witness of $i = \text{lift } p \ j$. Carrying j in the Some branch (rather than recomputing it from i and p) avoids any arithmetic juggling: we get the correct $'I_n.+1$ value with its bound proof for free, and can apply t directly. This is the universal idiom for constructing permutations on $'I_n.+2$ from a permutation on $'I_n.+1$ plus a special index.

Definition 5.14 (Extract-max). The inverse: given $\sigma \in \mathfrak{S}_{n+1}$ with $\sigma(p) = n + 1$ for some position p , remove that value and standardise to obtain $t \in \mathfrak{S}_n$.

[Rocq, eulerian.v:185-203]

```

Definition extract_max_fun (j : 'I_n.+1) : 'I_n.+1 :=
  odflt j (unlift ord_max (s (lift p j))).

Definition extract_max_perm : {perm 'I_n.+1} := perm
  extract_max_fun_inj.

```

Decoded. For $j \in \{0, \dots, n\}$: look at s at position j of the original (skipping p); the result is some value in $\{0, \dots, n + 1\}$ that is not $n + 1$ (because $\sigma(p) = n + 1$ and σ is a bijection); strip the high bit to land in $\{0, \dots, n\}$. The $\text{odflt } j \ (\dots)$ provides a default that is provably unreachable.

Theorem 5.15 (Insert-max is a bijection). *The map $(t, p) \mapsto \text{insert_max}(t, p)$ is a bijection $\mathfrak{S}_n \times \{1, \dots, n + 1\} \rightarrow \mathfrak{S}_{n+1}$.*

[Rocq, eulerian.v:437]

```

Lemma insert_max_perm_bij n :
   $\forall$  sigma : {perm 'I_n.+2},
   $\exists!$  tp : {perm 'I_n.+1} * 'I_n.+2,
    sigma = insert_max_perm tp.1 tp.2.

```

Decoded. Every $\sigma \in \mathfrak{S}_{n+2}$ arises uniquely as $\text{insert_max}(t, p)$ for some $(t, p) \in \mathfrak{S}_{n+1} \times \{0, \dots, n + 1\}$.

Proof sketch. Given σ , the unique p is $\sigma^{-1}(n + 1)$ and $t = \text{extract_max}(\sigma, p)$. The two roundtrips are direct calculations. $\square$

5.5 Eulerian recurrence

Theorem 5.16 (Eulerian recurrence). *For $n \geq 1$ and $0 \leq k \leq n$,*

$$A(n+1, k) = (k+1)A(n, k) + (n-k+1)A(n, k-1).$$

[Rocq, eulerian.v:458]

```
Theorem eulerian_rec n k :
  eulerian n.+1 k =
    (k.+1) * eulerian n k + (n.+1 - k) * eulerian n k.-1.
```

Decoded. $A(n+2, k) = (k+1)A(n+1, k) + (n+1-k)A(n+1, k-1)$. The $n.+1-k$ on the second term reads as “ $n+1-k$ ” under our parameterisation; compare to Stanley’s $(n+1-k)$ in $A(n+2, k) = (k+1)A(n+1, k) + (n+1-k)A(n+1, k-1)$.

Proof sketch. Using the insert-max bijection, classify each $\sigma \in \mathfrak{S}_{n+1}$ by the position p of $n+1$. There are $k+1$ ascent positions and $n-k$ descent positions where inserting $n+1$ preserves the descent count; inserting at the other $n-k+1$ slots increases it by one. Summing recovers the recurrence. $\square$

5.6 The set-refined count β

The descent count is a coarse invariant. The full descent *set* carries more information; the count of permutations with a fixed descent set is a finer object.

Definition 5.17 (β count). For a subset $S \subseteq \{1, \dots, n-1\}$, the count $\beta_n(S)$ is the number of permutations of $\{1, \dots, n\}$ whose descent set is exactly S :

$$\beta_n(S) = \#\{w \in \mathfrak{S}_n : D(w) = S\}.$$

[Rocq, beta.v:19]

```
Definition beta (n : nat) (D : {set 'I_n}) : nat :=
  #|[set sigma : {perm 'I_n.+1} | descent_set sigma == D]|.
```

Decoded. $\text{beta } n D = |\{\sigma \in \mathfrak{S}_{n+1} : D(\sigma) = D\}|$, the number of perms of length $n+1$ with descent set exactly D .

Off-by-one alert. As with `eulerian`, the Rocq parameter n is “perm size minus one.” So `beta n D` for $D : \{\text{set 'I}_n\}$ equals Stanley’s $\beta_{n+1}(D)$, with D interpreted as a subset of $\{0, \dots, n-1\}$ that corresponds to Stanley’s 1-indexed $S \subseteq \{1, \dots, n\}$.

Lemma 5.18 (β partitions Eulerian). *Summing $\beta_n(S)$ over all S of cardinality k recovers the Eulerian number:*

$$\sum_{|S|=k} \beta_n(S) = A(n, k).$$

[Rocq, beta.v:124]

```
Lemma beta_eulerian n k :
  \sum_ (D : {set 'I_n} | #|D| == k) beta D = eulerian n k.
```

Decoded. $\sum_{D \subseteq \{0, \dots, n-1\}, |D|=k} \beta_{n+1}(D) = A(n+1, k)$. The big-op syntax `\sum_ (D : {set 'I_n} | #|D| == k)` sums over all sets D of cardinality exactly k .

Proof sketch. The sets $\{\sigma : D(\sigma) = D\}$ for D ranging over subsets of cardinality k partition the set of permutations with k descents. The cardinalities sum. $\square$

Lemma 5.19 (β is reversal-invariant). For $S \subseteq \{1, \dots, n-1\}$, let $S^{\text{rev}} = \{n-i : i \in S\}$. Then $\beta_n(S^{\text{rev}}) = \beta_n(S^c)$, where S^c is the complement in $\{1, \dots, n-1\}$.

[Rocq, beta.v:140]

```
Lemma beta_rev n (D : {set 'I_n}) :
  beta [set rev_ord i | i in D] = beta (~: D).
```

Decoded. $\beta_{n+1}(\{n-1-i : i \in D\}) = \beta_{n+1}(\overline{D})$. The set comprehension $[\text{set rev_ord } i \mid i \text{ in } D]$ is the image of D under $i \mapsto n-1-i$; $\sim:D$ is the complement of D in $'I_n$.

Proof sketch. Reversal $w \mapsto w^{\text{rev}}$ is a bijection on $\mathfrak{S}_n$; by Lemma 5.8 it maps descent set D to $\{n-i : i \notin D\}$, which is the complement of $\{n-i : i \in D\}$. Counting on both sides matches. $\square$

5.7 What this chapter doesn't cover

This chapter establishes the basic objects $(D, \text{des}, A, \beta)$ and the foundational identities. The headline of the formalization,

$$\beta_n(S) \leq E_n \quad \text{with equality iff } S \text{ is alternating} \quad (\text{Stanley Cor 1.6.5})$$

requires the toggle-action machinery of §1.6.3 and is the subject of a future chapter.

5.8 Worpitzky's identity and the Eulerian generating function

Theorem 5.20 (Worpitzky). For all $m, n \geq 0$,

$$(m+1)^{n+1} = \sum_{k=0}^n A(n, k) \binom{m+n+1-k}{n+1},$$

where $A(n, k)$ counts the permutations of $n+1$ letters with k descents.

[Rocq, worpitzky.v:56]

```
Theorem worpitzky (n m : nat) :
  ((m.+1) ^ n.+1
   = \sum_ (k < n.+1) eulerian n k * 'C(m + n.+1 - k, n.+1))%N.
```

Decoded. A pure statement about natural numbers — axiom-free, in the combinatorial core. Proved by induction on n from the Eulerian recurrence `eulerian_rec`, with a Pascal-style splitting of $(m+1) \binom{m+n+1-k}{n+1}$ at each step.

Packaged as generating functions (Stanley §1.4): with $A_n(x) = \text{eul_pol } n$ the Eulerian polynomial,

$$\sum_{m \geq 0} (m+1)^{n+1} x^m = \frac{A_n(x)}{(1-x)^{n+2}}.$$

[Rocq, stanley_ogf.v:69]

```
Theorem stanley_1_4 n :
  pow_ogf n * ((1 - 'Xf) ^+ n.+2) = poly_fps (eul_pol n).
```

Decoded. An identity in `{fps int}` — note the coefficient ring: $\mathbb{Z}$ is not a field, and the inverse of $(1-x)^{n+2}$ exists because the `mathcomp_fps` unit theory only asks for an invertible constant coefficient. The coefficient-level content is exactly Worpitzky's identity combined with the negative-binomial expansion $(1-x)^{-(k+1)} = \sum_m \binom{m+k}{k} x^m$ (`coef_fps_geomXn`). This layer uses the classical axioms of `mathcomp-classical`; the Worpitzky identity itself does not.

Chapter 6

The q -analogues

This chapter formalizes the q -extensions of §1.4: the q -factorial $[n]_q!$ as the inversion generating function over $\mathfrak{S}_n$, and the bivariate q -Eulerian polynomial $A_n(q, t)$ as the joint generating function for (maj, des) . The two specializations $t = 1$ and $q = 1$ recover $[n]_q!$ and the classical Eulerian polynomial respectively.

6.1 q -integers and the q -factorial

Definition 6.1 (q -integer). $[n]_q = 1 + q + q^2 + \cdots + q^{n-1}$.

[Rocq, qfact.v:25]

```
Definition q_int (n : nat) : {poly int} := \sum_ (i < n) 'X^i.
```

Decoded. $[n]_q$ as a polynomial in the formal indeterminate $'X$, with integer coefficients. $\sum_{i < n} 'X^i$ sums X^i for $i \in \{0, \dots, n-1\}$.

Definition 6.2 (q -factorial). $[n]_q! = [1]_q \cdot [2]_q \cdots [n]_q$.

[Rocq, qfact.v:26]

```
Definition q_fact (n : nat) : {poly int} := \prod_ (k < n.+1) q_int k.+1.
```

Decoded. $\text{q_fact } n = [1]_q \cdot [2]_q \cdots [n+1]_q = [n+1]_q!$. The off-by-one is the same as in `eulerian`: our parameter is “size minus one,” so $\text{q_fact } n = [n+1]_q!$ in Stanley’s notation.

Theorem 6.3 (inv-generating function for $\mathfrak{S}_n$). $\sum_{w \in \mathfrak{S}_n} q^{\text{inv}(w)} = [n]_q!$

[Rocq, qfact.v:188]

```
Theorem inv_q_fact n :
  \sum_ (\sigma : {perm 'I_n.+1}) 'X^(inv \sigma) = q_fact n :> {poly int}.
```

Decoded. The polynomial $\sum_{\sigma \in \mathfrak{S}_{n+1}} X^{\text{inv}(\sigma)}$ is exactly $[n+1]_q!$.

Proof sketch. Induction on n : at each step, classify permutations of $\mathfrak{S}_{n+2}$ by where the maximum was inserted (using `insert_max_perm_bij`); inserting at position p adds exactly $n+1-p$ inversions, so the sum factors as $[n+1]_q! \cdot [n+2]_q$. $\square$

Theorem 6.4 (maj-generating function via Foata). $\sum_{w \in \mathfrak{S}_n} q^{\text{maj}(w)} = [n]_q!$

[Rocq, qfact.v:218]

```
Theorem maj_q_fact n :
  \sum_ (σ : {perm 'I_n.+1}) 'X^(maj σ) = q_fact n := {poly int}.
```

Decoded. Same identity as `inv_q_fact` but for `maj` instead of `inv` — a direct corollary of MacMahon’s equidistribution.

Proof sketch. Reindex the `maj`-sum along Foata’s bijection: $\sum_{\sigma} X^{\text{maj}(\sigma)} = \sum_{\sigma} X^{\text{inv}(\text{foata}(\sigma))}$ which equals $[n+1]_q!$ by the previous theorem. $\square$

6.2 The bivariate q -Eulerian polynomial

Definition 6.5 (q -Eulerian polynomial). $A_n(q, t) = \sum_{w \in \mathfrak{S}_n} q^{\text{maj}(w)} \cdot t^{\text{des}(w)}$.

[Rocq, qeul.v:37]

```
Definition q_eul_pol (n : nat) : {poly {poly int}} :=
  \sum_ (σ : {perm 'I_n.+1}) qpow (maj σ) * 'X^(des σ).
```

Decoded. $A_{n+1}(q, t) = \sum_{\sigma \in \mathfrak{S}_{n+1}} q^{\text{maj}(\sigma)} t^{\text{des}(\sigma)}$. Bivariate: outer indeterminate `'X` is t , inner (under `qpow`) is q .

Design choice: why bivariate `{poly {poly int}}`?

The natural type for “polynomial in t with coefficients polynomials in q ” is the iterated polynomial ring `{poly {poly int}}`. Mathcomp’s polynomial library is fully composable: `{poly R}` is itself a ring (in fact a commutative ring if R is), so `{poly {poly int}}` inherits all the algebra. The trade-off is that the syntax is heavier — the constant-poly embedding `_%:P`, the notation `qpow k` (for the inner q^k embedded as an outer constant), and `horner_eval` for substitution all need to be juggled. We name them in advance to keep the recurrence readable.

Theorem 6.6 (Specialize $t := 1$). $A_n(q, 1) = [n]_q!$

[Rocq, qeul.v:44]

```
Theorem q_eul_pol_t1 n :
  (q_eul_pol n).[1] = q_fact n.
```

Decoded. Evaluating $A_{n+1}(q, t)$ at $t = 1$ gives $\sum_{\sigma} q^{\text{maj}(\sigma)} \cdot 1^{\text{des}(\sigma)} = \sum_{\sigma} q^{\text{maj}(\sigma)} = [n+1]_q!$ (the previous theorem).

Definition 6.7 (Classical Eulerian polynomial). $A_n(t) = \sum_{k \geq 0} A(n, k) \cdot t^k$.

[Rocq, qeul.v:58]

```
Definition eul_pol (n : nat) : {poly int} :=
  \sum_ (k < n.+1) (eulerian n k)_%:R * 'X^k.
```

Decoded. $\text{eul_pol } n = \sum_{k=0}^n A(n+1, k) \cdot t^k$ (the off-by-one inherited from `eulerian`).

Theorem 6.8 (Specialize $q := 1$). $A_n(1, t) = A_n(t)$, the classical Eulerian polynomial.

[Rocq, qeul.v:84]

```
Theorem q_eul_pol_q1 n :
  q1_subst (q_eul_pol n) = eul_pol n.
```

Decoded. Substituting $q = 1$ in each coefficient of $A_{n+1}(q, t)$ recovers the classical Eulerian polynomial. The substitution is implemented by `q1_subst = map_poly (horner_eval 1)`, which evaluates each inner q -polynomial at $q = 1$.

Chapter 7

Longest alternating subsequence

This chapter formalizes Stanley §1.6.2: the longest alternating subsequence statistic $\text{as}(w)$, with the closed-form characterization

$$\text{as}(w) = \text{turn_count}(w) + 2,$$

where the right-hand side is two more than the number of “turning points” (positions where the direction of w changes).

7.1 Turning points

Definition 7.1 (Turning point). Position i is a *turning point* of w if the direction at position i differs from the direction at position $i + 1$: exactly one of $\{i, i + 1\}$ is a descent position.

[Rocq, altsub.v:41]

```
Definition is_turn s (i : 'I_n) : bool :=
  is_descent s (widen_ord (leqnSn _) i) (+) is_descent s (lift ord0
    i).
```

Decoded. $\text{is_turn } s \ i$ is the boolean XOR of two adjacent is_descent indicators. For $s : \{\text{perm } 'I_{n+2}\}$ and $i : 'I_n$, the two arguments of XOR are descent slots i and $i + 1$ (both in $'I_{n+1}$, via the same widen+lift idiom as Chapter 5).

[Rocq, altsub.v:44]

```
Definition turn_count s : nat :=
  #|[set i : 'I_n | is_turn s i]|.
```

Decoded. $\text{turn_count}(s) = \text{number of turning positions.}$

7.2 Alternating subsequences

Definition 7.2 (Alternating sequence of naturals). A seq $x_1, x_2, \dots, x_k$ is *alternating* if $x_1 < x_2 > x_3 < \dots$ or $x_1 > x_2 < x_3 > \dots$.

[Rocq, altsub.v:71]

```
Definition is_alt (xs : seq nat) : bool :=
  match xs with
  | [::] => true
  | [:: _] => true
  | x :: y :: xs' =>
```

```

      ((x < y) && alt_aux false y xs') || ((y < x) && alt_aux true
      y xs')
    end.

```

Decoded. Empty and singleton seqs are alternating. Otherwise we try both starting directions: “up first” ($x < y$ then `alt_aux false`) or “down first” ($y < x$ then `alt_aux true`); `is_alt` holds iff either branch succeeds. The auxiliary `alt_aux` alternates the boolean parity at each step.

Definition 7.3 (Permutation as a seq, subseq at chosen positions).

[Rocq, altsub.v:84]

```

Definition perm_seq n (s : {perm 'I_n.+2}) : seq nat :=
  [seq val (s i) | i <- enum 'I_n.+2].

```

Decoded. The one-line form of s : the seq $[s(0); s(1); \dots; s(n+1)]$ of natural numbers.

[Rocq, altsub.v:97]

```

Definition pick_seq n (s : {perm 'I_n.+2}) (I : {set 'I_n.+2}) :
  seq nat :=
  [seq val (s j) | j <- sort (fun a b => val a ≤ val b) (enum I)].

```

Decoded. For an index set $I \subseteq \{0, \dots, n+1\}$, sort the positions in I in increasing order and pick the corresponding values from s . This is “the subsequence of s at positions I , in order.”

7.3 The two definitions of as

Definition 7.4 (Bijective form: `as_perm_max`). $\text{as_perm_max}(s) = \max\{|I| : \text{pick_seq}(s, I) \text{ is alternating}\}$.

[Rocq, altsub.v:105]

```

Definition as_perm_max n (s : {perm 'I_n.+2}) : nat :=
  \max_(I : {set 'I_n.+2} | is_alt (pick_seq s I)) #|I|.

```

Decoded. The maximum cardinality over all index sets I whose sorted-value picked subseq is alternating. This is the geometric “length of the longest alternating subseq” definition.

Definition 7.5 (Closed-form: `as_perm`). $\text{as_perm}(s) = \text{turn_count}(s) + 2$.

[Rocq, altsub.v:109]

```

Definition as_perm n (s : {perm 'I_n.+2}) : nat := (turn_count
  s).+2.

```

Decoded. Two more than the number of turning positions. This is Stanley’s *closed-form* characterization, taken here as a definition; the equivalence with the bijective form is the headline theorem below.

7.4 The headline equivalence

Theorem 7.6 (Stanley’s characterization, §1.6.2). *For every $s \in \mathfrak{S}_{n+2}$, $\text{as_perm_max}(s) = \text{turn_count}(s) + 2$.*

[Rocq, altsub.v:2114]

```

Theorem as_perm_max_eq s : as_perm_max s = (turn_count s).+2.

```

Decoded. The longest alternating subsequence of s has length exactly $\text{turn_count}(s) + 2$.

■ Design choice: why this took five sessions

The proof has two halves — $\geq$ (existence of a long alt subseq) and $\leq$ (no longer one exists). The lower bound is constructive: pick the boundary endpoints plus one position per turning point. The upper bound is where the difficulty was: the naive “leftmost turn in $[\sigma_i, \sigma_{i+2})$ ” injection from sign-flips of the picked subseq to turning positions of s *is not injective* — adjacent sign-flips can share a single turning window. The proof in `altsub.v` pivots to a global bound via the seq-level statistic `flip_count`, which is monotone under taking subseqs: `flip_count(sign_seq(pick_seq s I)) ≤ flip_count(sign_seq(perm_seq s)) = turn_count(s)`. The detour is recorded in `docs/plans/ALTSUB_PLAN.md`.

Chapter 8

Toggle action and Stanley's Corollary

1.6.5

This chapter formalizes Stanley §1.6.3: the toggle action on descent sets, the ω -map, and the headline corollary that the alternating descent set strictly maximizes $\beta_n(S)$ — the central theorem of this project.

8.1 Symmetric difference and single-position toggle

Definition 8.1 (Symmetric difference, single-position toggle).

[Rocq, beta_omega.v:25]

```
Definition sym_diff (n : nat) (D E : {set 'I_n}) : {set 'I_n} :=  
  (D :\ E) :| (E :\ D).
```

Decoded. Standard set-theoretic symmetric difference $D \triangle E$.

[Rocq, beta_omega.v:28]

```
Definition toggle_at (n : nat) (D : {set 'I_n}) (i : 'I_n) : {set  
  'I_n} :=  
  sym_diff D [set i].
```

Decoded. $\text{toggle_at}(D, i) = D \triangle \{i\}$. Adds i if absent, removes it if present.

8.2 The ω -map

For a descent set $D \subseteq \{1, \dots, n-1\}$ in Stanley's $\mathfrak{S}_n$, $\omega(D)$ records the positions where adjacent membership in D changes.

[Rocq, beta_omega.v:57]

```
Definition omega_set (n : nat) (D : {set 'I_n.+1}) : {set 'I_n} :=  
  [set k : 'I_n | (widen_ord (leqnSn n) k \in D) != (lift_ord0 k  
    \in D)].
```

Decoded. $k \in \omega(D)$ iff membership of position k in D differs from membership of position $k+1$ in D . The two membership tests use the now-familiar widen+lift idiom to view k and $k+1$ inside the larger type.

8.3 The alternating descent set

Definition 8.2 (Alternating descent pattern). $D_{\text{alt}}^n = \{1, 3, 5, \dots\} \cap \{1, \dots, n-1\}$ (Stanley’s 1-indexed odd positions).

[Rocq, beta_swap.v:31]

```
Definition alt_desc_set (n : nat) : {set 'I_n} :=
  [set i : 'I_n | ~~ odd i].
```

Decoded. The set of *even-indexed* positions $\{0, 2, 4, \dots\}$ of $'I_n$. Under our 0-indexing, this matches Stanley’s 1-indexed odd positions $\{1, 3, 5, \dots\}$.

[Rocq, beta_swap.v:40]

```
Definition set_is_alt (n : nat) (D : {set 'I_n}) : bool :=
  [∀ i : 'I_n, [∀ j : 'I_n,
    (val j == (val i).+1) ⇒ ((i \in D) != (j \in D))]].
```

Decoded. $\text{set_is_alt}(D)$: every consecutive pair $(i, i+1)$ in $'I_n$ has differing membership in D . Equivalently, D is either D_{alt}^n or its complement (lemma `set_is_alt_classify` below).

8.4 Value-complement and β symmetry

[Rocq, beta_swap.v:128]

```
Definition compl_perm n (s : {perm 'I_n.+1}) : {perm 'I_n.+1} :=
  s * rev_perm_ord n.
```

Decoded. The value-complement of s : $\text{compl_perm}(s)i = n - s(i)$. Defined by post-composition with the value-reversal involution.

Lemma 8.3 (Descent set under value-complement). $D(\text{compl_perm}(s)) = \overline{D(s)}$ (*the complement in $'I_n$*).

[Rocq, beta_swap.v:160]

```
Lemma descent_set_compl n (s : {perm 'I_n.+1}) :
  descent_set (compl_perm s) = ~: descent_set s.
```

Decoded. Value-complementing every value flips ascents into descents and vice-versa, so the descent set goes to its complement.

Lemma 8.4 (β is complement-invariant). $\beta_n(D) = \beta_n(\overline{D})$.

[Rocq, beta_swap.v:166]

```
Lemma beta_compl n (D : {set 'I_n}) : beta D = beta (~: D).
```

Decoded. The number of perms with descent set D equals the number with descent set $\overline{D}$, by the bijection $s \mapsto \text{compl_perm}(s)$.

Lemma 8.5 (Set-alt classification). $\text{set_is_alt}(D)$ holds iff $D = D_{\text{alt}}^n$ or $D = \overline{D_{\text{alt}}^n}$.

[Rocq, beta_swap.v:182]

```
Lemma set_is_alt_classify n (D : {set 'I_n}) :
  set_is_alt D = (D == alt_desc_set n) || (D == ~: alt_desc_set n).
```

Decoded. Exactly two sets in $'I_n$ are set-alternating: the alternating pattern itself and its complement. This justifies “the alternating descent set” modulo the two-flavour ambiguity.

8.5 Stanley’s Proposition 1.6.4

Proposition 8.6 (ω -strictness, Stanley Prop 1.6.4). *If $\omega(D) \subsetneq \omega(E)$ then $\beta_n(D) < \beta_n(E)$.*

[Rocq, perm_seq_bridge.v:540]

```
Lemma omega_proper_beta_lt : ∀ m (D E : {set 'I_m.+1}),
  omega_set D \proper omega_set E →
  beta D < beta E.
```

Decoded. Strictly enlarging the ω -image strictly increases β . The proof goes through a deep “cd-index” machinery (min-max trees, ψ_i operators) developed in Stanley’s chapter and formalized in this project’s `psi_*.v` files; that material is not reproduced here.

8.6 The headline: Stanley’s Corollary 1.6.5

Corollary 8.7 (Alternating set strictly maximizes β , Stanley Cor 1.6.5). *For every set $D \subseteq \{1, \dots, n-1\}$ that is not alternating,*

$$\beta_n(D) < E_n,$$

where $E_n = \beta_n(D_{\text{alt}}^n)$ is the Euler number.

[Rocq, beta_swap.v:273]

```
Lemma beta_alt_max n (D : {set 'I_n}) :
  ~~ set_is_alt D → beta D < beta (alt_desc_set n).
```

Decoded. If D is not set-alternating, then $\beta(D)$ is strictly smaller than $\beta(D_{\text{alt}}^n)$ — the alternating descent set is the *unique* (up to complement) maximizer of β .

Proof sketch. If D is not alternating then there exist adjacent positions $i, i+1$ both in D or both not. Toggling at i (or at a fitting nearby position) strictly enlarges $\omega(D)$, and Proposition 8.6 gives a chain of strict β -increases that terminates at D_{alt}^n . (The full chain construction uses `block_left/block_right` from `beta_omega.v`.) $\square$

This is the headline theorem of the entire project. To verify axiom-freeness:

```
echo 'From mathcomp_eulerian Require Import beta_swap.
      Print Assumptions beta_alt_max.' \
| rocq top -R . mathcomp_eulerian
```

prints Closed under the global context.

Chapter 9

André's reflection method

Status. The headline recurrence $2E_{n+2} = \sum_k \binom{n+1}{k} E_k E_{n+1-k}$ is now *fully formalized* in `reflection.v`, which is part of the active build chain — 0 admits, 0 axioms (`Print Assumptions euler_rec` prints `Closed under the global context`). The companion document `docs/internal/PHASE_C_F` captures the earlier sessions of work and the (since resolved) structural obstacle; see the remark at the end of this chapter.

9.1 Euler numbers

Definition 9.1 (Euler numbers). $E_{n+1} = \beta_{n+1}(D_{\text{alt}}^n)$ (the count of alternating permutations of $\mathfrak{S}_{n+1}$). Set $E_0 = 1$.

[Rocq, `reflection.v:28`]

```
Definition euler (n : nat) : nat := beta (alt_desc_set n).

Definition eulerA (n : nat) : nat :=
  if n is k.+1 then euler k else 1.
```

9.2 The headline recurrence

Theorem 9.2 (André recurrence). *For every $n \geq 0$,*

$$2E_{n+2} = \sum_{k=0}^{n+1} \binom{n+1}{k} E_k E_{n+1-k}.$$

[Rocq, `reflection.v:1988`]

```
Theorem euler_rec n :
  2 * eulerA n.+2
  = \sum_ (k < n.+2) 'C(n.+1, k) * eulerA k * eulerA (n.+1 - k).
```

`euler_rec` is derived in one line from `two_eulerA_split` and the boundary cancellation lemma:

$$\text{sum_set_is_alt_eq_andre_sum} : \sum_t \sum_p \text{set_is_alt}(D(\text{insert_max}(t, p))) = \sum_k \binom{n+1}{k} E_k E_{n+1-k}.$$

Proof sketch. The left side counts the pairs (t, p) — equivalently, via the `insert_max_perm` bijection, the permutations $\sigma \in \mathfrak{S}_{n+2}$ — whose descent set is set-alternating of *either* flavour; by `beta_comp1` this is $2E_{n+2}$. The inner sum over the slot p of the maximum splits into three groups (`sum_set_is_alt_ord0`, `sum_set_is_alt_ord_max`, `interior_set_is_alt`):

- $p = 0$ forces a descent at slot 0, so only the `alt` flavour survives and the tail contributes E_{n+1} — the $k = n + 1$ term.
- $p = n + 1$ forces an ascent at slot n ; the prefix contributes E_{n+1} — the $k = 0$ term.
- An interior slot $c + 1$ forces an ascent at c and a descent at $c + 1$, which pins the flavour to the parity of c . The surviving condition on t is $D(t) \cup \{c\} = P_c$ for a parity-prescribed target P_c (`andre_target`), and — the key point — this *factors* through the `(image_left, perm_left, perm_right)` decomposition (`andre_union_eq_split`): left and right sub-permutations must each be alternating of a parity-prescribed flavour, while the boundary descent of t at slot c is absorbed by the union $\cup \{c\}$ and is *unconstrained*. Counting via the `assemble_perm` bijection and `card_draws` (`andre_interior_count`) yields $\binom{n+1}{c+1} E_{c+1} E_{n-c}$ — the $k = c + 1$ term. $\square$

The base case $n = 0$ ($2E_2 = 2$) is also proven directly in the same file (`euler_rec_n0_direct`).

Remark 9.3 (The structural obstacle, resolved). Earlier sessions stalled on the boundary descent at the position of the inserted maximum, which appears to couple (L, s_L, s_R) jointly: whether the last left value exceeds the first right value depends on all three components. The resolution is that the coupling never needs to be analysed: the inserted maximum forces the descent at slot $c + 1$ of σ *regardless* of the boundary comparison in t , so the alternation condition on σ constrains $D(t) \cup \{c\}$ — and membership of the boundary slot c in $D(t)$ is swallowed by the union. The indicator therefore *does* factor into independent s_L - and s_R -conditions, with no 4-to-1 analysis required.

9.3 Proposition 1.6.1: the generating function of the Euler numbers

Theorem 9.4 (Stanley, Proposition 1.6.1). *As formal power series,*

$$\sum_{n \geq 0} E_n \frac{x^n}{n!} = \sec x + \tan x.$$

[Rocq, stanley_egf.v:66]

Theorem stanley_1_6_1 : euler_egf = secf + tanf :> {fps rat}.

Corollary stanley_1_6_1_coef n :
(eulerA n)%:R = n'!%:R * ((secf + tanf : {fps rat})' _n).

Decoded. `euler_egf` is the exponential generating function $\sum_n E_n x^n / n!$ of the Euler numbers, an element of the ring `{fps rat}` of formal power series over $\mathbb{Q}$; the right-hand side is built from the formal series `sec` = $\cos^{-1}$ and `tan` = $\sin \cdot \sec$. Everything is formal: no analysis, no convergence.

Proof sketch. The André recurrence `euler_rec` above translates verbatim into the formal differential equation $2A' = 1 + A^2$, $A(0) = 1$ for $A = \text{euler_egf}$ (the translation is one application of the EGF workhorse lemma `egf_mul`: products of egfs are egfs of binomial convolutions). The series `sec + tan` satisfies the same equation (`sectan_ode`, using $\sin^2 + \cos^2 = 1$, itself proved by the derivative trick), and solutions of this quadratic equation are determined by their constant coefficient (`fps_quad_ode_uniq`: coefficient $n+1$ of a solution is computed from coefficients $\leq n$ since $2(n+1)$ is invertible in characteristic 0). $\square$

The FPS sub-library. The proof is carried by a new reusable library of formal power series (`fps/`, namespace `mathcomp_fps`, Apache-2.0, depending only on `mathcomp` and `mathcomp-classical`): `{fps R}` is the type of coefficient sequences with the full `mathcomp` algebra hierarchy — in particular a *com-unit-ring for any* `comUnitRingType R`, with f invertible iff f_0 is — plus formal derivatives (Leibniz rule) and the EGF calculus. Ring axioms are inherited from `{poly R}` through truncation (coefficient n of a product only sees coefficients $\leq n$), not re-proved.

Axiom note. This generating-function layer (and only it) uses the three standard classical axioms of `mathcomp-classical` (functional/propositional extensionality, indefinite description) — the same set as `mathcomp-analysis`. `Print Assumptions stanley_1_6_1` lists exactly those; the combinatorial chapters remain axiom-free.

Appendix A

Comprehensive lemma catalog

This appendix lists *every* named result in the active build chain — 1078 entries across 45 files. Each entry shows the kind (**Definition**, **Lemma**, ...), the file:line location (clickable to GitHub), and the `coqdoc` docstring extracted from the source. Entries narrated in the main chapters of this document carry a chapter/section badge.

This catalog is generated mechanically from the `.v` files; see `docs/formal/extract_catalog.py`. Adding a docstring to a `.v` file improves both the catalog and the deployed `coqdoc` HTML at the same time.

altsub.v

is_turn *Definition, line 43.* [Ch 7 §7.1]

`is_turn s i` is the boolean indicator that interior position `i : 'I_n` is a turning point of `s : {perm 'I_n.+2}`: the descent indicator at slot `widen i` differs (XOR) from the indicator at slot `lift ord0 i`. These two slots straddle position `i`.

turn_count *Definition, line 49.* [Ch 7 §7.1]

`turn_count s` is the number of turning points of `s`, i.e., the cardinality of `[set i : 'I_n | is_turn s i]`. Stanley $\backslash S \{ \}$ 1.6.2: the "turn count" governing the longest alternating subsequence length.

alt_aux *Fixpoint, line 69.* [aux]

`alt_aux b x xs` checks alternation of the seq `x :: xs` given the expected direction `b` of the first comparison (`true` = up, `false` = down). Successive directions flip at each step.

is_alt *Definition, line 79.* [Ch 7 §7.2]

`is_alt xs` holds when `xs` is alternating: comparisons of consecutive elements strictly alternate between `<` and `>`. Two starts are possible (up-down or down-up); both are accepted by disjunction.

perm_seq *Definition, line 93.* [Ch 7 §7.2]

`perm_seq s` is the one-line word `[s 0; s 1; ...; s (n+1)]` of `s : {perm 'I_n.+2}`, as a `seq nat`.

size_perm_seq *Lemma, line 97.* [aux]

`perm_seq s` always has length `n.+2`.

pick_seq *Definition, line 108.* [Ch 7 §7.2]

`pick_seq s I` is the subseq of `perm_seq s` indexed by the position set `I : {set 'I_n.+2}` in ascending order: sort `I` by underlying `val`, then read off the `s`-values.

as_perm_max *Definition, line 114.* [Ch 7 §7.3]

as_perm_max *s* is the bijective definition of **as**(*s*) from Stanley \S{}1.6.2: the maximum cardinality of a position set *I* whose ordered image **pick_seq s I** is alternating.

as_perm *Definition, line 120.* [Ch 7 §7.3]

as_perm *s* is the DIRECT definition (Path X) of **as**(*s*) as **(turn_count s).+2**. The headline **as_perm_max_eq** proves these two definitions agree.

is_alt_cons2 *Lemma, line 164.* [aux]

Defining equation of **is_alt** on a 2-element prefix *x :: y :: xs*: either start with an ascent or a descent. Used as a rewrite.

turn_count_le *Lemma, line 186.* [aux]

Cardinality bound: the turn-set of *s* : **{perm 'I_n.+2}** has at most *n* turning points.

as_permE *Lemma, line 192.* [aux]

Definitional unfolding **as_perm s = (turn_count s).+2**; rewrite rule.

turn_count_id *Lemma, line 200.* [aux]

The identity permutation has no descents, hence no turning points: **turn_count 1 = 0**.

turn_count_alt_desc *Lemma, line 231.* [aux]

When *s* has the alternating descent pattern, every interior position is a turning point: **turn_count s = n**.

as_perm_alt_desc *Lemma, line 245.* [aux]

When *s* has the alternating descent pattern, **as_perm s** attains its maximum value **n.+2**.

sign_seq *Definition, line 346.* [aux]

sign_seq xs is the boolean direction sequence at each adjacent pair: **xs[i] < xs[i+1]** for *i* = 0, ..., **size xs - 2**. Empty for **xs = nil**.

size_sign_seq *Lemma, line 352.* [aux]

Size of **sign_seq xs**: one less than **size xs**.

flip_count *Definition, line 361.* [aux]

flip_count ss is the number of adjacent disagreements (XOR flips) in a boolean seq **ss**. Counts indices *i* with **ss[i] != ss[i+1]**.

turn_count_seq *Definition, line 370.* [aux]

turn_count_seq xs is the seq-level turn count: number of direction flips in **xs**, i.e., **flip_count (sign_seq xs)**. The seq analogue of **turn_count** for permutations.

size_sign_seq_perm *Lemma, line 428.* [aux]

The sign sequence of **perm_seq s** has length **n.+1**, one slot per pair of consecutive positions.

enum_ord_split *Lemma, line 435.* [aux]

Decomposition of **enum 'I_n.+2** as **ord0** followed by the **lift ord0**-image of **enum 'I_n.+1**. Used to align indexing of **perm_seq** with **sign_seq**.

not_is_descentE *Lemma, line 454.* [aux]

Negated descent indicator as a strict less-than: **~~ is_descent s i** iff **val (s (widen i)) < val (s (lift ord0 i))**.

sign_seq_perm_seq *Lemma, line 472.* [aux]

Bridge identifying **sign_seq (perm_seq s)** with the (negated) descent indicator over **enum 'I_n.+1**. Connects the seq-level flip-count machinery to the perm-level descent/turn structure.

as_perm_max_le_size *Lemma, line 513. [aux]*
 Trivial size bound: $\text{as_perm_max } s \leq n.+2$, the size of the underlying $\text{perm_seq } s$.

sorted_val_enum_ord *Lemma, line 530. [aux]*
 $\text{enum } 'I_m$ is sorted by val ascending. Used to identify $\text{pick_seq } s$ $[\text{set: } \dots]$ with $\text{perm_seq } s$.

pick_seq_setT *Lemma, line 542. [aux]*
 Picking the full position set yields the full one-line word: $\text{pick_seq } s$ $[\text{set: } 'I_n.+2] = \text{perm_seq } s$.

is_alt_three *Lemma, line 563. [aux]*
 An alternating seq of length at least 3 cannot have its first three entries strictly monotone in either direction.

slot_iv *Definition, line 599. [aux]*
 $\text{slot_iv } i \ j$ is the set of descent slots $k : 'I_n.+1$ whose underlying nat satisfies $\text{val } i \leq \text{val } k < \text{val } j$. Indexes the "slot interval" between positions i and j .

mem_slot_iv *Lemma, line 603. [aux]*
 Membership in $\text{slot_iv } i \ j$ unfolded to the val inequalities.

slot_descent_const *Lemma, line 619. [aux]*
 Constancy of the descent indicator on a turn-free slot interval: if no turning point of s lies in slot range $[\text{val } i, \text{val } j)$, then $\text{is_descent } s \ k_1 = \text{is_descent } s \ k_2$ for all slots k_1, k_2 in the range. Used to derive monotone behavior of s on turn-free zones.

constant_descent_monotone *Lemma, line 655. [aux]*
 Constant-descent implies monotonicity: if $\text{is_descent } s \ k = b$ for all slots k in $[\text{val } p, \text{val } q)$, then s is monotone on positions (p, q) : ascending if $b = \text{false}$, descending if $b = \text{true}$.

inter_turn_monotone *Lemma, line 727. [aux]*
 Direction transport across a turn-free slot range: if no turning point lies in $[\text{val } i, \text{val } j)$, then the s -comparison direction on any sub-pair (p, q) inside $[i, j]$ matches the boundary direction at (i, j) .

turn_iv *Definition, line 834. [aux]*
 $\text{turn_iv } s \ a \ b$ is the set of turning points $t : 'I_n$ of s whose witness position $(\text{val } t).+1$ lies in the half-open interval $[a, b)$.

turn_iv_subset *Lemma, line 839. [aux]*
 $\text{turn_iv } s \ a \ b$ is a subset of the global turn set of s .

bool_triangle *Lemma, line 923. [aux]*
 Hamming-style triangle inequality on three booleans: $a \text{ XOR } c \leq (a \text{ XOR } b) + (b \text{ XOR } c)$. Building block for the flip-count monotonicity arguments.

triangle_xor_nat *Lemma, line 931. [aux]*
 Mirror variant of $\text{triangle_xor_nat_g}$ with the middle position on the left. Used for "insert at front" reductions in flip count.

flip_count_pairmap_insert *Lemma, line 950. [aux]*
 Front-insertion monotonicity for flip_count on a pairmap : inserting one element at the front never decreases the flip count. Step 1 building block for $\text{flip_count_pairmap_insert_anywhere}$.

flip_step_helper *Lemma, line 967.* [aux]

Inductive step helper for **flip_count_pairmap_insert_anywhere**: combines the IH bound $s1 \leq \dots + s2$ with a structural matching condition to handle the four $(x < z) = (y < z)$ cases uniformly.

flip_count_pairmap_insert_anywhere *Lemma, line 1002.* [aux]

Insertion-anywhere monotonicity for **flip_count** on a **pairmap**: inserting one element at any position never decreases the flip count. Generalizes **flip_count_pairmap_insert** to interior positions.

flip_count_sign_seq_insert_front *Lemma, line 1027.* [aux]

Sign-seq front-insertion monotonicity: inserting y at the front of xs never decreases **flip_count** (**sign_seq** $_$).

subseq_drop_extra *Lemma, line 1080.* [aux]

Strict-subseq dropping: if xs is a proper subseq of ys , some index i of ys can be removed while still containing xs .

size_take_drop_skip *Lemma, line 1100.* [aux]

Size of ys with element at position i removed: $(\text{size } ys) - 1$.

flip_count_pairmap_le_subseq *Lemma, line 1114.* [aux]

Subseq monotonicity for **flip_count** on a **pairmap**: if xs is a subseq of ys , its flip count is at most that of ys . Iterates **flip_count_pairmap_insert_anywhere** across the size difference.

flip_count_sign_seq_le_subseq *Lemma, line 1151.* [aux]

Subseq monotonicity specialized to **sign_seq**: if xs is a subseq of ys , then **flip_count** (**sign_seq** xs) $\leq$ **flip_count** (**sign_seq** ys). Used in **as_perm_max_upper** to bound subseq alternation by the global flip count.

is_alt_bool_aux *Fixpoint, line 1183.* [aux]

is_alt_bool_aux a xs checks alternation of a boolean seq with a leading "previous" element a : every adjacent pair must XOR to true.

is_alt_bool *Definition, line 1191.* [aux]

is_alt_bool xs holds when the boolean seq xs is fully alternating: every adjacent pair differs.

flip_count_is_alt_bool *Lemma, line 1199.* [aux]

Fully alternating boolean seqs maximize flip count: **flip_count** $xs = (\text{size } xs) - 1$ when **is_alt_bool** xs holds.

sign_seq_is_alt *Lemma, line 1214.* [aux]

The sign seq of an **is_alt** nat seq is itself fully alternating as a boolean seq: every adjacent direction pair differs.

is_alt_flip_count *Lemma, line 1243.* [aux]

Flip-count formula for alternating seqs: an **is_alt** seq of size at least 2 has **flip_count** (**sign_seq** xs) $= (\text{size } xs) - 2$. Used in **as_perm_max_upper** to reduce alternation to flip-count bounds.

sort_enum_strict_sorted *Lemma, line 1263.* [aux]

Sorting the enumeration of a $\{\text{set } 'I_n.+2\}$ by **val** gives a strictly sorted seq, since the underlying enumeration is duplicate-free.

turn_inj *Definition, line 1307.* [aux]

`turn_inj t` injects an interior position `t : 'I_n` into `{ 'I_n.+2 }` at the offset `(val t).+1`. Used to construct the witness set `turn_witness` for the lower-bound construction.

val_turn_inj *Lemma, line 1311.* [aux]

Underlying value of `turn_inj t`: `(val t).+1`.

turn_inj_inj *Lemma, line 1315.* [aux]

`turn_inj` is injective.

turn_witness *Definition, line 1325.* [aux]

`turn_witness s` is the explicit alternating-subseq witness: `{ord0; ord_max} $\{\}\cup$\{turn_inj t | t turning point of s\}`. Has cardinality `(turn_count s).+2`; its `pick_seq` is the alternating subsequence of length `(turn_count s).+2`.

card_turn_image *Lemma, line 1329.* [aux]

Image of `turn_inj` on the turn set has cardinality `turn_count s`.

turn_image_lt_max *Lemma, line 1335.* [aux]

Range bound: any element of the `turn_inj`-image of the turn set has $0 < \text{val } x \leq n$; in particular, neither `ord0` nor `ord_max`.

ord0_notin_turn_image *Lemma, line 1345.* [aux]

`ord0` is not in the `turn_inj`-image of the turn set.

ord_max_notin_turn_image *Lemma, line 1353.* [aux]

`ord_max` is not in the `turn_inj`-image of the turn set.

ord0_neq_ord_max *Lemma, line 1363.* [aux]

Trivial distinction: `ord0` and `ord_max` differ in `{'I_n.+2}`.

card_turn_witness *Lemma, line 1368.* [aux]

Cardinality of the lower-bound witness set: `##|turn_witness s| = (turn_count s).+2`.

slot_descent_const_strict *Lemma, line 1381.* [aux]

Strict-left variant of `slot_descent_const`: hypothesis only excludes turns with witness STRICTLY greater than `val i`. Used in the lower-bound construction where `i` itself may be a turn position.

inter_turn_monotone_strict *Lemma, line 1416.* [aux]

Strict-left variant of `inter_turn_monotone`: monotone direction transport with a strict lower-bound hypothesis on turn witnesses.

dir_left_of_turn *Lemma, line 1454.* [aux]

Direction LEFT of a turn witness: when `b = turn_inj t` and no interior turn lies in `(val a, val b)`, the direction `s a` vs `s b` equals `~~ is_descent s (widen_ord t)` (the "left half" descent indicator at slot `t`).

dir_right_of_turn *Lemma, line 1484.* [aux]

Direction RIGHT of a turn witness: dual of `dir_left_of_turn`; the direction `s b` vs `s c` equals `~~ is_descent s (lift ord0 t)` (the "right half" descent indicator at slot `t`).

sign_flip_at_turn *Lemma, line 1517.* [aux]

Key adjacency lemma for the lower bound: at an interior witness `b = turn_inj t` flanked by neighbors `a, c` with no interior turns in the half-open intervals, the directions across `(a, b)` and `(b, c)` DIFFER (since `t` is a turning point). Used in `triple_flip_pos_seq` to verify the alternating property of `pick_seq s (turn_witness s)`.

uniq_adj *Definition, line 1543.* [aux]

`uniq_adj xs` holds when every adjacent pair in `xs` consists of distinct elements. Weaker than `uniq`, used to characterize when `is_alt` follows from `sign_seq` alternation.

uniq_adj_head_neq *Lemma, line 1552.* [aux]

Head distinctness extracted from `uniq_adj`.

alt_aux_from_sign *Lemma, line 1561.* [aux]

Key reverse direction: an alternating boolean `sign seq` plus adjacent distinctness implies the `alt_aux` predicate. Building block for `is_alt_from_sign`.

sign_seq_alt_of_triple_flip *Lemma, line 1590.* [aux]

Triple-flip sufficient condition for `sign_seq` alternation: if every adjacent triple $(js[i], js[i+1], js[i+2])$ has flipping `s`-direction, then the `sign seq` of the `s`-image is alternating as a boolean `seq`.

uniq_adj_of_uniq *Lemma, line 1613.* [aux]

`uniq_adj` follows from full `uniq` for `nat seqs`.

is_alt_from_sign *Lemma, line 1625.* [aux]

Main reverse characterization: `is_alt xs` follows from `is_alt_bool (sign_seq xs)` together with adjacent distinctness. Used to construct alternating subseqs from `sign-flip` witnesses.

pos_seq *Definition, line 1648.* [aux]

`pos_seq s` is the `turn_witness s` enumerated in ascending `val` order: the sorted `seq` of position indices in the lower-bound witness construction.

size_pos_seq *Lemma, line 1652.* [aux]

`pos_seq s` has length $(\text{turn_count } s) + 2$.

pos_seq_uniq *Lemma, line 1656.* [aux]

`pos_seq s` is duplicate-free.

pick_seq_pos_seq *Lemma, line 1661.* [aux]

Definitional unfolding: `pick_seq s (turn_witness s)` equals the map of `pos_seq s` under `val (s _)`.

pos_seq_strict_sorted *Lemma, line 1666.* [aux]

`pos_seq s` is strictly sorted by `val`.

mem_pos_seq *Lemma, line 1671.* [aux]

Membership transfer: $x \in \text{pos_seq } s$ iff $x \in \text{turn_witness } s$.

pos_seq_val_lt *Lemma, line 1677.* [aux]

Strict-sorted property: index-order in `pos_seq s` matches `val`-order.

pos_seq_val_lt_inv *Lemma, line 1689.* [aux]

Converse strict-sorted property: `val`-order forces index-order in `pos_seq s`.

no_inner_in_pos_seq *Lemma, line 1702.* [aux]

Gap property: no element of `pos_seq s` has `val` strictly between consecutive entries `pos_seq s i` and `pos_seq s i.+1`.

pos_seq_nth0 *Lemma, line 1719.* [aux]

First element of `pos_seq s` is `ord0`.

pos_seq_last_val *Lemma, line 1734.* [aux]

Last element of `pos_seq s` has value `n.+1` (i.e., is `ord_max`).

interior_is_turn_inj *Lemma, line 1759.* [aux]

Interior elements of `pos_seq s` (indices $0 < i < \text{size} - 1$) are `turn_inj`-images of turning points: they correspond to actual turns of `s`.

triple_flip_pos_seq *Lemma, line 1792.* [aux]

Triple-flip property for adjacent triples in `pos_seq s`: directions across consecutive `s`-comparisons differ. The interior witness in each triple is a turning point, so `sign_flip_at_turn` applies.

is_alt_pick_turn_witness *Theorem, line 1846.* [aux]

Witness alternation: `pick_seq s` (`turn_witness s`) is alternating. Combines `triple_flip_pos_seq` with the sign-seq characterization of `is_alt`.

as_perm_max_lower *Theorem, line 1863.* [aux]

Headline lower bound for `as_perm_max`: $(\text{turn_count } s).+2 \leq \text{as_perm_max } s$. Witnessed by `turn_witness s`, which has $(\text{turn_count } s).+2$ elements and induces an alternating `pick_seq` (Stanley \S{1.6.2}).

sort_enum_subseq_enum *Lemma, line 1883.* [aux]

Sorted enumeration of a subset is a subseq of the sorted full enumeration: `sort by val (enum I)` is a subseq of `enum 'I_n.+2`.

pick_seq_subseq_perm_seq *Lemma, line 1904.* [aux]

`pick_seq s I` is a subseq of `perm_seq s` for any position set `I`.

flip_count_pick_le_perm *Lemma, line 1913.* [aux]

Flip-count comparison for `pick_seq` vs `perm_seq`: any pick has `flip_count` at most that of the full perm seq.

flip_count_as_sum *Lemma, line 1923.* [aux]

`flip_count` expressed as an indexed sum of consecutive XOR-pairs. Used to align with the `turn_count` sum over `is_turn`.

neg_add_neg *Lemma, line 1941.* [aux]

XOR is invariant under double negation: $\sim\sim a (+) \sim\sim b = a (+) b$.

flip_count_perm_seq_eq_turn_count *Lemma, line 1948.* [aux]

Bridge identity: $\text{flip_count } (\text{sign_seq } (\text{perm_seq } s)) = \text{turn_count } s$. The seq-level flip count of the full word equals the perm-level turn count, via `sign_seq_perm_seq` and the descent-XOR characterization of `is_turn`.

as_perm_max_upper *Lemma, line 1986.* [aux]

Headline upper bound for `as_perm_max`: $\text{as_perm_max } s \leq (\text{turn_count } s).+2$. Combines the alternating `flip_count` formula `is_alt_flip_count` with the subseq monotonicity `flip_count_pick_le_perm` and the bridge `flip_count_perm_seq_eq_turn_count`.

as_perm_max_eq *Theorem, line 2009.* [Ch 7 §7.4]

HEADLINE THEOREM (Stanley \S{1.6.2}): the longest alternating subsequence count of `s` : $\{\text{perm 'I}_n.+2\}$ equals $(\text{turn_count } s).+2$. That is, $\text{as}(w) = (\text{turn_count } w).+2$, proving the two definitions `as_perm_max` (bijective max) and `as_perm` (turn count) coincide. Combines `as_perm_max_upper` and `as_perm_max_lower`.

beta.v

beta *Definition, line 24. [Ch 5 §5.6]*

beta n D is the set-refined descent count: the number of permutations of $'I_n.+1$ whose descent set equals exactly D . Stanley's **beta** $_n(D)$ (Stanley EC1, S1.4); summing **beta** D over $|D| = k$ recovers **eulerian** n k .

descent_set_rev_perm *Lemma, line 33. [aux]*

Descent set of the reversal: **D**(**rev_perm** s) is the **rev_ord**-image of the complement of **D**(s). Set-level refinement of **is_descent_rev**.

descent_set_rev_perm_ord *Lemma, line 47. [aux]*

The reversal permutation $n, n-1, \dots, 0$ has descent set equal to all positions.

imset_rev_ord_inv *Lemma, line 61. [aux]*

rev_ord is involutive on subsets: applying **imset** **rev_ord** twice is the identity.

imset_rev_ord_compl *Lemma, line 71. [aux]*

rev_ord-image commutes with set complement.

beta0 *Lemma, line 91. [aux]*

beta $_n(\text{emptyset}) = 1$: only the identity permutation has empty descent set.

descent_set_full_rev *Lemma, line 102. [aux]*

Only the strictly-decreasing permutation **rev_perm_ord** has descent set equal to all positions.

beta_full *Lemma, line 118. [aux]*

beta $_n(\text{top}) = 1$: only the reversal permutation has full descent set.

sum_beta_eq_fact *Lemma, line 132. [aux]*

$\text{sum}_D \text{beta}_n(D) = (n+1)!$: the **beta** partition of $\{\text{perm } 'I_n.+1\}$ by descent set covers all permutations.

beta_eulerian *Lemma, line 147. [Ch 5 §5.6]*

Connection to Eulerian numbers: summing **beta** D over all D of cardinality k recovers the Eulerian number $A(n+1, k)$ (Stanley EC1, S1.4).

beta_rev *Lemma, line 166. [Ch 5 §5.6]*

Reversal-complement symmetry: **beta** $_n(D) = \text{beta}_n(\text{rev_ord}(\text{complement } D))$. Set-level refinement of **eulerian_symm**, proved via the **rev_perm** involution.

beta_bridge.v

set_to_seq *Definition, line 29. [aux]*

set_to_seq D – convert the finset $D : \{\text{set } 'I_n\}$ to the sorted **seq** **nat** of underlying positions; bridges the finset descent world to the **seq**-based **psi_cdindex** machinery.

mem_set_to_seq *Lemma, line 35. [aux]*

mem_set_to_seq – membership in **set_to_seq** D equals membership in the unsorted underlying value list (sorting is order-preserving on memberships).

mem_set_to_seq_iff *Lemma, line 41. [aux]*

mem_set_to_seq_iff – alternative form of membership in **set_to_seq** D as a **has** over **enum** D .

mem_set_to_seq_ord *Lemma, line 53. [aux]*

`mem_set_to_seq_ord` – ordinal-level membership bridge: `val i` is in `set_to_seq D` iff `i` is in `D`.

uniq_set_to_seq *Lemma, line 67. [aux]*

`uniq_set_to_seq` – the seq-of-positions `set_to_seq D` has no duplicates.

omega_seq_local *Definition, line 83. [aux]*

`omega_seq_local s` – the omega map at the seq level (mirrors `psi_cdindex.omega_seq`): the list of `k` in `iota 0 (max s) + 1` such that exactly one of `k`, `k+1` belongs to `s`.

foldr_maxn_set_to_seq_lb *Lemma, line 94. [aux]*

`foldr_maxn_set_to_seq_lb` – any element of `set_to_seq D` is bounded above by `foldr_maxn 0` over the same list (used to ensure the `iota` bound in `omega_seq_local` covers the relevant range).

omega_set_seq_local_bridge *Lemma, line 106. [aux]*

`omega_set_seq_local_bridge` – key bridge: `k \in omega_set D` (finset side) iff `val k \in omega_seq_local (set_to_seq D)` (seq side). Connects the two formulations of Stanley’s omega map.

beta_omega.v

sym_diff *Definition, line 28. [Ch 8 §8.1]*

`sym_diff D E` – symmetric difference $(D \setminus E) \cup (E \setminus D)$ of two finsets over $\{\text{set } I_n\}$.

toggle_at *Definition, line 33. [Ch 8 §8.1]*

`toggle_at D i` – single-position toggle: flips membership of `i` in the descent set `D` by symmetric difference with `[set i]`.

toggle_at_in *Lemma, line 38. [aux]*

`toggle_at_in` – membership in `toggle_at D i` is xor of $(i == j)$ and $(j \in D)$.

toggle_atK *Lemma, line 47. [aux]*

`toggle_atK` – toggling at the same position twice is the identity.

toggle_at_self *Lemma, line 56. [aux]*

`toggle_at_self` – at the toggled position, membership is flipped: $i \in \text{toggle_at } D \iff i \notin D$.

toggle_at_other *Lemma, line 62. [aux]*

`toggle_at_other` – away from the toggled position `i`, membership in `D` is preserved.

omega_set *Definition, line 73. [Ch 8 §8.2]*

`omega_set D` – Stanley’s omega-map (Stanley EC1 \S{1.6}) at the finset level: $k \in \text{omega_set } D$ iff exactly one of `k` and `k+1` belongs to `D` (xor on consecutive positions).

mem_omega_set *Lemma, line 78. [aux]*

`mem_omega_set` – unfolds membership of `k` in `omega_set D` to the underlying xor of consecutive `D`-memberships.

toggle_at_omega_bit_i_new *Lemma, line 86. [aux]*

`toggle_at_omega_bit_i_new` – when consecutive `i, j` are both in `D`, toggling `i` turns the omega-bit at the connecting position `k` from off to on; produces the witness ordinal `k`.

toggle_at_omega_strict_superset *Lemma, line 111.* [aux]

toggle_at_omega_strict_superset – key omega-monotonicity step (left toggle): if i, j both in D with $j = i+1$ and the predecessor of i also lies in D when defined, then toggling out i strictly enlarges **omega_set**. Feeds into **beta_swap** case analysis.

toggle_at_j_omega_bit_i_new *Lemma, line 155.* [aux]

toggle_at_j_omega_bit_i_new – right-toggle analogue of **toggle_at_omega_bit_i_new**: toggling j in the consecutive pair i, j turns the omega-bit at k from off to on.

toggle_at_j_omega_strict_superset *Lemma, line 181.* [aux]

toggle_at_j_omega_strict_superset – right-toggle omega monotonicity (case A of beta-swap): if i, j both in D with $j = i+1$ and the successor of j is also in D (when defined), then toggling j strictly enlarges **omega_set**.

left_ok *Definition, line 225.* [aux]

left_ok $s\ i\ l$ – candidate left endpoint test: $l \leq i$ and every position between l and i is a descent of s .

right_ok *Definition, line 231.* [aux]

right_ok $s\ i\ r$ – candidate right endpoint test: $i \leq r$ and every position between i and r is a descent of s .

left_ok_self *Lemma, line 237.* [aux]

left_ok_self – a descent position i is its own valid left endpoint candidate.

right_ok_self *Lemma, line 246.* [aux]

right_ok_self – a descent position i is its own valid right endpoint candidate.

block_left *Definition, line 256.* [aux]

block_left $s\ i$ – left endpoint of the maximal Foata descent block containing i : if i is a descent, the smallest $l \leq i$ such that all positions in $[l..i]$ are descents; otherwise i .

block_right *Definition, line 263.* [aux]

block_right $s\ i$ – right endpoint of the maximal Foata descent block containing i : if i is a descent, the largest $r \geq i$ such that all positions in $[i..r]$ are descents; otherwise i .

block_left_left_ok *Lemma, line 269.* [aux]

block_left_left_ok – the chosen **block_left** $s\ i$ satisfies the **left_ok** predicate (when i is a descent).

block_right_right_ok *Lemma, line 277.* [aux]

block_right_right_ok – the chosen **block_right** $s\ i$ satisfies the **right_ok** predicate (when i is a descent).

block_left_min *Lemma, line 286.* [aux]

block_left_min – minimality of **block_left** $s\ i$ among all valid left-endpoint candidates.

block_right_max *Lemma, line 295.* [aux]

block_right_max – maximality of **block_right** $s\ i$ among all valid right-endpoint candidates.

block_left_le *Lemma, line 304.* [aux]

block_left_le – the left endpoint of the block does not exceed i .

block_right_ge *Lemma, line 310.* [aux]

block_right_ge – the right endpoint of the block is at least i .

block_left_descent *Lemma, line 316.* [aux]

block_left_descent – every position from **block_left** $s\ i$ up to i is a descent of s .

block_right_descent *Lemma, line 326.* [aux]

block_right_descent – every position from i up to **block_right** $s\ i$ is a descent of s .

block_descent_chain *Lemma, line 336.* [aux]

block_descent_chain – every position in the full block [**block_left** $s\ i$.. **block_right** $s\ i$] is a descent of s .

block_chain_step *Lemma, line 358.* [aux]

block_chain_step – single-step value comparison: for any k in the block, s strictly decreases from position k to $k+1$.

block_chain_values *Lemma, line 367.* [aux]

block_chain_values – chain-of-values: the values $s\ p$ are strictly decreasing across the entire Foata block (from **block_left** $s\ i$ through to **block_right** $s\ i + 1$).

beta_swap.v

alt_desc_set *Definition, line 34.* [Ch 8 §8.3]

alt_desc_set n – the canonical alternating descent set on $\{\text{set } 'I_n\}$: the set of even positions $\{i : \sim\{\} \sim\{\} \text{ odd } i\}$. The descent set that maximises **beta** (Stanley Cor 1.6.5).

mem_alt_desc_set *Lemma, line 39.* [aux]

mem_alt_desc_set – membership in **alt_desc_set** n reduces to parity: $i \in \text{alt_desc_set } n$ iff $\sim\{\} \sim\{\} \text{ odd } i$.

set_is_alt *Definition, line 45.* [Ch 8 §8.3]

set_is_alt D – decidable predicate: D is "set-alternating" iff every consecutive ordinal pair has differing D -memberships.

set_not_altP *Lemma, line 53.* [aux]

set_not_altP – reflection: if D is not set-alternating then there exist consecutive i, j with the same D -membership.

toggle_at_compl *Lemma, line 69.* [aux]

toggle_at_compl – the **toggle_at** operation commutes with complementation: **toggle_at** $(\sim : D)\ i = \sim : \text{toggle_at } D\ i$.

compl_perm *Definition, line 99.* [Ch 8 §8.4]

compl_perm s – value-complement permutation: post-compose s with **rev_perm_ord** n so that **compl_perm** $s\ i = \text{rev_ord } (s\ i)$. Bijection used to prove **beta_compl** (set complement invariance).

compl_permE *Lemma, line 104.* [aux]

compl_permE – evaluation rule for **compl_perm** s : it sends i to **rev_ord** $(s\ i)$.

compl_perm_inj *Lemma, line 110.* [aux]

compl_perm_inj – **compl_perm** is injective on permutations (right-cancellation by **rev_perm_ord**).

compl_perm_involutive *Lemma, line 115.* [aux]

compl_perm_involutive – **compl_perm** is an involution: applying it twice is the identity (since **rev_perm_ord** is).

is_descent_compl *Lemma, line 126.* [aux]

is_descent_compl – value-complementing flips every descent bit: $\text{is_descent } (\text{compl_perm } s) \text{ } i = \sim \sim \text{is_descent } s \text{ } i$.

descent_set_compl *Lemma, line 141.* [Ch 8 §8.4]

descent_set_compl – value-complementing complements the descent set: $\text{descent_set } (\text{compl_perm } s) = \sim : \text{descent_set } s$.

beta_compl *Lemma, line 150.* [Ch 8 §8.4]

beta_compl – the count $\text{beta } D$ is invariant under set complementation: $\text{beta } D = \text{beta } (\sim : D)$. Proven via the value-complement involution **compl_perm**.

set_is_alt_classify *Lemma, line 170.* [Ch 8 §8.4]

set_is_alt_classify – classification of set-alternating sets (Stanley $\backslash S \{ \}$ 1.6): exactly two such sets exist, **alt_desc_set** n and its complement. Together with **beta_compl** this implies all set-alternating sets share the same beta value.

beta_set_is_alt_eq *Lemma, line 206.* [aux]

beta_set_is_alt_eq – any set-alternating D' has the same beta as **alt_desc_set** n (consequence of **set_is_alt_classify** and **beta_compl**).

not_set_is_alt_n_ge2 *Lemma, line 219.* [aux]

not_set_is_alt_n_ge2 – if D fails **set_is_alt** then $n \geq 2$ (a violating consecutive pair requires at least two positions).

omega_set_alt_full *Lemma, line 231.* [aux]

omega_set_alt_full – the omega-set of **alt_desc_set** $m.+1$ is the full set [**set**: 'I_m]: every consecutive position pair has differing memberships, so every omega-bit is on.

not_set_is_alt_omega_not_full *Lemma, line 249.* [aux]

not_set_is_alt_omega_not_full – a non-set-alternating D has **omega_set** D strictly smaller than the full set; the offending consecutive pair witnesses an "off" omega bit.

carlitz.v

q_onesub_prod *Definition, line 37.* [aux]

The product $\text{prod_}(i \leq N) (1 - q^i x)$, the Carlitz denominator.

q_stair *Definition, line 41.* [aux]

The generating function of the Gaussian binomials $m+N$, N_q .

q_stair0 *Lemma, line 44.* [aux]

(no docstring)

coef_fpsXfM *Lemma, line 48.* [aux]

Coefficient shift under multiplication by x (local copy).

q_stair_step *Lemma, line 58.* [aux]

Peeling one factor: $(1 - q^{(N+1)} x) \text{ } m+N+1$, $N+1$ -series = $m+N$, N -series.

q_staircase *Theorem, line 75.* [aux]

(no docstring)

q_onesub_prod_unit *Lemma, line 85.* [aux]

Corollary in division form: the inverse of the Carlitz denominator.

q_stairE *Corollary, line 91.* [aux]

(no docstring)

cycles.v

cycle_count *Definition, line 24.* [Ch 2 §2.1]

`cycle_count s` is the number of cycles in the disjoint-cycle decomposition of `s`, equivalently the cardinality of `porbits s`, the set of orbits of `s` acting on `T` by iteration. Stanley's $c(w)$.

stirling_c *Definition, line 34.* [Ch 2 §2.2]

`stirling_c n k` is the number of permutations of $'I_n$ with exactly k cycles in their disjoint-cycle decomposition. This is Stanley's $c(n, k)$, the signless Stirling number of the first kind (Stanley EC1 \S{1.3.2}).

cycle_count_le *Lemma, line 45.* [Ch 2 §2.1]

Each cycle is non-empty, so `cycle_count s` $\leq$ `##|T|`. Cycle analogue of `des_le` for descents.

porbit_id_singleton *Lemma, line 54.* [aux]

Under the identity permutation each `porbit` is the singleton $\{x\}$. Used in `cycle_count_id`.

cycle_count_id *Lemma, line 65.* [Ch 2 §2.1]

The identity permutation on `T` has exactly `##|T|` cycles, one singleton per element (Stanley \S{1.3.1}: each fixed point is a 1-cycle).

stirling_c_row_sum *Lemma, line 86.* [aux]

Row sum: summing `stirling_c n k` over all cycle counts $k \leq n$ recovers the total number of permutations of $'I_n$. Mirrors `eulerian_row_sum` in `eulerian.v`.

stirling_c_row_sum_fact *Lemma, line 100.* [Ch 2 §2.2]

Row sum, factorial form: $\sum_k c(n, k) = n!$. Stanley's identity derived from `stirling_c_row_sum` via `card_Sn`.

cycles_rec.v

lift_perm_id_iter *Lemma, line 55.* [aux]

Iterating `perm.lift_perm ord_max ord_max s` at `ord_max` stays at `ord_max`. Used in `porbit_lift_perm_id_max`.

porbit_lift_perm_id_max *Lemma, line 64.* [aux]

The `porbit` of `ord_max` under `perm.lift_perm ord_max ord_max s` is the singleton $\{\text{ord_max}\}$.

lift_perm_id_iter_lift *Lemma, line 76.* [aux]

Iteration on lifted points commutes with `lift ord_max`: applying $(\text{perm.lift_perm ord_max ord_max } s)^{+i}$ at `lift ord_max k` is the lift of $(s^{+i}) k$.

porbit_lift_perm_id_lift *Lemma, line 86.* [aux]

The `porbit` of `lift ord_max k` under the lifted `perm` is the `lift ord_max`-image of the `porbit` of `k` under `s`.

porbits_lift_perm_id *Lemma, line 100.* [aux]

Decomposition of `porbits (perm.lift_perm ord_max ord_max s)` into the singleton orbit $\{\text{ord_max}\}$ and the `lift ord_max`-images of the orbits of `s`. Used in `cycle_count_lift_perm_id`.

cycle_count_lift_perm_id *Lemma, line 124.* [aux]

H1 proper: extending `s` with `ord_max` as a fixed point adds exactly one cycle. Building block for the $j = \text{ord_max}$ class of the Stirling fiber bijection (see `stirling_fiber.v`).

descent.v

is_descent *Definition, line 20.* [Ch 5 §5.1]

`is_descent s i` holds when `s` descends at position `i`, i.e. when `s i > s (i+1)`. Stanley's predicate "`i` is a descent of `w`" (Stanley EC1, S1.4).

descent_set *Definition, line 25.* [Ch 5 §5.1]

`descent_set s` is the descent set of `s`, Stanley's $D(w)$ (Stanley EC1, S1.4).

des *Definition, line 29.* [Ch 5 §5.1]

`des s` is the descent number of `s`, Stanley's $d(w) = \#D(w)$ (Stanley EC1, S1.4).

asc *Definition, line 32.* [Ch 5 §5.1]

`asc s = n - des s` is the ascent number of `s`.

mem_descent_set *Lemma, line 37.* [aux]

Membership in `descent_set s` is just `is_descent s i`; rewrites `inE` form.

des_le *Lemma, line 41.* [Ch 5 §5.1]

Descent number is bounded by `n`, the number of consecutive positions.

des_add_asc *Lemma, line 45.* [aux]

Descents and ascents partition the `n` consecutive positions.

des_id *Lemma, line 49.* [Ch 5 §5.1]

The identity permutation has no descents.

rev_perm_ord *Definition, line 78.* [Ch 5 §5.2]

`rev_perm_ord` is the reversal permutation $i \mapsto n - i$ of $'I_{n+1}$.

rev_perm_ordE *Lemma, line 81.* [aux]

Equivalence with the underlying `rev_ord` for rewriting.

rev_perm *Definition, line 87.* [Ch 5 §5.2]

`rev_perm s` is the reversal of `s` (compose with `rev_perm_ord` on the left). It corresponds to reading the one-line notation of `s` backwards.

rev_permE *Lemma, line 90.* [aux]

Defining equation: `rev_perm s j = s (rev_ord j)`.

is_descent_rev *Lemma, line 95.* [Ch 5 §5.2]

Reversal swaps descents and ascents pointwise: `i` is a descent of `rev_perm s` iff `rev_ord i` is not a descent of `s`.

des_rev_perm *Lemma, line 113.* [aux]

Descent count under reversal: `des (rev_perm s) = n - des s`. This is the involution underlying the symmetry `eulerian n k = eulerian n (n-k)`.

eulerian.v

eulerian *Definition, line 16.* [Ch 5 §5.3]

`eulerian n k` is the Eulerian number $A(n+1, k)$: the number of permutations of $'I_{n+1}$ with exactly `k` descents. Stanley's $A(n+1, k)$ (Stanley EC1, S1.4).

eulerian_row_sum *Lemma, line 27.* [aux]

Row sum: summing `eulerian n k` over `k <= n` gives the cardinality of $\{\text{perm } 'I_{n+1}\}$. Partition of permutations by descent count.

eulerian_row_sum_fact *Lemma, line 38.* [Ch 5 §5.3]

Row sum equals $(n+1)!$: combines `eulerian_row_sum` with $|S_{\{n+1\}}| = (n+1)!$.

eulerian_out_of_range *Lemma, line 43.* [Ch 5 §5.3]

Out-of-range vanishing: `eulerian n k = 0` when $k > n$, since no permutation has more than n descents.

des0_id *Lemma, line 51.* [Ch 5 §5.3]

Only the identity has zero descents. Used to prove `eulerian n 0 = 1`.

eulerian_n_0 *Lemma, line 81.* [aux]

Boundary value: `eulerian n 0 = 1`, counting only the identity.

rev_perm_inj *Lemma, line 95.* [aux]

Reversal is injective on permutations (left cancellation by `rev_perm_ord`).

rev_perm_involutive *Lemma, line 99.* [aux]

Reversal is an involution: `rev_perm (rev_perm s) = s`.

eulerian_symm *Lemma, line 104.* [aux]

Symmetry of Eulerian numbers: $A(n+1, k) = A(n+1, n-k)$ (Stanley EC1, S1.4). Proved via the reversal involution on $\{\text{perm } 'I_{n.+1}\}$.

eulerian_n_n *Lemma, line 117.* [aux]

Boundary value: `eulerian n n = 1`, counting only the reversal permutation.

widenSn_inj *Lemma, line 137.* [aux]

`widen_ord (leqnSn n.+1) : 'I_{n.+1} \rightarrow 'I_{n.+2}` is injective.

widenSn_neq_ord_max *Lemma, line 141.* [aux]

Widening from $'I_{n.+1}$ to $'I_{n.+2}$ never reaches `ord_max`.

insert_max_fun *Definition, line 156.* [Ch 5 §5.4]

Underlying function of `insert_max_perm t p`: sends p to `ord_max` and elsewhere lifts t via `widen_ord`.

insert_max_fun_p *Lemma, line 163.* [aux]

`insert_max_fun` sends position p to `ord_max`.

insert_max_fun_lift *Lemma, line 167.* [aux]

`insert_max_fun` on lifted positions widens t .

insert_max_fun_inj *Lemma, line 172.* [aux]

`insert_max_fun` is injective: combine injectivity of `widen` and t .

insert_max_perm *Definition, line 185.* [Ch 5 §5.4]

`insert_max_perm t p` is the permutation of $'I_{n.+2}$ obtained from $t : \{\text{perm } 'I_{n.+1}\}$ by inserting the value `ord_max` at position p . Underlies the bijection $\{\text{perm } 'I_{n.+2}\} \sim \{\} = \{\text{perm } 'I_{n.+1}\} \times 'I_{n.+2}$.

insert_max_permE *Lemma, line 188.* [aux]

Equivalence with the underlying `insert_max_fun` for rewriting.

insert_max_perm_at_p *Lemma, line 192.* [aux]

`insert_max_perm t p` sends p to `ord_max`, the inserted maximum value.

insert_max_perm_lift *Lemma, line 196.* [aux]

`insert_max_perm t p` on lifted positions widens t .

extract_max_ne *Fact, line 212.* [aux]

Off-position values of s avoid ord_max when $s\ p = \text{ord_max}$, by injectivity.

extract_max_fun *Definition, line 221.* [Ch 5 §5.4]

Underlying function of **extract_max_perm**: removes the value ord_max (located at position p) and reindexes the remaining values onto $'I_n.+1$.

extract_max_funE *Lemma, line 225.* [aux]

Defining equation: widening **extract_max_fun** j recovers s ($\text{lift } p\ j$).

extract_max_fun_inj *Lemma, line 235.* [aux]

extract_max_fun is injective.

extract_max_perm *Definition, line 244.* [Ch 5 §5.4]

extract_max_perm sp is the permutation of $'I_n.+1$ obtained from $s : \{\text{perm } 'I_n.+2\}$ (with $s\ p = \text{ord_max}$) by deleting the value ord_max .

extract_max_permE *Lemma, line 247.* [aux]

Equivalence with the underlying **extract_max_fun** for rewriting.

extract_max_widen *Lemma, line 251.* [aux]

Defining equation: widening **extract_max_perm** j recovers s ($\text{lift } p\ j$).

insert_extract_max *Lemma, line 268.* [aux]

Right inverse: inserting the extracted max back at position p recovers s .

des_insert_max_ord0 *Lemma, line 288.* [aux]

Inserting ord_max at position 0 adds exactly one descent (a fresh descent is created at position 0 from the new max value).

des_insert_max_ord_max *Lemma, line 317.* [aux]

Inserting ord_max at position ord_max (the rightmost position) does not change the descent count.

des_insert_max_interior *Lemma, line 362.* [aux]

Inserting ord_max at an interior position above j adds a descent iff j was an ascent in t . Key combinatorial lemma for the Eulerian recurrence.

insert_max_perm_fiber *Lemma, line 434.* [aux]

The inverse of **insert_max_perm** $t\ p$ sends ord_max back to p : identifies the fiber of position-of-max under insertion.

sum_descent *Lemma, line 450.* [aux]

The descent count **des** t equals the sum of descent indicators over positions.

eulerian_rec *Lemma, line 493.* [Ch 5 §5.5]

Eulerian recurrence (Stanley EC1, S1.4): $A(n+2, k+1) = (k+2) A(n+1, k+1) + (n+1-k) A(n+1, k)$. Proved via the insert-max bijection.

worpitzky_binom_id *Lemma, line 538.* [aux]

Key algebraic identity for Worpitzky's induction step (valid for $k \leq n$): $x * C(x+k, n+1) = (k+1) C(x+k, n+2) + (n+1-k) C(x+k+1, n+2)$.

worpitzky *Lemma, line 555.* [aux]

Worpitzky's identity (Stanley EC1, S1.4): $x^{(n+1)} = \sum_{(k < n+1)} A(n+1, k) * C(x+k, n+1)$. Proved by induction on n using **eulerian_rec** and **worpitzky_binom_id**.

binS *Lemma, line 593.* [aux]

Signed Pascal extension: a saturated-arithmetic-friendly form of Pascal's rule, valid for all t (including $t = 0$, where $t.-1 = 0$).

aux_id_step *Lemma, line 598.* [aux]

Key recurrence for the alternating binomial sum: $g(N+1, u) = g(N, u.-1)$. Proved via Pascal applied to both binomial factors plus a reindex.

aux_id *Lemma, line 626.* [aux]

Alternating binomial convolution identity: $\sum_j (-1)^j C(n+2, j) C(t-j, n+1) = [t == n+1]$. Used in the inversion step of **eulerian_explicit**.

eulerian_explicit *Lemma, line 669.* [aux]

Eulerian explicit formula (Stanley EC1, S1.4): $A(n+1, k) = \sum_{j \leq k} (-1)^j C(n+2, j) (k+1-j)^{(n+1)}$. Proved by Worpitzky inversion against **aux_id**.

foata.v

cyc_last_to_front *Definition, line 48.* [aux]

cyc_last_to_front s moves the last letter of s to the front. Empty sequence stays empty. Building block for **foata_step**.

split_blocks_aux *Fixpoint, line 55.* [aux]

split_blocks_aux P cur s is the tail-recursive worker for **split_blocks**, threading the in-progress block cur . A boundary occurs after each P -letter.

split_blocks *Definition, line 67.* [aux]

split_blocks P s cuts s into maximal blocks each ending with a P -letter (except possibly the last block). Concatenating the blocks recovers s .

foata_step *Definition, line 74.* [Ch 4 §4.1]

foata_step a u is one step of the Foata bijection: compare the last letter of u to a , split u into blocks via **split_blocks** (with predicate $< a$ or $a <$ accordingly), cyclically rotate each block via **cyc_last_to_front**, and append a . See Stanley EC1 \S{1.3.4}.

foata *Definition, line 86.* [Ch 4 §4.1]

foata w is Foata's first fundamental bijection at the word level: iterate **foata_step** left-to-right starting from the empty word. Satisfies $\text{inv_seq}(\text{foata } w) = \text{maj_seq } w$ (Theorem **foata_inv_eq_maj**).

is_desc_seq *Definition, line 95.* [Ch 4 §4.1]

is_desc_seq w k holds iff position k is a descent of w , i.e. $w_k > w_{k+1}$.

maj_seq *Definition, line 100.* [Ch 4 §4.1]

maj_seq w is the major index of w at the seq level: sum of 1-indexed descent positions $k+1$ over **is_desc_seq** w k .

inv_seq *Definition, line 105.* [Ch 4 §4.1]

inv_seq w is the inversion count of w at the seq level: number of pairs (i, j) with $i < j < \text{size } w$ and $w_i > w_j$.

count_gt *Definition, line 110.* [aux]

count_gt a w is the number of letters of w strictly greater than a .

cyc_last_to_front_perm_eq *Lemma, line 143.* [aux]

cyc_last_to_front preserves the multiset of letters.

split_blocks_aux_flatten *Lemma, line 155.* [aux]

Flattening the blocks of **split_blocks_aux** yields $\text{cur} ++ s$; the accumulator-aware version of **split_blocks_flatten**.

split_blocks_flatten *Lemma, line 166.* [aux]

Concatenating the blocks of **split_blocks** $P\ s$ recovers s .

perm_eq_flatten_map_cyc *Lemma, line 172.* [aux]

Rotating each block via **cyc_last_to_front** preserves the multiset of the flattened sequence.

foata_step_perm_eq *Lemma, line 181.* [aux]

foata_step $a\ u$ is a permutation of **rcons** $u\ a$: the step rearranges letters but preserves the multiset (with a inserted at the end).

foata_perm_eq *Lemma, line 202.* [aux]

foata w is a permutation of w : the bijection preserves the multiset of letters.

foata_size *Lemma, line 217.* [aux]

foata preserves length.

foata_uniq *Lemma, line 221.* [aux]

foata preserves uniqueness.

inv_seq_rcons *Lemma, line 232.* [aux]

Classical recursion: appending letter a at the end of w increases **inv_seq** by **count_gt** $a\ w$, the number of letters of w above a .

count_gt_perm_eq *Lemma, line 262.* [aux]

count_gt only depends on the multiset (preserved under **perm_eq**).

maj_seq_rcons *Lemma, line 278.* [aux]

Appending letter a to nonempty w adds **size** w to **maj_seq** iff the previous last letter exceeds a (creating a new descent at position **size** $w - 1$, with 1-indexed contribution **size** w); otherwise **maj_seq** is unchanged.

foata_step_last *Lemma, line 311.* [aux]

foata_step $a\ u$ always ends with a .

foata_rcons *Lemma, line 321.* [aux]

foata commutes with right-extension: **foata** (**rcons** $w\ a$) = **foata_step** a (**foata** w). Used for induction on w from the right.

foata_last *Lemma, line 331.* [aux]

Last letter of **foata** ($a :: w$) equals last letter of $a :: w$ (i.e. **last** $a\ w$).

foata_last_eq *Lemma, line 350.* [aux]

General form for the **rcons** induction: when w is non-empty, **last** (**foata** w) = **last** w for any default.

cross_inv *Definition, line 364.* [aux]

cross_inv $s1\ s2$ counts cross-inversions between $s1$ and $s2$: pairs (i, j) with $s1_i > s2_j$. Equals the inversions of $s1 ++ s2$ that straddle the join point.

cross_inv_nil_r *Lemma, line 368.* [aux]

cross_inv is zero on the right when $s2$ is empty.

cross_inv_rcons *Lemma, line 373. [aux]*

`cross_inv` recursion when an element is appended on the right.

inv_seq_cat *Lemma, line 380. [aux]*

Decomposition of `inv_seq` over concatenation: left inversions, right inversions, and cross-inversions.

count_gt_cat *Lemma, line 398. [aux]*

`count_gt` is additive under concatenation.

cross_inv_perm_eq_r *Lemma, line 405. [aux]*

`cross_inv` only depends on the multiset of its right argument.

cross_inv_perm_eq_l *Lemma, line 415. [aux]*

`cross_inv` only depends on the multiset of its left argument (since `count_gt` does).

count_lt *Definition, line 424. [aux]*

`count_lt a w` is the number of letters of `w` strictly less than `a`.

perm_eq_flatten_map_pred *Lemma, line 431. [aux]*

If each block is `perm_eq` to its image under `f`, the flattened images are `perm_eq` to the flattened originals.

inv_seq_flatten_swap_eq *Lemma, line 444. [aux]*

Block-rotation accounting: if every block is replaced by a `perm_eq` image, the change in `inv_seq` of the flatten equals the sum of per-block changes (cross-inversions are unaffected since they only see multisets).

cyc_last_to_front_rcons *Lemma, line 477. [aux]*

Computational form: `cyc_last_to_front (rcons b' l) = l :: b'`.

inv_seq_cons_eq_rcons_shift *Lemma, line 488. [aux]*

Inversions shift when `l` moves from the end to the front of `b'`: `inv_seq (l :: b') - inv_seq (rcons b' l) = count_lt l b' - count_gt l b'`.

cyc_diff_block_lt *Lemma, line 511. [aux]*

Per-block inversion change in the `last u < a` case: rotating drops `size b'` inversions when the small letter `l < a` moves to the front past the `b' > a` block. Used by `foata_step_inv_lt`.

cyc_diff_block_gt *Lemma, line 535. [aux]*

Per-block inversion change in the `a < last u` case: rotating adds `size b'` inversions when the large letter `a < l` moves to the front past the `b' < a` block. Used by `foata_step_inv_gt`.

split_blocks_aux_all_nonempty *Lemma, line 558. [aux]*

Every block produced by `split_blocks_aux` is non-empty.

wf_block *Definition, line 583. [aux]*

`wf_block P b` holds when `b` is non-empty, its last letter satisfies `P`, and all non-last letters do not. Captures the well-formedness of blocks output by `split_blocks` (except possibly the last).

wf_block_rcons *Lemma, line 591. [aux]*

`wf_block P (rcons b' l)` is equivalent to `P l && all (~~ P) b'`.

split_blocks_aux_wf *Lemma, line 602. [aux]*

Structural invariant of `split_blocks_aux`: when `cur` is all not-`P` and the last letter of `s` satisfies `P`, every produced block is a `wf_block`.

split_blocks_wf *Lemma, line 631. [aux]*

Every block of **split_blocks** P s is a **wf_block** when s is non-empty and its last letter satisfies P .

split_blocks_aux_size_when_last_P *Lemma, line 638. [aux]*

Number of blocks of **split_blocks_aux** P cur s equals **count** P s , when s is non-empty and ends with a P -letter.

split_blocks_size_eq *Lemma, line 666. [aux]*

Number of blocks of **split_blocks** P s equals **count** P s (assuming last letter of s satisfies P).

wf_block_decomp *Lemma, line 673. [aux]*

Decomposition: any **wf_block** P b is of the form $rcons$ b' l with P l and b' all not- P .

size_count_lt_gt *Lemma, line 688. [aux]*

For **uniq** u with $a \notin u$, size of u splits as **count_lt** a u + **count_gt** a u (no letter equals a).

sum_size_belast_wf *Lemma, line 705. [aux]*

Sum of $(size\ b) - 1$ over **wf_blocks** equals the total length minus the number of blocks.

sum_inv_cyc_lt_blocks *Lemma, line 735. [aux]*

Aggregated per-block inversion change in the **last** $u < a$ case ($P = (y < a)$): summed cyc-rotation drop equals total internal sizes. Used by **foata_step_inv_lt**.

sum_inv_cyc_gt_blocks *Lemma, line 772. [aux]*

Aggregated per-block inversion change in the **a < last** u case ($P = (a < y)$): summed cyc-rotation gain equals total internal sizes. Used by **foata_step_inv_gt**.

split_blocks_lt_strict *Lemma, line 808. [aux]*

Strong form of **split_blocks_wf** in the $< a$ case: under **uniq** u and $a \notin u$, every non-last letter in every block is strictly $> a$ (not just not $< a$).

split_blocks_gt_strict *Lemma, line 844. [aux]*

Strong form of **split_blocks_wf** in the $a <$ case: every non-last letter is strictly $< a$.

foata_step_inv_lt *Lemma, line 880. [aux]*

Case **last** $u < a$ of **foata_step_inv**: appending a when last letter is smaller leaves **inv_seq** unchanged.

foata_step_inv_gt *Lemma, line 950. [aux]*

Case $a < \mathbf{last}$ u of **foata_step_inv**: appending a when last letter is larger increases **inv_seq** by **size** u .

foata_step_inv *Lemma, line 1014. [aux]*

Combined per-step invariant: **foata_step** adds **size** u to **inv_seq** iff a new descent appears (i.e. $a < \mathbf{last}$ u). Mirrors **maj_seq_rcons**.

foata_inv_eq_maj *Theorem, line 1038. [aux]*

Headline equidistribution at the seq level: **inv_seq** (**foata** w) = **maj_seq** w for **uniq** w . Proven by induction on w from the right using **foata_step_inv** and **maj_seq_rcons**.

cyc_first_to_back *Definition, line 1070. [aux]*

cyc_first_to_back s moves the first letter of s to the back. Inverse of **cyc_last_to_front**.

cyc_first_to_back_perm_eq *Lemma, line 1078.* [aux]

cyc_first_to_back preserves the multiset.

cyc_first_to_backK *Lemma, line 1087.* [aux]

cyc_first_to_back is the left inverse of cyc_last_to_front.

split_blocks_inv_aux *Fixpoint, line 1097.* [aux]

split_blocks_inv_aux P cur s is the tail-recursive worker for split_blocks_inv: cuts before each P-letter rather than after.

split_blocks_inv *Definition, line 1112.* [aux]

split_blocks_inv P s cuts s into blocks each STARTING with a P-letter (cutting right before each P-letter after the first). Inverse-direction analogue of split_blocks.

split_blocks_inv_aux_flatten *Lemma, line 1116.* [aux]

Accumulator-aware flatten property for split_blocks_inv_aux.

split_blocks_inv_aux_app_notP *Lemma, line 1162.* [aux]

When a prefix nots is all not-P, it gets absorbed into the current block accumulator.

split_blocks_inv_cyc_wf *Lemma, line 1189.* [aux]

Big cancellation: applying split_blocks_inv P to the flatten of rotated wf_blocks recovers the rotated blocks themselves. Key step in proving foata_step_undoK.

foata_step_undo *Definition, line 1224.* [aux]

foata_step_undo s inverts a single foata_step: peels the trailing letter a and rotates each split_blocks_inv block back via cyc_first_to_back.

split_blocks_inv_aux_eq *Lemma, line 1235.* [aux]

split_blocks_inv_aux only depends on the predicate extensionally.

split_blocks_inv_eq *Lemma, line 1245.* [aux]

split_blocks_inv only depends on the predicate extensionally.

foata_step_undoK *Lemma, line 1254.* [aux]

Cancellation: foata_step_undo is the left inverse of foata_step on uniq inputs with a fresh letter.

foata_inv_aux *Fixpoint, line 1360.* [aux]

foata_inv_aux n s inverts n iterations of foata_step by repeatedly applying foata_step_undo.

foata_inv *Definition, line 1370.* [aux]

foata_inv s is the seq-level inverse of foata: repeatedly peels foata_steps for size s iterations.

foata_invK_aux *Lemma, line 1375.* [aux]

Length-indexed cancellation: foata_inv_aux n (foata w) = w for size w = n and uniq w.

foata_invK *Lemma, line 1404.* [aux]

foata_inv is the left inverse of foata on uniq sequences.

foata_inj_uniq *Lemma, line 1413.* [aux]

foata is injective on uniq sequences (immediate from foata_invK).

maj_eq_maj_seq *Lemma, line 1429.* [aux]

Bridge: perm-level maj equals seq-level maj_seq of perm_to_seq.

foata_perm_to_seq_size *Lemma, line 1450.* [aux]

foata (perm_to_seq s) has size $n.+1$; needed to build foata_perm.

foata_perm_to_seq_uniq *Lemma, line 1455.* [aux]

foata (perm_to_seq s) is uniq; needed to build foata_perm.

foata_perm_to_seq_bnd *Lemma, line 1461.* [aux]

foata (perm_to_seq s) is bounded by $n.+1$; needed to build foata_perm.

foata_perm *Definition, line 1471.* [Ch 4 §4.2]

foata_perm s is the Foata bijection lifted to $\{\text{perm } 'I_n.+1\}$: apply foata at the seq level, then re-package as a permutation.

perm_to_seq_foata_perm *Lemma, line 1477.* [aux]

Commutation: perm_to_seq of foata_perm s is foata of perm_to_seq s.

inv_eq_inv_seq *Lemma, line 1484.* [aux]

Bridge: perm-level inv equals seq-level inv_seq of perm_to_seq.

foata_perm_inv_maj *Lemma, line 1516.* [Ch 4 §4.3]

Headline perm-level identity: inv (foata_perm s) = maj s, i.e. foata_perm sends maj-classes to inv-classes. Direct lift of foata_inv_eq_maj via maj_eq_maj_seq and inv_eq_inv_seq.

foata_perm_inj *Lemma, line 1527.* [Ch 4 §4.3]

foata_perm is injective; follows from foata_inj_uniq (cancellation via foata_invK). On the finite set $\{\text{perm } 'I_n.+1\}$ this implies surjective, used to prove inv_maj_eqidistr.

inv_maj_eqidistr *Theorem, line 1542.* [Ch 4 §4.3]

MacMahon's equidistribution theorem (Stanley EC1 \S{}1.3.4): the statistics inv and maj are equidistributed on $\{\text{perm } 'I_n.+1\}$. Proved via the bijection foata_perm, which satisfies inv (foata_perm s) = maj s and is injective hence bijective.

fps/fps.v

fpsP *Lemma, line 76.* [aux]

Coefficientwise equality is equality (uses functional extensionality).

fps_zero *Definition, line 92.* [aux]

(no docstring)

fps_add *Definition, line 93.* [aux]

(no docstring)

fps_opp *Definition, line 94.* [aux]

(no docstring)

fps_addA *Fact, line 96.* [aux]

(no docstring)

fps_addC *Fact, line 99.* [aux]

(no docstring)

fps_addOf *Fact, line 102.* [aux]

(no docstring)

fps_addNf *Fact, line 105.* [aux]
 (no docstring)

coef_fps0 *Lemma, line 111.* [aux]
 (no docstring)

coef_fpsD *Lemma, line 114.* [aux]
 (no docstring)

coef_fpsN *Lemma, line 117.* [aux]
 (no docstring)

coef_fpsB *Lemma, line 120.* [aux]
 (no docstring)

coef_fps_sum *Lemma, line 123.* [aux]
 (no docstring)

fps_one *Definition, line 138.* [aux]
 (no docstring)

fps_mul *Definition, line 140.* [aux]
 (no docstring)

fps_trunc *Definition, line 144.* [aux]
 Polynomial truncation: the first m+1 coefficients.

coef_fps_trunc *Lemma, line 146.* [aux]
 (no docstring)

coef_fpsM_trunc *Lemma, line 152.* [aux]
 (no docstring)

coefM_eqr *Lemma, line 163.* [aux]
 (no docstring)

coefM_eql *Lemma, line 169.* [aux]
 (no docstring)

fps_mula *Fact, line 175.* [aux]
 (no docstring)

fps_mulC *Fact, line 187.* [aux]
 (no docstring)

fps_trunc_one *Lemma, line 193.* [aux]
 (no docstring)

fps_mul1f *Fact, line 200.* [aux]
 (no docstring)

fps_mulD1 *Fact, line 207.* [aux]
 (no docstring)

fps_oner_neq0 *Fact, line 213.* [aux]
 (no docstring)

coef_fps1 *Lemma, line 223.* [aux]
 (no docstring)

coef_fpsM *Lemma, line 226.* [aux]
(no docstring)

coef0_fpsM *Lemma, line 230.* [aux]
Coefficient 0 is a ring morphism.

fps_scale *Definition, line 237.* [aux]
(no docstring)

fps_scaleA *Fact, line 239.* [aux]
(no docstring)

fps_scale1f *Fact, line 242.* [aux]
(no docstring)

fps_scaleDr *Fact, line 245.* [aux]
(no docstring)

fps_scaleDl *Fact, line 248.* [aux]
(no docstring)

coef_fpsZ *Lemma, line 255.* [aux]
(no docstring)

fps_scaleAl *Fact, line 258.* [aux]
(no docstring)

fps_scaleAr *Fact, line 267.* [aux]
(no docstring)

poly_fps *Definition, line 286.* [aux]
(no docstring)

coef_poly_fps *Lemma, line 288.* [aux]
(no docstring)

poly_fps_inj *Lemma, line 291.* [aux]
(no docstring)

poly_fps_is_zmod_morphism *Fact, line 294.* [aux]
(no docstring)

poly_fps_is_monoid_morphism *Fact, line 297.* [aux]
(no docstring)

poly_fpsC *Lemma, line 310.* [aux]
(no docstring)

inv_pref *Fixpoint, line 331.* [aux]
(no docstring)

size_inv_pref *Lemma, line 337.* [aux]
(no docstring)

fps_inv_coef *Definition, line 340.* [aux]
(no docstring)

inv_prefE *Lemma, line 342.* [aux]
(no docstring)

fps_inv_coef0 *Lemma, line 350.* [aux]
(no docstring)

fps_inv_coefS *Lemma, line 353.* [aux]
(no docstring)

fps_unit *Definition, line 361.* [aux]
(no docstring)

fps_inv *Definition, line 363.* [aux]
(no docstring)

fps_mulVf *Fact, line 367.* [aux]
(no docstring)

fps_unitP *Fact, line 383.* [aux]
(no docstring)

fps_inv_out *Fact, line 389.* [aux]
(no docstring)

fps_unitE *Lemma, line 401.* [aux]
The headline characterization of units.

fps_geom *Definition, line 414.* [aux]
The all-ones series $1 + x + x^2 + \dots$

coef_fps_onesubX *Lemma, line 416.* [aux]
(no docstring)

onesubX_mul_geom *Lemma, line 428.* [aux]
(no docstring)

fps_onesubX_unit *Lemma, line 440.* [aux]
(no docstring)

fps_inv_onesubX *Lemma, line 446.* [aux]
The acceptance test: the inverse of $1 - x$ is the geometric series.

fps/fps_comp.v

fps_comp *Definition, line 41.* [aux]
(no docstring)

coef_fps_comp *Lemma, line 44.* [aux]
(no docstring)

coef_fpsXn_small *Lemma, line 52.* [aux]
(no docstring)

coef_fpsXn_trunc *Lemma, line 67.* [aux]
(no docstring)

coef_polyXn_small *Lemma, line 80.* [aux]
(no docstring)

coef_comp_poly_wide *Lemma, line 89.* [aux]
(no docstring)

coef_comp_poly_eqp *Lemma, line 102.* [aux]
(no docstring)

coef_fps_comp_trunc *Lemma, line 118.* [aux]
(no docstring)

fps_compD *Lemma, line 139.* [aux]
(no docstring)

fps_compZ *Lemma, line 146.* [aux]
(no docstring)

fps_compN *Lemma, line 153.* [aux]
(no docstring)

fps_comp0 *Lemma, line 159.* [aux]
(no docstring)

fps_comp1 *Lemma, line 164.* [aux]
(no docstring)

fps_compC *Lemma, line 170.* [aux]
(no docstring)

coef0_fps_comp *Lemma, line 173.* [aux]
(no docstring)

fps_compX *Lemma, line 182.* [aux]
(no docstring)

fps_compM *Lemma, line 192.* [aux]
(no docstring)

fps_comp_coef_eqf *Lemma, line 214.* [aux]
Coefficient n of f \o g only sees f's coefficients <= n.

fps_comp_deriv_poly *Lemma, line 223.* [aux]
The chain rule for polynomial heads, by induction on the polynomial.

fps_comp_deriv *Theorem, line 239.* [aux]
(no docstring)

fps_comp_unit *Lemma, line 268.* [aux]
(no docstring)

fps_compV *Lemma, line 272.* [aux]
(no docstring)

fps/fps_deriv.v

fps_deriv *Definition, line 37.* [aux]
(no docstring)

coef_fps_deriv *Lemma, line 39.* [aux]
(no docstring)

fps_derivD *Lemma, line 42.* [aux]
(no docstring)

fps_derivN *Lemma, line 45.* [aux]
 (no docstring)

fps_derivB *Lemma, line 48.* [aux]
 (no docstring)

fps_derivZ *Lemma, line 51.* [aux]
 (no docstring)

fps_deriv0 *Lemma, line 54.* [aux]
 (no docstring)

fps_deriv1 *Lemma, line 57.* [aux]
 (no docstring)

fps_derivC *Lemma, line 60.* [aux]
 (no docstring)

fps_deriv_sum *Lemma, line 63.* [aux]
 (no docstring)

fps_derivM *Lemma, line 71.* [aux]
 The Leibniz rule.

fps_deriv_poly *Lemma, line 95.* [aux]
 Compatibility with the polynomial derivative.

fps_derivX *Lemma, line 101.* [aux]
 (no docstring)

natf_neq0_char0 *Lemma, line 115.* [aux]
 (no docstring)

natSf_neq0 *Lemma, line 118.* [aux]
 (no docstring)

fps_prim *Definition, line 121.* [aux]
 (no docstring)

coef_fps_prim0 *Lemma, line 124.* [aux]
 (no docstring)

coef_fps_primS *Lemma, line 127.* [aux]
 (no docstring)

fps_derivK *Lemma, line 131.* [aux]
 Differentiating a primitive recovers the series.

fps_deriv_eq0 *Lemma, line 138.* [aux]
 A series with zero derivative is its constant coefficient.

fps_deriv_inj0 *Lemma, line 149.* [aux]
 (no docstring)

fps/fps_egf.v

egf *Definition, line 35.* [aux]
(no docstring)

coef_egf *Lemma, line 37.* [aux]
(no docstring)

factf_neq0 *Lemma, line 40.* [aux]
(no docstring)

binf_neq0 *Lemma, line 43.* [aux]
(no docstring)

egf_eq *Lemma, line 47.* [aux]
egf is injective, coefficientwise.

eq_egf *Lemma, line 54.* [aux]
egf respects pointwise equality (via functional extensionality).

egfD *Lemma, line 57.* [aux]
(no docstring)

egfZ *Lemma, line 60.* [aux]
(no docstring)

egf_dirac *Lemma, line 64.* [aux]
The egf of the Kronecker delta is 1.

egf_mul *Lemma, line 71.* [aux]
THE workhorse: products of egfs are egfs of binomial convolutions.

egf_deriv *Lemma, line 87.* [aux]
Differentiation of an egf is the index shift.

fps/fps_explog.v

factU *Lemma, line 40.* [aux]
(no docstring)

fps_e *Definition, line 50.* [aux]
(no docstring)

fps_l *Definition, line 52.* [aux]
(no docstring)

fps_e0 *Lemma, line 55.* [aux]
(no docstring)

fps_l0 *Lemma, line 58.* [aux]
(no docstring)

fps_e_deriv *Lemma, line 61.* [aux]
(no docstring)

onepX_unit *Lemma, line 71.* [aux]
(no docstring)

onepX_mul_altgeom *Lemma, line 76.* [aux]
(no docstring)

fps_l_deriv *Lemma, line 89.* [aux]
(no docstring)

fps_exp *Definition, line 102.* [aux]
(no docstring)

fps_log *Definition, line 103.* [aux]
(no docstring)

coef0_fps_exp *Lemma, line 105.* [aux]
(no docstring)

fps_exp_unit *Lemma, line 108.* [aux]
(no docstring)

coef0_fps_log *Lemma, line 111.* [aux]
(no docstring)

fps_unit1 *Lemma, line 114.* [aux]
(no docstring)

fps_exp_deriv *Lemma, line 117.* [aux]
(no docstring)

fps_log_deriv *Lemma, line 124.* [aux]
(no docstring)

fps_lin_ode_uniq *Lemma, line 141.* [aux]
Solutions of the linear equation $y' = a y$ are determined by $y(0)$.

fps_deriv_inj0U *Lemma, line 158.* [aux]
Equal derivatives and equal constant coefficients force equality.

fps_expD *Lemma, line 172.* [aux]
(no docstring)

fps_logK *Lemma, line 185.* [aux]
 $\exp(\log u) = u$.

fps_expK *Lemma, line 196.* [aux]
 $\log(\exp f) = f$.

fps_exp_log_geom *Lemma, line 209.* [aux]
(no docstring)

fps/fps_ode.v

fps_quad_ode_uniq *Lemma, line 32.* [aux]
Solutions of $2 y' = c + y^2$ are determined by their constant term.

fps/fps_ogf.v

sum_bin_stair *Lemma, line 30.* [aux]

$\sum_{j \leq m} C(m - j + k, k) = C(m + k + 1, k + 1).$

coef_fps_geomXn *Lemma, line 48.* [aux]

The negative binomial expansion: $1/(1-x)^{(k+1)} = \sum_m C(m+k,k) x^m.$

onesubXn_mul_geomn *Lemma, line 59.* [aux]

Powers of $1 - x$ and of the geometric series are mutually inverse.

fps/fps_trig.v

expf *Definition, line 34.* [aux]

(no docstring)

sinf *Definition, line 35.* [aux]

(no docstring)

cosf *Definition, line 37.* [aux]

(no docstring)

secf *Definition, line 39.* [aux]

(no docstring)

tanf *Definition, line 40.* [aux]

(no docstring)

expf0 *Lemma, line 46.* [aux]

(no docstring)

sinf0 *Lemma, line 49.* [aux]

(no docstring)

cosf0 *Lemma, line 52.* [aux]

(no docstring)

cosf_unit *Lemma, line 55.* [aux]

(no docstring)

cos_mul_sec *Lemma, line 58.* [aux]

(no docstring)

secf0 *Lemma, line 61.* [aux]

(no docstring)

tanf0 *Lemma, line 67.* [aux]

(no docstring)

expf_deriv *Lemma, line 74.* [aux]

(no docstring)

sinf_deriv *Lemma, line 77.* [aux]

(no docstring)

cosf_deriv *Lemma, line 85.* [aux]

(no docstring)

sin2cos2 *Lemma, line 97.* [aux]
(no docstring)

sec2f *Lemma, line 108.* [aux]
 $\sec^2 = 1 + \tan^2$.

secf_deriv *Lemma, line 121.* [aux]
(no docstring)

tanf_deriv *Lemma, line 132.* [aux]
(no docstring)

sectan0 *Lemma, line 142.* [aux]
(no docstring)

sectan_ode *Theorem, line 145.* [aux]
(no docstring)

inversions.v

is_inv *Definition, line 27.* [Ch 3 §3.1]
 $\text{is_inv } s \ i \ j$ holds when the pair (i, j) is an inversion of s , i.e. $i < j$ but $s \ i > s \ j$.
Stanley EC1 \S{}1.3.3.

inv_set *Definition, line 32.* [Ch 3 §3.1]
 $\text{inv_set } s$ is the set of inversions of s : pairs (i, j) with $i < j$ but $s \ i > s \ j$.

inv *Definition, line 37.* [Ch 3 §3.1]
 $\text{inv } s$ is the inversion count of s , i.e. $\#\text{inv_set } s$. Stanley's $\text{inv}(w)$.

inv_id *Lemma, line 41.* [Ch 3 §3.1]
The identity permutation has no inversions: $\text{inv } 1 = 0$.

inv_double_sum *Lemma, line 51.* [aux]
Bridge lemma: $\text{inv } s$ expressed as the nested double-sum $\sum_j \sum_{(i < j) (s \ j < s \ i)}$. Bridges the inv_set cardinality form to the seq-level double-sum form of inv_seq , used in `foata.v` to prove inv_eq_inv_seq .

maj *Definition, line 82.* [Ch 3 §3.2]
 $\text{maj } s$ is the major index: the sum of the (1-indexed) descent positions of s . Stanley EC1 \S{}1.3.3. Our $\text{descent_set } s$ is 0-indexed in $'I_n$; we shift by 1 to recover Stanley's 1-indexed positions in $\{1, \dots, n\}$.

maj_id *Lemma, line 85.* [aux]
The identity permutation has zero major index: $\text{maj } 1 = 0$.

card_ord_pair *Lemma, line 101.* [aux]
The number of strictly-ordered pairs in $'I_m * 'I_m$ is the binomial coefficient $'C(m, 2)$.
Used to bound inv and maj .

inv_le *Lemma, line 142.* [Ch 3 §3.1]
Bound: $\text{inv } s \leq 'C(n+1, 2)$ for $s : \{\text{perm } 'I_{n+1}\}$, since inversions are a subset of all strictly-ordered pairs.

sum_succ_ord_eq_bin2 *Lemma, line 154.* [aux]
Helper: $\sum_{(i : 'I_n) (val \ i) . +1} = 'C(n+1, 2)$. Used to bound maj .

maj_le *Lemma, line 168.* [Ch 3 §3.2]

Bound: $\text{maj } s \leq 'C(n+1, 2)$ for $s : \{\text{perm } 'I_n.+1\}$, by comparing descent-position sum to the full sum of $(\text{val } i).+1$ over $'I_n$.

coinv_set *Definition, line 180.* [Ch 3 §3.1]

coinv_set s is the co-inversion set of s : pairs (i, j) with $i < j$ and $s\ i < s\ j$. Since s is a permutation, every strictly-ordered pair is either an inversion or a co-inversion.

rev_ord_lt *Lemma, line 185.* [aux]

Order-reversal property of **rev_ord**: $\text{rev_ord } a < \text{rev_ord } b$ iff $b < a$. Used in the **inv** reversal identity.

card_split_ord_pair *Lemma, line 194.* [aux]

Split: every strictly-ordered pair is either an inversion or a co-inversion, so $\text{inv } s + \#\#|\text{coinv_set } s| = 'C(n+1, 2)$.

inv_rev_perm_eq_coinv *Lemma, line 214.* [aux]

The inversions of **rev_perm** s are in bijection with the co-inversions of s via $(i, j) \mapsto (\text{rev_ord } j, \text{rev_ord } i)$.

inv_rev_perm *Lemma, line 238.* [Ch 3 §3.1]

Reversal identity for **inv**: $\text{inv } (\text{rev_perm } s) = 'C(n+1, 2) - \text{inv } s$. Stanley EC1 \S{}1.3.3.

maj_rev_perm *Lemma, line 260.* [aux]

Reversal identity for **maj**: $\text{maj } (\text{rev_perm } s) + (n+1) * \text{des } s = 'C(n+1, 2) + \text{maj } s$. The naive $\text{maj } (\text{rev_perm } s) = 'C(n+1, 2) - \text{maj } s$ is FALSE; the correct identity carries a $(n+1) * \text{des } s$ offset (Stanley EC1).

mmtree.v

mmtree *Inductive, line 39.* [aux]

mmtree T is the labelled binary tree datatype underlying Stanley's min-max tree construction $M(w)$ for sequences with labels in T .

mmtree_to_seq *Fixpoint, line 48.* [aux]

mmtree_to_seq t is the in-order traversal of t : left subtree, then root, then right subtree.

min_pos *Definition, line 56.* [aux]

min_pos s is the least index j in s at which the minimum of s occurs; returns 0 on the empty sequence (vacuous case).

min_in *Lemma, line 61.* [aux]

min_in states that **foldr** **minn** $a\ s$ always lies in the sequence $a :: s$; used to bound **min_pos** within range.

min_pos_lt *Lemma, line 74.* [aux]

min_pos_lt : on a nonempty sequence the split index **min_pos** s is in range, ensuring the recursion in **mmtree_of_seq_fuel** is well-defined.

mmtree_of_seq_fuel *Fixpoint, line 85.* [aux]

mmtree_of_seq_fuel $\text{fuel } s$ is the fuel-bounded M1 tree construction: splits s at **min_pos** s , recursing on the take/drop halves. Fuel equal to **size** s suffices since each recursion strictly shrinks.

mmtree_of_seq *Definition, line 101.* [aux]

`mmtree_of_seq s` runs `mmtree_of_seq_fuel` with fuel size `s`, the M1 (minimum-only) variant of Stanley's min-max tree construction.

mmtree_of_seq_fuel_correct *Lemma, line 107.* [aux]

`mmtree_of_seq_fuel_correct` : the fuel-bounded construction round-trips in-order, i.e. `mmtree_to_seq` inverts `mmtree_of_seq_fuel` when fuel bounds the sequence size.

mmtree_of_seqK *Theorem, line 136.* [aux]

`mmtree_of_seqK` is the M1 round-trip theorem: in-order traversal inverts `mmtree_of_seq` for every input sequence.

ex_seq *Definition, line 148.* [aux]

`ex_seq` is the concrete test sequence used to demonstrate that the construction yields a genuinely branching tree.

ordinal_reindex.v

leq_lift *Lemma, line 18.* [aux]

Weak monotonicity of `lift k` : $'I_n \rightarrow 'I_{n+1}$: lifting preserves $\leq$.

ltn_lift *Lemma, line 22.* [aux]

Strict monotonicity of `lift k` : $'I_n \rightarrow 'I_{n+1}$: lifting preserves $<$.

perm_compress.v

drop_fun_ne *Fact, line 23.* [aux]

`s` sends `ord_max` and `lift ord_max i` to distinct values, by injectivity.

drop_fun *Definition, line 29.* [aux]

`drop_fun i` is the unique $j : 'I_n$ such that $s (\text{lift } \text{ord_max } i) = \text{lift } (s \text{ ord_max}) j$. This is `s` restricted to $'I_{n+1} \setminus \{\text{ord_max}\}$ and reindexed onto $'I_n$.

drop_funE *Lemma, line 33.* [aux]

Defining equation for `drop_fun`: $s (\text{lift } \text{ord_max } i)$ equals $\text{lift } (s \text{ ord_max}) (\text{drop_fun } i)$.

drop_fun_inj *Lemma, line 37.* [aux]

`drop_fun` is injective, inheriting injectivity from `s`.

drop_perm *Definition, line 45.* [aux]

`drop_perm s` is the permutation of $'I_n$ obtained by deleting position `ord_max` from `s` and collapsing around the value `s ord_max`.

drop_permE *Lemma, line 48.* [aux]

Equivalence with the underlying `drop_fun` for rewriting.

lift_drop_permE *Lemma, line 53.* [aux]

Defining equation: lifting `drop_perm` back through `s ord_max` recovers `s` on lifted positions. Used to relate descents of `s` and `drop_perm s`.

lift_fun *Definition, line 68.* [aux]

Underlying function of `lift_perm k t`: sends `ord_max` to `k` and lifts `t` elsewhere via `lift k`.

lift_fun_ord_max *Lemma, line 75.* [aux]
 lift_fun sends ord_max to k.

lift_fun_lift *Lemma, line 79.* [aux]
 lift_fun on lifted positions equals lift k (t .).

lift_fun_inj *Lemma, line 83.* [aux]
 lift_fun is injective: combine injectivity of lift and t.

lift_perm *Definition, line 95.* [aux]
 lift_perm k t is the permutation of 'I_n.+1 obtained from t : {perm 'I_n} by inserting a new maximum position ord_max sent to k. Inverse of drop_perm.

lift_permE *Lemma, line 98.* [aux]
 Equivalence with the underlying lift_fun for rewriting.

lift_perm_ord_max *Lemma, line 102.* [aux]
 lift_perm k t sends ord_max to k.

lift_perm_lift *Lemma, line 106.* [aux]
 lift_perm k t on lifted positions equals lift k (t .).

lift_perm_drop_perm *Lemma, line 125.* [aux]
 Right inverse: lifting a dropped permutation at the dropped value recovers the original.
 Establishes the bijection {perm 'I_n.+1} $\sim$ 'I_n.+1 x {perm 'I_n}.

perm_seq_basics.v

is_descent_seq *Definition, line 35.* [aux]
 is_descent_seq w k is the descent predicate: k is a descent of w iff $w_k > w_{k+1}$.
 Sequence-level mirror of Stanley's $\text{Des}(w) = \{ i : w_i > w_{i+1} \}$. Moved here from psi_descent_v2.v so \S{}1.4 files can use it without importing the cd-index machinery.

perm_to_seq *Definition, line 44.* [aux]
 perm_to_seq s – bridge from {perm 'I_n} to seq nat: the list [val (s 0); val (s 1); ...; val (s (n-1))].

perm_to_seq_size *Lemma, line 49.* [aux]
 perm_to_seq_size – the seq view of a permutation has length n.

perm_to_seq_uniq *Lemma, line 54.* [aux]
 perm_to_seq_uniq – the seq view of a permutation has no duplicates (since s is injective).

nth_perm_to_seq *Lemma, line 62.* [aux]
 nth_perm_to_seq – accessing perm_to_seq s at index k yields val (s (Ordinal Hk)).

perm_to_seq_inj *Lemma, line 72.* [aux]
 perm_to_seq_inj – the perm_to_seq map is injective: distinct permutations yield distinct value lists.

is_descent_perm_seq *Lemma, line 90.* [aux]
 is_descent_perm_seq – descent equivalence: is_descent s i (perm side) equals is_descent_seq (perm_to_seq s) (val i) (seq side).

descent_to_bvec *Definition, line 107.* [aux]
 descent_to_bvec D – boolean-vector representation of a descent set, indexed by enum 'I_n: the i-th entry is (i \in D).

size_descent_to_bvec *Lemma, line 111.* [aux]

size_descent_to_bvec – the boolean vector has length n.

nth_descent_to_bvec *Lemma, line 117.* [aux]

nth_descent_to_bvec – accessing descent_to_bvec D at k yields (Ordinal Hk \{}in D).

descent_to_bvec_inj *Lemma, line 127.* [aux]

descent_to_bvec_inj – two descent sets giving the same boolean vector must be equal.

desc_positions_bvec *Lemma, line 145.* [aux]

desc_positions_bvec – the descent positions extracted from the boolean vector descent_to_bvec D coincide with set_to_seq D (both sorted-asc lists of positions of D).

seq_nth_bound *Lemma, line 183.* [aux]

seq_nth_bound – under the section hypotheses (uniqueness, bound, size), every entry of w read at an ordinal index is itself bounded by n.

seq_to_fun *Definition, line 191.* [aux]

seq_to_fun i – the underlying function $i \mapsto \text{nth } 0 \ w \ (\text{val } i)$ on ordinals, used to lift a uniq bounded seq to a permutation.

seq_to_fun_inj *Lemma, line 195.* [aux]

seq_to_fun_inj – seq_to_fun is injective (uses uniqueness of w).

seq_to_perm *Definition, line 208.* [aux]

seq_to_perm – the permutation built from seq_to_fun; inverse of perm_to_seq for uniq, n-bounded seqs of length n.

perm_to_seq_bnd *Lemma, line 218.* [aux]

perm_to_seq_bnd – every entry of perm_to_seq s is strictly less than n.

perm_to_seq_seq_to_perm *Lemma, line 225.* [aux]

perm_to_seq_seq_to_perm – round-trip identity: perm_to_seq (seq_to_perm w) = w for uniq, bounded seqs of size n.

perm_seq_bridge.v

char_mono_perm_to_seq *Lemma, line 44.* [aux]

char_mono_perm_to_seq – the seq-side char_mono of perm_to_seq s equals the perm-side boolean vector of descent_set s. Bridges psi_cdindex descent patterns to perm descent sets.

psi_apply_psis_comm *Lemma, line 70.* [aux]

psi_apply_psis_comm – the single psi i commutes with the iterated apply_psis ss (on uniq seqs), via psi_comm.

apply_psis_rev *Lemma, line 83.* [aux]

apply_psis_rev – since each psi i is involutive and they commute, apply_psis is independent of the order of the operations: rev ss gives the same result as ss.

apply_psis_revK *Lemma, line 94.* [aux]

apply_psis_revK – applying ss then rev ss cancels back to w (each psi i is involutive).

apply_psis_cancel *Lemma, line 107. [aux]*

`apply_psis_cancel` – `apply_psis ss` is its own inverse: applying `ss` twice cancels back to `w`.

powerset_internal_apply_psis *Lemma, line 119. [aux]*

`powerset_internal_apply_psis` – the internal-vertex powerset used to enumerate the M-class is invariant under `apply_psis`; consequence of `internal_vertices_apply_psis`.

class_char_monos_uniq *Lemma, line 132. [aux]*

`class_char_monos_uniq` – the descent patterns of the M-class members of `w` are pairwise distinct (combines `fact3` with `uniq_expand_cde`).

char_mono_class_inj *Lemma, line 146. [aux]*

`char_mono_class_inj` – M-class injectivity: within the M-class of `w`, two class members with the same descent pattern are actually the same sequence. Follows from `fact3` and `uniq_expand_cde`.

all_bnd_apply_psis *Lemma, line 203. [aux]*

`all_bnd_apply_psis` – the boundedness predicate `all (< n)` is preserved by `apply_psis` (uses `perm_eq_apply_psis`).

apply_psis_size_eq *Lemma, line 213. [aux]*

`apply_psis_size_eq` – `apply_psis` preserves size: if `size w = n` then `size (apply_psis ss w) = n`.

uniq_expand_cde *Lemma, line 223. [aux]*

`uniq_expand_cde` – `expand_cde` produces a list of pairwise distinct boolean vectors (key for M-class injectivity).

nil_in_powerset_internal *Lemma, line 243. [aux]*

`nil_in_powerset_internal` – the empty list is always a member of `powerset_internal w` (the M-class always contains `w` itself).

char_mono_in_expand_cde *Lemma, line 256. [aux]*

`char_mono_in_expand_cde` – the descent pattern of `w` itself appears in `expand_cde (phi_w w)` (corresponding to `ss = nil` in the `fact3` enumeration).

find_ss *Definition, line 276. [aux]*

`find_ss w bv` – search for an `ss` in `powerset_internal w` such that `apply_psis ss w` has descent pattern `bv`; returns the first match (or `::` if none).

find_ss_spec *Lemma, line 282. [aux]*

`find_ss_spec` – whenever `bv` is a descent pattern realized in the M-class of `w`, `find_ss w bv` returns a valid witness: in `powerset_internal w` and yielding the requested pattern.

class_map *Definition, line 312. [aux]*

`class_map bv sigma` – map a permutation `sigma` to the M-class member with descent pattern `bv`, packaged back as a perm. The core injection used in the proof of `omega_proper_beta_lt`.

perm_to_seq_class_map *Lemma, line 322. [aux]*

`perm_to_seq_class_map` – `perm_to_seq` of `class_map bv sigma` is exactly `apply_psis (find_ss ... bv) (perm_to_seq sigma)`.

omega_seq_mem_eq *Lemma, line 334. [aux]*

`omega_seq_mem_eq` – the `psi_cdindex omega_seq` and the local `omega_seq_local` (in `beta_bridge`) agree on memberships (definitionally identical).

omega_set_seq_bridge_bounded *Lemma, line 341.* [aux]

omega_set_seq_bridge_bounded – bridge in the bounded form: for $k < m$, $k \setminus \{\} \text{in } \text{omega_seq } (\text{set_to_seq } D)$ iff $\text{Ordinal } Hkm \setminus \{\} \text{in } \text{omega_set } D$.

omega_seq_subset_bounded *Lemma, line 352.* [aux]

omega_seq_subset_bounded – a subset relation on omega-sets transports to membership at the seq level for any list of bounded indices.

index_lt_sorted *Lemma, line 373.* [aux]

index_lt_sorted – in a uniq sorted list, order on values coincides with order on positions: $x < y$ iff $\text{index } x \text{ s} < \text{index } y \text{ s}$.

S_w_seq_all_lt *Lemma, line 403.* [aux]

S_w_seq_all_lt – elements of **S_w_seq** w (the support indices of the omega map) are bounded by $(\text{size } w) - 2$.

omega_proper_beta_lt *Lemma, line 445.* [Ch 8 §8.5]

omega_proper_beta_lt – Stanley EC1 (2nd ed.) Proposition 1.6.4 at the finset level: a strict inclusion $\text{omega_set } D \setminus \{\} \text{proper } \text{omega_set } E$ of omega-sets implies the strict inequality $\text{beta } D < \text{beta } E$ of descent counts. Headline result of this file; proof injects $\{\sigma \mid \text{descent } D\}$ into $\{\tau \mid \text{descent } E\}$ via the M-class **class_map** and exhibits a strict witness via **strict_witness_exists**.

psi_cdindex_core.v

window_size_apply_psis *Lemma, line 20.* [aux]

apply_psis preserves **window_size** at every index.

has_left_child_apply_psis *Lemma, line 29.* [aux]

apply_psis preserves **has_left_child** at every index.

is_internal_apply_psis *Lemma, line 38.* [aux]

apply_psis preserves **is_internal**.

internal_vertices_apply_psis *Lemma, line 47.* [aux]

apply_psis leaves the list of internal vertices invariant.

phi_w_apply_psis *Lemma, line 58.* [aux]

Key invariance: the cd-index **phi_w** is constant on the M-class generated by repeated **psi** applications.

phi_w_as_map *Lemma, line 72.* [aux]

phi_w w expressed as a map over **internal_vertices** w : each internal vertex contributes **D_letter** (if it has a left child) or **C_letter**.

leq_seqb_total *Lemma, line 89.* [aux]

Lex order on **seq bool** is total.

leq_seqb_trans *Lemma, line 97.* [aux]

Lex order on **seq bool** is transitive.

leq_seqb_anti *Lemma, line 105.* [aux]

Lex order on **seq bool** is antisymmetric.

sort_perm_eq_leq_seqb *Lemma, line 116.* [aux]

Sorting by `leq_seqb` is a `perm_eq`-invariant canonical form on multisets of bit-strings, used to compare M-class `char_monos` to `expand_cde` outputs.

descent_psi_effect *Lemma, line 131.* [aux]

Bit-level effect of `psi v` on the descent sequence at any position `k`: R-vertex toggles bit `v`; LR-vertex swaps bits `v.-1` and `v`. Underlies the `char_mono`-recovery proofs.

nth_char_mono *Lemma, line 178.* [aux]

`char_mono w` indexed by `k` returns the descent bit at position `k`.

size_char_mono *Lemma, line 188.* [aux]

`char_mono` has length `(size w).-1` (one bit per descent position).

is_internal_lt *Lemma, line 194.* [aux]

Internal vertices live strictly inside the descent index range, i.e. $v < (\text{size } w).-1$.

char_mono_psi_effect *Lemma, line 207.* [aux]

Bit-level effect of `psi v` on `char_mono`: a C-vertex (no left child) toggles bit at position `v`; a D-vertex (has left child) swaps bits at positions `v.-1` and `v`.

index_rcons_eq_size *Lemma, line 235.* [aux]

Helper: if `index x (rcons s y) = size s` then $x \notin s$.

mm_pos_rcons_lt *Lemma, line 248.* [aux]

Helper: `mm_pos` is never at the trailing position of a non-empty prefix, because that would force the last element to be both min and max, contradicting the presence of any other element.

has_left_child_t_last_valid *Lemma, line 344.* [aux]

Tree-level: in any valid mmtree, the last in-order position has no left child. Proved by structural induction on the tree.

has_left_child_last *Lemma, line 379.* [aux]

Seq-level corollary: the last vertex never has a left child.

has_left_child_last_fuel *Lemma, line 393.* [aux]

Fuel-bounded variant of `has_left_child_last`.

has_left_child_is_internal *Lemma, line 425.* [aux]

`has_left_child` implies `is_internal` (window size > 1). Tree induction on the mmtree of `w`.

size_powerset_of *Lemma, line 470.* [aux]

Generic powerset-by-fold has size $2^{\text{size } \text{ivs}}$.

size_powerset_internal *Lemma, line 488.* [aux]

`powerset_internal w` has cardinality $2^{\text{internal_vertices } w}$.

uniq_powerset_of *Lemma, line 493.* [aux]

Generic powerset-by-fold is `uniq` when its input is `uniq`.

uniq_powerset_internal *Lemma, line 529.* [aux]

`powerset_internal w` is duplicate-free.

subset_powerset_of *Lemma, line 533.* [aux]

Each element of the generic powerset is a subset of the input.

subset_powerset_internal *Lemma, line 563.* [aux]

Each element of `powerset_internal w` is a subset of `internal_vertices w`.

subseq_powerset_of *Lemma, line 568.* [aux]

Each element of the generic powerset is a subseq of the input.

subseq_powerset_internal *Lemma, line 597.* [aux]

Each element of `powerset_internal w` is a subseq of `internal_vertices w`, hence is uniq when `internal_vertices w` is.

char_mono_apply_psis_C_bit *Lemma, line 604.* [aux]

Bit-level recovery for a C-vertex: bit `v` of `char_mono (apply_psis ss w)` equals the original descent bit XOR the indicator of `v \{\}` in `ss`. Used to recover `ss` from `char_monos`.

char_mono_apply_psis_D_bit_pred *Lemma, line 650.* [aux]

Bit-level recovery for a D-vertex at the predecessor position `v.-1`: the bit equals the original descent bit at `v` (if `v \{\}` in `ss`) or at `v.-1` (otherwise) – encoding the swap behaviour.

char_mono_apply_psis_D_bit_self *Lemma, line 723.* [aux]

Bit-level recovery for a D-vertex at its self position `v`: the bit equals the original descent bit at `v.-1` (if `v \{\}` in `ss`) or at `v` (otherwise) – companion to `_D_bit_pred`.

size_expand_cde *Lemma, line 791.* [aux]

`expand_cde m` has 2^k elements, where k is the number of non-E letters in `m`.

size_expand_cde_phi_w *Lemma, line 807.* [aux]

`expand_cde (phi_w w)` has cardinality $2^{|internal_vertices w|}$, matching the size of `powerset_internal w` for the M-class.

check_fact3 *Definition, line 824.* [aux]

`check_fact3 w` is the boolean Fact #3 statement: the multiset of M-class `char_monos` (sorted) equals the multiset `expand_cde (phi_w w)` (sorted).

check_fact3P *Lemma, line 833.* [aux]

Reflection between `check_fact3 w` and the equality of the sorted multisets it encodes.

check_fact3_size1 *Lemma, line 845.* [aux]

Base case for `check_fact3`: sequences of size at most 1 satisfy it trivially (no internal vertices, empty cd-index).

psi_cdindex_defs.v

is_internal *Definition, line 26.* [aux]

`is_internal i w` holds iff vertex `i` is non-endpoint in the min-max tree of `w` (its window has size > 1).

apply_psis *Definition, line 31.* [aux]

`apply_psis ops w` applies the sequence of `psi` operators left-to-right, threading `psi i` for each `i` in `ops` through `w`.

char_mono *Definition, line 36.* [aux]

`char_mono w` is the descent bit-string of `w` (length $(size w) - 1$): bit `k` is `true` iff there is a descent at position `k`.

cde *Inductive, line 41. [aux]*

The cd-alphabet for classifying vertices: **C_letter** (right child only), **D_letter** (both children), **E_letter** (endpoint).

classify_vertex_cde *Definition, line 46. [aux]*

classify_vertex_cde *i w* returns the cd-letter for vertex *i* of *w*: **E_letter** if endpoint, **D_letter** if it has a left child, else **C_letter**.

phi'_w *Definition, line 53. [aux]*

phi'_w *w* is the full cde-classification string, one letter per vertex of *w* (length **size w**).

phi_w *Definition, line 58. [aux]*

phi_w *w* is the cd-index of *w*: **phi'_w w** with all **E_letter** (endpoints) stripped.

internal_vertices *Definition, line 63. [aux]*

internal_vertices *w* enumerates internal-vertex positions of *w* in ascending order.

expand_cde *Fixpoint, line 69. [aux]*

expand_cde *letters* expands a cd-word to the multiset of bit-strings: *c = a+b* yields one bit, *d = ab+ba* yields two bits in either order; **E_letter** contributes nothing.

powerset_internal *Definition, line 84. [aux]*

powerset_internal *w* enumerates all subsequences of **internal_vertices w**, used to index the M-equivalence class of *w*.

leq_seqb *Fixpoint, line 90. [aux]*

leq_seqb is the lexicographic order on **seq bool**, used to canonicalize the multiset of expanded cd-monomials by sorting.

apply_psis_nil *Lemma, line 102. [aux]*

Empty operator list acts as identity.

size_apply_psis *Lemma, line 107. [aux]*

apply_psis preserves the length of *w*.

uniq_apply_psis *Lemma, line 114. [aux]*

apply_psis preserves uniqueness of *w*.

perm_eq_apply_psis *Lemma, line 122. [aux]*

apply_psis ops w is a permutation of *w* (each **psi** is a transposition).

apply_psis_cat *Lemma, line 129. [aux]*

Concatenation of operator lists corresponds to function composition.

psi_cdindex_support.v

S_w_seq_bound *Lemma, line 41. [aux]*

Every element of **S_w_seq w** is bounded: $k < (\text{size } w) - 2$. Holds because the last vertex of *w* is always an endpoint, hence cannot contribute a D letter.

classify_vertex_cde_psi *Lemma, line 71. [aux]*

Vertex classification is invariant under **psi j**.

mm_pos_lt_pred *Lemma, line 89. [aux]*

mm_pos s $< (\text{size } s) - 1$ for **uniq s** of size at least 2. Ensures the root of the min-max tree is an internal vertex.

classify_vertex_left *Lemma, line 156.* [aux]

Classification splits across `mm_pos`: for $i < j = \text{mm_pos}$ s, use classification on the left subtree take j s.

classify_vertex_right *Lemma, line 178.* [aux]

Classification on the right side of `mm_pos`: for $i > j$, shift index by $j.+1$ and classify in the right subtree drop $j.+1$ s.

classify_vertex_mm *Lemma, line 207.* [aux]

Classification at the root $j = \text{mm_pos}$ s of a uniq seq of size at least 2: `D_letter` if the root has a left subtree ($j > 0$), else `C_letter`.

phi_w_cons_mm0 *Lemma, line 232.* [aux]

Cons decomposition of `phi_w` when `mm_pos` = 0 (root has no left subtree): the root contributes a `C_letter` and recursion peels into the tail.

phi_w_decomp_mm *Lemma, line 260.* [aux]

Decomposition of `phi_w` at `mm_pos` > 0: `phi_w` s equals the cd-index of the left subtree concatenated with `D_letter` then the cd-index of the right subtree.

D_offsets_cons_C *Lemma, line 304.* [aux]

Cons decomposition of `D_offsets` over a `C` head: every offset shifts by 1 bit.

D_offsets_cat_D *Lemma, line 323.* [aux]

Concatenation decomposition of `D_offsets` across a `D` inserted between `m1` and `m2`: left offsets, then the `D` at offset `cde_total_width m1`, then right offsets shifted by `cde_total_width m1 + 2`.

S_w_seq_decomp_mm *Lemma, line 396.* [aux]

Decomposition of `S_w_seq` at `mm_pos` > 0, parallel to `phi_w_decomp_mm`: left positions, then $j.-1$ (the `D`-position contributed by the root), then right positions shifted by $j.+1$.

cde_total_width_phi_w_all *Lemma, line 481.* [aux]

Total bit-width of `phi_w w` equals $(\text{size } w).-1$ for any uniq w (proved by structural induction on size with `mm_pos` decomposition). Captures that `expand_cde (phi_w w)` elements are descent-bit strings.

cde_total_width_phi_w *Lemma, line 566.* [aux]

Sized variant of `cde_total_width_phi_w_all`; states the bit-width identity under the usable hypothesis $2 \leq \text{size } w$.

D_offsets_phi_w_eq_S_w_seq *Lemma, line 581.* [aux]

Headline structural identity: the `D`-offsets of `phi_w w` coincide with `S_w_seq w`. Proved by structural induction on size with `mm_pos` decomposition.

uniq_expand_cde *Lemma, line 697.* [aux]

`expand_cde m` enumerates each cd-monomial expansion exactly once.

perm_eq_from_subset *Lemma, line 712.* [aux]

Two uniq sequences of the same size with one a subset of the other are permutations of each other.

check_fact3_of_perm_eq *Lemma, line 730.* [aux]

`perm_eq` of the M-class `char_monos` with `expand_cde (phi_w w)` implies the boolean Fact #3 statement.

D_vertex_descent_transition *Lemma, line 743. [aux]*

At a D-vertex (LR-internal, `has_left_child i w`), the descent bits at `i.-1` and `i` always differ – the LR-swap behaviour of `psi`.

char_mono_self_mem *Lemma, line 759. [aux]*

Self-support: `char_mono w` is itself a member of `expand_cde (phi_w w)`, using `phi_w_support_general` applied to `X = char_mono w`.

char_mono_apply_psis_mem *Lemma, line 801. [aux]*

Membership: every M-class element has its `char_mono` inside `expand_cde (phi_w w)`, by combining `phi_w_apply_psis` and `char_mono_self_mem`.

in_internal_vertices *Lemma, line 815. [aux]*

Membership in `internal_vertices` is exactly `is_internal`.

uniq_map_char_mono_powerset *Lemma, line 897. [aux]*

The list of M-class `char_monos` (indexed by `powerset_internal w`) is duplicate-free, by `char_mono_apply_psis_inj`.

check_fact3_true *Lemma, line 911. [aux]*

Boolean Fact #3 holds for every `uniq w`: combines `uniq` + same-size + membership to derive `perm_eq`, then the sorted equality.

fact3 *Lemma, line 932. [aux]*

Headline Fact #3 (Stanley EC1 \S{}1.6.3, Theorem 1.6.3): the multiset of `char_monos` over the M-class of `w`, canonicalised by `sort leq_seqb`, equals the canonicalised expansion `expand_cde (phi_w w)`. Direct consequence of `check_fact3_true` via `check_fact3P`.

psi_cdindex_support_defs.v

cde_width *Definition, line 21. [aux]*

`cde_width 1` is the number of bits a single cd-letter contributes to `expand_cde`: C contributes 1, D contributes 2, E contributes 0.

cde_total_width *Definition, line 26. [aux]*

`cde_total_width m` is the bit-length of every element of `expand_cde m`; equals the sum of `cde_width` over `m`.

cde_offset *Definition, line 31. [aux]*

`cde_offset m i` is the cumulative bit-offset where the `i`-th letter of `m` starts in any element of `expand_cde m`.

D_offsets *Definition, line 37. [aux]*

`D_offsets m` enumerates the bit positions where each D in `m` starts; these are exactly the indices at which an expansion must have a transition (the two bits of `d` differ).

has_transition *Definition, line 42. [aux]*

`has_transition X k` holds iff bits `k` and `k.+1` of `X` differ.

all_D_transitions *Definition, line 47. [aux]*

`all_D_transitions m X` holds iff `X` has a bit transition at every D-offset of `m`; characterizes membership in `expand_cde m`.

size_in_expand_cde *Lemma, line 51. [aux]*

Every element of `expand_cde m` has length `cde_total_width m`.

has_transition_cons *Lemma, line 68.* [aux]

Cons-shift for **has_transition**: bumping the index by 1 corresponds to dropping a leading bit.

has_transition_cons2 *Lemma, line 73.* [aux]

Cons-shift by 2 for **has_transition**: dropping two leading bits.

cde_offset_C_succ *Lemma, line 78.* [aux]

Cons-unfolding for **cde_offset** over a C head: shift by 1 bit.

cde_offset_D_succ *Lemma, line 83.* [aux]

Cons-unfolding for **cde_offset** over a D head: shift by 2 bits.

expand_cde_mem_transitions *Lemma, line 89.* [aux]

Forward direction of the transition characterisation: every **X** in **expand_cde m** has the required bit transitions at all D-offsets.

transitions_expand_cde_mem *Lemma, line 141.* [aux]

Backward direction: any bit-string of the right length with D-offset transitions is a member of **expand_cde m**.

expand_cde_mem_iff *Lemma, line 220.* [aux]

Combined characterisation: when **X** has the right length, membership in **expand_cde m** is equivalent to having a transition at every D-offset. Pivot lemma used by **phi_w_support_general**.

mem_filter_iota_nth *Lemma, line 235.* [aux]

Membership in the descent set **filter (nth X) (iota 0 m)** reduces to the bit at that index.

foldr_maxn_ge *Lemma, line 245.* [aux]

Every element of **s** is bounded above by **foldr maxn 0 s**.

mem_omega_seq *Lemma, line 255.* [aux]

Bool form of **omega_seq** membership: the XOR criterion plus the bound $k \leq \max s$.

has_transition_omega_seq *Lemma, line 263.* [aux]

Bridge: a bit transition in **X** at index $k < m - 1$ is equivalent to $k \in \text{omega_seq}$ of **X**'s descent set. Pivot used in the support proof to translate D-offset transitions into **omega_seq** membership.

cde_total_width_cat *Lemma, line 295.* [aux]

cde_total_width is additive over concatenation.

psi_cdindex_tree.v

window_size_last_fuel *Lemma, line 44.* [aux]

Fuel-bounded variant: **window_size** at the last index of a non-empty **w** equals 1. Used to prove the leaf-status of the last vertex.

window_size_last *Lemma, line 73.* [aux]

Last vertex has window size 1, i.e. is always an endpoint.

window_size_t *Fixpoint, line 85.* [aux]

window_size_t i t is the tree-level analogue of **window_size**: recursion on the **mmtree t** using the in-order index **i**.

has_left_child_t *Fixpoint, line 96.* [aux]
 has_left_child_t i t is the tree-level analogue of has_left_child: direct structural recursion on the mmtree.

is_internal_t *Definition, line 106.* [aux]
 Tree-level analogue of is_internal.

window_size_t_Node_lt *Lemma, line 115.* [aux]
 Cons-style equation: index in left subtree forwards to 1.

window_size_t_Node_eq *Lemma, line 121.* [aux]
 At the root index, the window covers the root and the right subtree.

window_size_t_Node_gt *Lemma, line 127.* [aux]
 Index above the root forwards to the right subtree with shifted index.

has_left_child_t_Node_lt *Lemma, line 141.* [aux]
 Cons-style equation for has_left_child_t: index in left subtree.

has_left_child_t_Node_eq *Lemma, line 147.* [aux]
 Root has a left child iff the left subtree is non-empty.

has_left_child_t_Node_gt *Lemma, line 153.* [aux]
 Index above the root forwards has_left_child_t to right subtree.

size_mmtree_to_seq_node *Lemma, line 167.* [aux]
 In-order seq of Node l x r has size |l| + |r| + 1.

has_left_child_t_0 *Lemma, line 173.* [aux]
 Position 0 (leftmost vertex) never has a left child.

window_size_t_eq *Lemma, line 188.* [aux]
 Bridge: tree-level window_size_t agrees with seq-level window_size on the in-order traversal of any valid mmtree.

has_left_child_t_eq *Lemma, line 207.* [aux]
 Bridge: tree-level and seq-level has_left_child agree.

is_internal_t_eq *Lemma, line 225.* [aux]
 Bridge: tree-level and seq-level is_internal agree.

endpoint_implies_next_t *Lemma, line 237.* [aux]
 Tree-level: an endpoint at position k is immediately followed (at k.+1) by a vertex with a left child. Used to bound S_w_seq.

LR_pred_is_endpoint_t *Lemma, line 304.* [aux]
 Tree-level: the predecessor of an LR-vertex (D-vertex) is an endpoint. Key structural property used in the bit-recovery lemmas.

endpoint_implies_next_has_left_child *Lemma, line 394.* [aux]
 Seq-level corollary of endpoint_implies_next_t via the mmtree bridge.

LR_pred_is_endpoint *Lemma, line 412.* [aux]
 Seq-level corollary of LR_pred_is_endpoint_t: D-vertex predecessor is always an endpoint. Used by the bit-recovery proofs in psi_cdindex_core.v.

psi_cdindex_tree_shape.v

mmtree_shape_fuel *Fixpoint, line 29.* [aux]

`mmtree_shape_fuel fuel s` is the fuel-bounded shape encoding of the min-max tree of `s`:
`head = mm_pos s`, `tail` = shape of left subtree concatenated with shape of right subtree.

mmtree_shape *Definition, line 48.* [aux]

`mmtree_shape s` is the canonical shape encoding of the min-max tree of `s`, obtained by saturating the fuel at `size s`. Two sequences with equal shape have isomorphic min-max trees.

mmtree_shape_fuel_monotone *Lemma, line 54.* [aux]

`mmtree_shape_fuel` is independent of the fuel once it is large enough.

mmtree_shape_nil *Lemma, line 78.* [aux]

Shape of the empty sequence is empty.

mmtree_shape_cons *Lemma, line 83.* [aux]

Cons unfolding: shape of `a :: s0 = mm_pos` then shapes of left and right subtrees, mirroring the tree-construction recursion.

size_mmtree_shape *Lemma, line 106.* [aux]

Shape encoding has the same length as the underlying sequence.

mmtree_shape_order_iso *Lemma, line 134.* [aux]

Headline structural lemma: order-isomorphic sequences (same size, both uniq, same comparison pattern) produce the same tree shape. This is the single heavy proof underlying the per-property invariance lemmas below.

seq_cat_left_eq *Lemma, line 189.* [aux]

Left-cancellation for `++` when prefixes have equal size.

seq_cat_right_eq *Lemma, line 200.* [aux]

Right-cancellation for `++` when prefixes have equal size.

mmtree_shape_decompose *Lemma, line 212.* [aux]

Inverse of `mmtree_shape_cons`: equal shapes split into equal roots and equal subtree shapes, providing a structural recursion principle.

has_left_child_of_shape *Lemma, line 244.* [aux]

`has_left_child` depends only on the tree shape: equal shapes give equal `has_left_child` at every index.

window_size_of_shape *Lemma, line 291.* [aux]

`window_size` depends only on the tree shape.

has_left_child_order_iso *Lemma, line 345.* [aux]

Compatibility wrapper used downstream: order-isomorphic uniq sequences give equal `has_left_child` at every index. Composes `has_left_child_of_shape` with `mmtree_shape_order_iso`.

mmtree_shape_psi *Lemma, line 367.* [aux]

Heavy lemma: applying `psi j` does not change the tree shape (for uniq `w`). Proved by case-splitting on `j` vs `m = mm_pos w` and reducing the root case to `mmtree_shape_order_iso` on the rotated tail.

has_left_child_psi *Lemma, line 486.* [aux]

Headline corollary: `psi j` preserves `has_left_child`. Five-line proof composing `has_left_child_of_shape` with `mmtree_shape_psi`.

psi_cdindex_witness.v

omega_seq *Definition, line 30.* [aux]

ω_{seq} s is the seq-level ω map: positions k such that exactly one of k , $k+1$ belongs to s . Mirrors ω_{set} from `beta_swap.v` on raw `seq nat` (no finset).

is_D_letter *Definition, line 39.* [aux]

is_D_letter l tests whether l is the D cd-letter.

S_w_seq *Definition, line 45.* [aux]

S_w_seq w is the d-position set of w : predecessors of D-vertices in the cd-classification. This is the support set S_w from Stanley's Prop 1.6.4.

omega_monotone_class_count *Lemma, line 104.* [aux]

Weak monotonicity: if ω_{seq} S is contained in ω_{seq} T , then every M-class whose d-positions are covered by ω_{seq} S is also covered by ω_{seq} T . Half of Stanley's Prop 1.6.4.

witness_perm *Definition, line 125.* [aux]

$witness_perm$ n k is the permutation $1; \dots; k; k+2; k+1; k+3; \dots; n$: a single inversion at position k embedded in a sorted backbone. Realises a cd-word with $S_w = \{k\}$, used in `strict_witness_exists`.

leq_maxn *Lemma, line 129.* [aux]

Local arithmetic helper: $maxn$ a $b = a$ when $b \leq a$.

geq_maxn *Lemma, line 137.* [aux]

Local arithmetic helper: $maxn$ a $b = b$ when $a \leq b$.

geq_minn *Lemma, line 141.* [aux]

Local arithmetic helper: $minn$ a $b = a$ when $a \leq b$.

path_leq_last *Lemma, line 149.* [aux]

A leq-path from a over s gives $a \leq last\ a\ s$.

foldr_maxn_path *Lemma, line 158.* [aux]

On a sorted (leq-path) sequence, `foldr maxn` reduces to its last element.

foldr_minn_ge *Lemma, line 178.* [aux]

`foldr minn` a $s = a$ when every element of s is at least a .

all_le_iota *Lemma, line 187.* [aux]

Every element of $iota\ m.+1\ n$ is at least m .

min_pos_iota *Lemma, line 194.* [aux]

min_pos of an ascending $iota$ is 0 (minimum is the head).

path_iota *Lemma, line 201.* [aux]

$iota\ m.+1\ n$ is a leq-path with starting point m .

last_iota *Lemma, line 205.* [aux]

Last element of $m :: iota\ m.+1\ n$ is $m + n$.

foldr_maxn_iota *Lemma, line 212.* [aux]

`foldr maxn` of an ascending $iota$ equals its last element.

max_pos_iota *Lemma, line 222.* [aux]

max_pos of an ascending $iota$ is the last index.

mm_pos_iota *Lemma, line 239. [aux]*
 mm_pos of an ascending **iota** is 0 (root is the minimum).

size_witness_perm *Lemma, line 245. [aux]*
 witness_perm **n** **k** has length **n** when $k + 2 < n$.

witness_perm_uniq *Lemma, line 256. [aux]*
 witness_perm **n** **k** is duplicate-free when $k + 2 < n$.

check_strict_witness *Definition, line 297. [aux]*
 Boolean witness: witness_perm **n** **k** is uniq, of length **n**, and has **S_w_seq** equal to $[: : k]$. Used for vm_compute sanity checks.

check_strict_witness_correct *Lemma, line 323. [aux]*
 check_strict_witness reflects strict_witness_exists for that **n**,**k**.

has_left_child_iota *Lemma, line 335. [aux]*
 Helper: ascending sequences (**iota**) have no left children in their min-max tree, since mm_pos = 0 at every recursion depth.

foldr_minn_all_gt *Lemma, line 348. [aux]*
 Helper: foldr minn **a** **s** = **a** when **a** is strictly less than every element of **s**.

mm_pos_min_first *Lemma, line 359. [aux]*
 mm_pos (**a** :: **s**) = 0 when **a** is the strict minimum of **a** :: **s**.

min_pos_core *Lemma, line 372. [aux]*
 min_pos of the witness core $[k+2; k+1] ++ \text{iota } k.+3 \text{ m}$ is 1 (the **k+1** at position 1 is the unique minimum).

max_pos_core_gt0 *Lemma, line 386. [aux]*
 max_pos of the witness core is positive when the suffix $\text{iota } k.+3 \text{ m}$ is non-empty.

mm_pos_core *Lemma, line 401. [aux]*
 mm_pos of the witness core is 1 (position of **k+1**) when the suffix is non-empty.

hlc_core_not1 *Lemma, line 418. [aux]*
 In the witness core, only position 1 has a left child; all other positions are leaves of the min-max tree.

hlc_core_1 *Lemma, line 437. [aux]*
 Position 1 in the witness core has a left child (the **k+2** vertex).

ws_core_1 *Lemma, line 450. [aux]*
 window_size at the mm_pos position 1 in the witness core is **m.+1** (covers **k+1** and the entire ascending suffix).

S_w_seq_core *Lemma, line 466. [aux]*
 S_w_seq of the witness core is $[: : 0]$: the only D-position is at index 1, contributing 0 to the d-position set.

S_w_seq_witness_k0 *Lemma, line 501. [aux]*
 Base case **k** = 0: witness_perm **n** 0 is exactly the witness core, so **S_w_seq** = $[: : 0]$.

classify_skip_mm0 *Lemma, line 514. [aux]*
 When the head **a** is at the root (mm_pos = 0), classifying any positive position **i** in **a** :: rest reduces to classifying **i-1** in the right subtree rest.

S_w_seq_shift *Lemma, line 537. [aux]*

Index-shift property: when the head is at the root, **S_w_seq** of the cons equals the **S_w_seq** of the tail with every entry incremented.

drop1_witness_map_succ *Lemma, line 581. [aux]*

Tail of **witness_perm n k.+1** equals the +1-shift of **witness_perm n.-1 k**; provides the inductive step for **k**.

classify_map_succ *Lemma, line 606. [aux]*

classify_vertex_cde is invariant under the +1-shift on a uniq sequence (since cd-classification depends only on order).

S_w_seq_map_succ *Lemma, line 634. [aux]*

S_w_seq is invariant under the +1-shift on a uniq sequence.

mm_pos_witness *Lemma, line 647. [aux]*

mm_pos of **witness_perm n k** is 0 for $k \geq 1$: the 1 at the head is the strict minimum of the permutation.

strict_witness_exists *Lemma, line 698. [aux]*

Strict-witness existence (Stanley Prop 1.6.4, strict half): for every $k < n.-2$ there is a uniq permutation **w** of length **n** with **S_w_seq w** = $[: : k]$. The witness is **witness_perm n k**.

psi_comm.v

sorted_uniq_nth_ltn *Lemma, line 15. [aux]*

sorted_uniq_nth_ltn : on a uniq sorted sequence, value order at two indices is the same as index order; ranking helper for **psi_comm**.

shift_preserves_ltn *Lemma, line 36. [aux]*

shift_preserves_ltn : the modular shift by **delta** preserves the $<$ relation on ranks within the appropriate non-extremum sub-range. Modular-arithmetic core of interior order preservation.

rank_shift_preserves_interior_order *Lemma, line 79. [aux]*

rank_shift_preserves_interior_order : at non-head positions **p**, **q** the rank-shift on **L** is order-preserving (no wrap-around). Used to propagate descent comparisons through **psi**.

window_size_psi *Lemma, line 163. [aux]*

window_size_psi : applying **psi j** does not change **window_size i** at any other position **i** – generalization of **window_size_psi_self** to cross positions, foundational for **psi_comm**.

window_at_psi_disjoint *Lemma, line 324. [aux]*

window_at_psi_disjoint : **psi j** preserves **window_at i** verbatim when the windows at **i** and **j** are disjoint (i.e. $i + \text{ws}_i \leq j$).

psi_comm_disjoint_lr *Lemma, line 354. [aux]*

psi_comm_disjoint_lr : disjoint commutativity in the directed form $i + \text{ws}_i \leq j$; combinatorially trivial since the windows act on non-overlapping slices of **w**.

psi_comm_disjoint *Lemma, line 446. [aux]*

psi_comm_disjoint : symmetric form of **psi_comm_disjoint_lr** – **psi i** and **psi j** commute whenever the two M-windows are disjoint.

window_size_psi_ancestor *Lemma, line 479.* [aux]

`window_size_psi_ancestor` : nested specialization of `window_size_psi` – when `W_j` sits inside `W_i`, applying `psi i` preserves `window_size j`.

sort_map_mono *Lemma, line 513.* [aux]

`sort_map_mono` : `sort leq` commutes with a monotone injection on the elements of `L`; structural helper for `psi_map_comm`.

index_map_inj_in *Lemma, line 551.* [aux]

`index_map_inj_in` : `index` commutes with a locally injective map `f`, a structural helper for the rank-shift transfer.

mono_inj_in *Lemma, line 574.* [aux]

`mono_inj_in` : a $\leftarrow$ -monotone function on a finite set is injective there; trivial corollary used in `psi_map_comm`.

rank_shift_map_comm *Lemma, line 590.* [aux]

`rank_shift_map_comm` : `rank_shift_seq` commutes with any monotone injection on the elements of `L`; algebraic core of `psi_map_comm`.

psi_map_comm *Lemma, line 657.* [aux]

`psi_map_comm` : `psi k` commutes with any comparison-preserving relabelling `f`. Reduces nested commutativity to interior-only rank-shift commutativity.

rank_shift_psi_comm *Lemma, line 720.* [aux]

`rank_shift_psi_comm` : when `d` is rooted at the head (`mm_pos d = 0`) and `k > 0`, `rank_shift_seq` commutes with `psi k`. Algebraic core of `psi_comm_nested`.

psi_comm_nested *Lemma, line 833.* [aux]

`psi_comm_nested` : nested commutativity – when `W_j` sits inside `W_i`, `psi i` and `psi j` commute. Combines window-stability and `rank_shift_psi_comm`.

psi_comm *Theorem, line 1351.* [aux]

`psi_comm` is the M3 commutativity theorem (Stanley Fact #1, half 1): the operators `psi i` all pairwise commute on `uniq` inputs. Proved by `window_trichotomy` dispatching to disjoint or nested commutativity.

psi_core.v

max_pos *Definition, line 18.* [aux]

`max_pos s` is the least index in `s` at which the maximum of `s` occurs, the dual of `min_pos`.

mm_pos *Definition, line 23.* [aux]

`mm_pos s` is Stanley’s min-or-max split index: the least index at which either the minimum or the maximum of `s` occurs.

max_in *Lemma, line 28.* [aux]

`max_in` is the dual of `min_in`: `foldr maxn a s` always lies in `a :: s`.

max_pos_lt *Lemma, line 39.* [aux]

`max_pos_lt` : on a nonempty sequence the index `max_pos s` is in range.

mm_pos_lt *Lemma, line 47.* [aux]

`mm_pos_lt` : on a nonempty sequence the min-or-max split index is in range, ensuring `mmtree_of_seq_mm_fuel` recursion is well-defined.

mmtree_of_seq_mm_fuel *Fixpoint, line 56.* [aux]

`mmtree_of_seq_mm_fuel fuel s` is the Stanley-correct min-max tree construction (M2 variant): splits `s` at `mm_pos s` rather than `min_pos s`, producing the alternating-extremum tree shape.

mmtree_of_seq_mm *Definition, line 72.* [aux]

`mmtree_of_seq_mm s` is Stanley's min-max tree $M(w)$ for $s = w$: the canonical M-class representative used to define `psi`.

mmtree_of_seq_mm_fuel_correct *Lemma, line 77.* [aux]

`mmtree_of_seq_mm_fuel_correct` : the M2 fuel construction round-trips in-order, the analogue of `mmtree_of_seq_fuel_correct`.

mmtree_of_seq_mmK *Theorem, line 101.* [aux]

`mmtree_of_seq_mmK` is the M2 round-trip theorem: in-order traversal inverts the Stanley min-max construction `mmtree_of_seq_mm`.

window_size_fuel *Fixpoint, line 115.* [aux]

`window_size_fuel fuel i s` computes the size of the M-window at in-order position `i` (i.e. $1 + |\text{right subtree of vertex } i|$) by fuel-bounded recursion mirroring `mmtree_of_seq_mm_fuel`.

window_size *Definition, line 132.* [aux]

`window_size i s` is $1 + |\text{right subtree of vertex at in-order position } i|$ in $M(s)$, i.e. the size of the M-window on which `psi i` acts; 0 when $i \geq \text{size } s$.

window_at *Definition, line 138.* [aux]

`window_at i w` is the in-order labels of the M-window at position `i`: the contiguous slice `w_i, w_{i+1}, ..., w_{i+ws-1}` where `ws = window_size i w`.

window_size_fuel_bound *Lemma, line 149.* [aux]

`window_size_fuel_bound` : combined invariant for the fuel version – the window is positive iff `i` is in range, and is bounded by the available suffix length.

window_size_gt0 *Lemma, line 212.* [aux]

`window_size_gt0` : the M-window at an in-range position is nonempty.

window_size_bound *Lemma, line 221.* [aux]

`window_size_bound` : the M-window at `i` never exceeds the suffix `size w - i`, so `window_at i w` is a slice of `w`.

window_size_oor *Lemma, line 228.* [aux]

`window_size_oor` : out-of-range positions have empty window, by which `psi i` degenerates to the identity.

rank_shift_seq *Definition, line 245.* [aux]

`rank_shift_seq L` is the rank-shift on the M-window: replace the head with the opposite extremum (max if head = min, else min) and rotate the other ranks accordingly. This is the algebraic core of `psi_i`.

size_rank_shift_seq *Lemma, line 253.* [aux]

`size_rank_shift_seq` : `rank_shift_seq` preserves length (definitional).

psi *Definition, line 265.* [aux]

`psi i w` is Stanley's `psi_i` operator on the M-class representative `w`: it fixes the prefix of length `i`, applies the rank-shift to the M-window `window_at i w`, and concatenates the unchanged suffix. Total function; identity outside the window or when $i \geq \text{size } w$.

map_nth_iota_sorted *Lemma, line 272. [aux]*
 map_nth_iota_sorted : mapping nth 0 sorted across iota 0 (size sorted) reproduces sorted. Helper for the rank-shift permutation analysis.

map_mod_iota_rot *Lemma, line 286. [aux]*
 map_mod_iota_rot : the modular shift $r \mapsto (r + \text{shift}) \% k$ on iota 0 k equals a rotation of iota 0 k; key step in proving rank_shift_perm_eq.

rank_shift_perm_eq *Lemma, line 320. [aux]*
 rank_shift_perm_eq : rank_shift_seq L is a permutation of L, so psi i preserves the multiset of labels in w.

sort_rank_shift_seq *Lemma, line 362. [aux]*
 sort_rank_shift_seq : the rank-shift commutes with sort leq, a consequence of rank_shift_perm_eq.

uniq_rank_shift_seq *Lemma, line 375. [aux]*
 uniq_rank_shift_seq : the rank-shift preserves uniqueness, by permutation.

size_rank_shift_seq2 *Lemma, line 382. [aux]*
 size_rank_shift_seq2 : size preservation derived from perm_eq, convenient form for downstream rewrites.

psi_perm_eq *Lemma, line 390. [aux]*
 psi_perm_eq is the M2 multiset preservation theorem: psi i w is a permutation of w for every i and w.

rank_shift_seqE *Lemma, line 423. [aux]*
 rank_shift_seqE : explicit map characterization of rank_shift_seq L in terms of the sort and the head's extremum role.

nth_rank_shift_seq *Lemma, line 439. [aux]*
 nth_rank_shift_seq : positionwise formula for rank_shift_seq L, used in the involutivity and interior-order proofs.

head_rank_shift_seq *Lemma, line 452. [aux]*
 head_rank_shift_seq : the head of rank_shift_seq L is the formula above instantiated at $n = 0$.

rank_shift_seq_involutive *Lemma, line 467. [aux]*
 rank_shift_seq_involutive : applying the rank-shift twice is the identity, provided the head of L is the global min or max. Algebraic heart of the psi_involutive theorem.

window_size_fuel_monotone *Lemma, line 540. [aux]*
 window_size_fuel_monotone : the fuel-bounded window_size is independent of fuel as long as fuel covers the input size.

window_size_cons *Lemma, line 568. [aux]*
 window_size_cons : structural unfolding of window_size on a cons by the mm_pos split, mirroring the recursion of mmtree_of_seq_mm.

psi_id oor *Lemma, line 592. [aux]*
 psi_id oor : psi i w = w when i is out of range, the edge convention from M2_PSI_INFORMAL.md.

psi_id_trivial *Lemma, line 603. [aux]*
 psi_id_trivial : psi i w = w on a trivial window (size ≤ 1 , i.e. a leaf or single-vertex right subtree).

window_at_cons *Lemma, line 624.* [aux]
 window_at_cons : structural unfolding of window_at on a cons, by the mm_pos split.
 Companion of window_size_cons.

size_psi *Lemma, line 665.* [aux]
 size_psi : psi i preserves length, derived from psi_perm_eq.

uniq_psi *Lemma, line 669.* [aux]
 uniq_psi : psi i preserves uniqueness, derived from psi_perm_eq.

take_psi *Lemma, line 674.* [aux]
 take_psi : psi j does not touch positions strictly before j, so any prefix of length k ≤ j is unchanged.

foldr_minn_aux *Lemma, line 698.* [aux]
 foldr_minn_aux : foldr minn a s bounds a and every element of s from above; technical helper for the extremum reasoning.

foldr_minn_le *Lemma, line 711.* [aux]
 foldr_minn_le : convenient form of foldr_minn_aux – the fold-min bounds every element of a :: s from below.

foldr_maxn_aux *Lemma, line 718.* [aux]
 foldr_maxn_aux : dual of foldr_minn_aux for maxn.

foldr_maxn_ge *Lemma, line 730.* [aux]
 foldr_maxn_ge : dual of foldr_minn_le for maxn.

min_val_perm_eq *Lemma, line 738.* [aux]
 min_val_perm_eq : the global min as computed by foldr minn depends only on the multiset, so psi i preserves the minimum value.

max_val_perm_eq *Lemma, line 751.* [aux]
 max_val_perm_eq : dual of min_val_perm_eq for the global max.

window_fits_left *Lemma, line 767.* [aux]
 window_fits_left : when i sits in the left subtree of the root (i < mm_pos w), the M-window at i does not cross the root.

drop_psi *Lemma, line 786.* [aux]
 drop_psi : psi i does not touch positions at or after the window end, so the suffix from i + window_size i w is unchanged.

nth_psi_left *Lemma, line 805.* [aux]
 nth_psi_left : positionwise version of take_psi – entries before the window are fixed.

nth_psi_right *Lemma, line 814.* [aux]
 nth_psi_right : positionwise version of drop_psi – entries after the window are fixed.

nth_psi_inside *Lemma, line 824.* [aux]
 nth_psi_inside : within the window, psi i w reads off rank_shift_seq (window_at i w) at the offset k - i.

take_psi_perm *Lemma, line 849.* [aux]
 take_psi_perm : when j sits past the window end, the prefix take j (psi i w) is a permutation of take j w; used in mm_pos_psi_eq to argue invariance of the global extremum positions.

notin_take_mm *Lemma, line 886. [aux]*
notin_take_mm : neither the global min nor the global max appears in the prefix take (mm_pos w) w, by minimality of mm_pos.

min_val_drop *Lemma, line 904. [aux]*
min_val_drop : if the global min is not in take j w, the min of drop j w equals the global min of w.

max_val_drop *Lemma, line 929. [aux]*
max_val_drop : dual of min_val_drop for the global max.

mm_pos_char *Lemma, line 958. [aux]*
mm_pos_char : the mm_pos s index is uniquely characterized by: no extremum sits in take j s and the extremum sits at nth 0 s j. Used to prove mm_pos is preserved by psi.

nth_w_mm_pos *Lemma, line 996. [aux]*
nth_w_mm_pos : the entry at mm_pos w is exactly the global min or the global max of w, by definition of mm_pos.

sorted_head_le *Lemma, line 1011. [aux]*
sorted_head_le : on a sorted sequence, the head bounds every element from below.

sorted_last_ge *Lemma, line 1024. [aux]*
sorted_last_ge : on a sorted sequence, the last element bounds every element from above.

min_eq_nth_sort_0 *Lemma, line 1040. [aux]*
min_eq_nth_sort_0 : the foldr minn minimum equals the first entry of the sorted list – bridge from value-based to rank-based statements.

max_eq_nth_sort_last *Lemma, line 1060. [aux]*
max_eq_nth_sort_last : dual of min_eq_nth_sort_0 – the foldr maxx maximum equals the last entry of the sorted list.

rank_shift_head_min_to_max *Lemma, line 1087. [aux]*
rank_shift_head_min_to_max : if the head of L is the minimum, the rank-shift sends it to the maximum; this is one half of the head-flip.

rank_shift_head_max_to_min *Lemma, line 1106. [aux]*
rank_shift_head_max_to_min : the dual head-flip – if the head of L is the maximum, the rank-shift sends it to the minimum.

mm_pos_psi_eq *Lemma, line 1137. [aux]*
mm_pos_psi_eq : psi i preserves the global root index mm_pos, by the head-flip combined with extremum-preservation away from the window. Key step toward psi_involutive.

ws_lt_size *Lemma, line 1266. [aux]*
ws_lt_size : a nontrivial window forces i < size w; used to discharge in-range hypotheses cheaply.

take_mm_psi *Lemma, line 1276. [aux]*
take_mm_psi : when i is in the left subtree, psi i commutes with take (mm_pos w), i.e. acts on the left subtree's labels alone. Recursive descent step for window_size_psi_self and window_at_psi_self.

drop_mm_psi *Lemma, line 1309.* [aux]

drop_mm_psi : dual of **take_mm_psi** – when i is in the right subtree, $\text{psi } i$ commutes with $\text{drop } (\text{mm_pos } w) . +1$ and re-indexes i by subtracting the left-plus-root size.

window_size_psi_self *Lemma, line 1358.* [aux]

window_size_psi_self : applying $\text{psi } i$ to w does not change the window size at the same position i . Proved by induction on size using **take_mm_psi**/**drop_mm_psi**.

window_at_psi_self *Lemma, line 1418.* [aux]

window_at_psi_self : the window at i after one application of $\text{psi } i$ is exactly the rank-shifted original window. Required by the second $\text{psi } i$ application in **psi_involutive**.

window_head_extremum *Lemma, line 1495.* [aux]

window_head_extremum : the head of **window_at** i w is always an extremum (min or max) of the window – Stanley’s structural fact (F2) transferred to the window. Discharge condition for **rank_shift_seq_involutive**.

psi_involutive *Theorem, line 1559.* [aux]

psi_involutive is the M2 involutivity theorem: $\text{psi } i (\text{psi } i w) = w$ for every i and uniq w . Stanley fact about **psi_i**.

window_trichotomy *Lemma, line 1617.* [aux]

window_trichotomy : two M-windows in the same word are either disjoint (left or right) or one is nested in the other – the binary tree geometry that drives the **psi_comm** proof.

foldr_minn_le_nth *Lemma, line 1821.* [aux]

foldr_minn_le_nth : positionwise version of **foldr_minn_le**.

foldr_maxn_ge_nth *Lemma, line 1827.* [aux]

foldr_maxn_ge_nth : positionwise version of **foldr_maxn_ge**.

mm_pos_order_iso *Lemma, line 1836.* [aux]

mm_pos_order_iso : **mm_pos** depends only on the comparison structure of the sequence – two order-isomorphic sequences have the same **mm_pos**. Foundational for transferring tree shape across the rank-shift.

order_iso_take *Lemma, line 1943.* [aux]

order_iso_take : the order-isomorphism property descends to **take** m of both sequences; helper for the recursive case in **window_size_order_iso**.

order_iso_drop *Lemma, line 1961.* [aux]

order_iso_drop : the order-isomorphism property descends to **drop** $m . +1$ of both sequences; companion of **order_iso_take**.

window_size_order_iso *Lemma, line 1978.* [aux]

window_size_order_iso : **window_size** is a function only of comparison structure – two order-isomorphic sequences have the same window sizes at every position. Lifts **mm_pos_order_iso** recursively.

mm_pos_drop_mm *Lemma, line 2045.* [aux]

mm_pos_drop_mm : after dropping the prefix up to the root, the new head is itself the root of its subtree, so its **mm_pos** is 0.

psi_0_eq *Lemma, line 2077.* [aux]

psi_0_eq : when **mm_pos** $s = 0$ (the head is the root), **psi 0 s** reduces to **rank_shift_seq s** – the action on the whole sequence.

drop_psi_above *Lemma, line 2102.* [aux]

drop_psi_above : psi j does not affect any suffix drop k w when k is past the window end – generalizes drop_psi.

psi_descent_thms.v

window_at_uniq *Lemma, line 14.* [aux]

window_at_uniq : uniqueness propagates from w to window_at i w.

size_window_at *Lemma, line 20.* [aux]

size_window_at : in-range, size (window_at i w) = window_size i w; the slice fits exactly.

nth_window_at *Lemma, line 31.* [aux]

nth_window_at : indexing into window_at i w is offset by i relative to indexing into w.

elem_in_range *Lemma, line 42.* [aux]

elem_in_range : every entry of L is bounded between the sorted extremes of L. Tag-along helper for cmp_out_of_range.

head_min_not_descent *Lemma, line 81.* [aux]

head_min_not_descent : if the window head is the window-min, then i is not a descent of w. Connects head-extremum to descent polarity.

head_max_is_descent *Lemma, line 104.* [aux]

head_max_is_descent : dual of head_min_not_descent – if the window head is the window-max, then i is a descent of w.

elem_rs_in_range *Lemma, line 147.* [aux]

elem_rs_in_range : every entry of rank_shift_seq L is bounded between the sorted extremes of L; rank-shift counterpart of elem_in_range.

rs_head_max_descent *Lemma, line 198.* [aux]

rs_head_max_descent : after the rank-shift sends the head to the max, position 0 becomes a descent. Used in the psi_R_add and psi_LR_swap1 cases of the descent-effect theorems.

rs_head_min_no_descent *Lemma, line 229.* [aux]

rs_head_min_no_descent : dual of rs_head_max_descent – after rank-shift sends head to min, position 0 is no longer a descent.

cmp_out_of_range *Lemma, line 261.* [aux]

cmp_out_of_range : when v lies outside the sorted range of L, the comparison nth L j > v is invariant under replacing L by rank_shift_seq L. Used at right-boundary descent positions.

cmp_out_of_range_left *Lemma, line 285.* [aux]

cmp_out_of_range_left : the cmp_out_of_range variant with v on the left of >; used at the left boundary i-1.

descent_psi_interior *Lemma, line 325.* [aux]

descent_psi_interior : at strictly interior window positions k, psi i preserves the descent value (interior order is preserved by rank-shift).

descent_psi_rboundary *Lemma, line 372.* [aux]

descent_psi_rboundary : at the right boundary k = i + ws - 1, psi i preserves the descent value, by post_window_extremum + cmp_out_of_range.

descent_psi_lboundary_R *Lemma, line 418.* [aux]

descent_psi_lboundary_R : at the left boundary $i-1$, for an R vertex, **psi i** preserves the descent value. Uses **pre_window_extremum_R** + **cmp_out_of_range_left**.

descent_psi_R_add *Lemma, line 450.* [aux]

descent_psi_R_add : R-vertex case where i is not a descent – applying **psi i** adds i to the descent set, leaving all other positions unchanged. Half of Stanley’s Fact #2 (R case).

descent_psi_R_remove *Lemma, line 502.* [aux]

descent_psi_R_remove : R-vertex case where i is a descent – applying **psi i** removes i from the descent set. Other half of Stanley’s Fact #2 (R case).

descent_psi_LR_swap1 *Lemma, line 554.* [aux]

descent_psi_LR_swap1 : LR-vertex case where i is not a descent (hence $i-1$ is) – applying **psi i** swaps the descent from $i-1$ to i . Stanley’s Fact #2 (LR case, swap1).

descent_psi_LR_swap2 *Lemma, line 623.* [aux]

descent_psi_LR_swap2 : LR-vertex case where i is a descent (hence $i-1$ is not) – applying **psi i** swaps the descent from i to $i-1$. Stanley’s Fact #2 (LR case, swap2).

psi_descent_v2.v

has_left_child_fuel *Fixpoint, line 40.* [aux]

has_left_child_fuel fuel i s tests, by fuel-bounded recursion on **mm_pos**, whether the vertex at in-order position i of $M(s)$ has a left child (equivalent to it being an LR vertex).

has_left_child *Definition, line 57.* [aux]

has_left_child i w : the vertex at in-order position i of $M(w)$ has a left child. Used to dispatch the descent-effect of **psi_i** into LR (with left child) and R (right-child only) cases.

has_left_child_fuel_0 *Lemma, line 62.* [aux]

has_left_child_fuel_0 : the vertex at in-order position 0 never has a left child (it is the leftmost vertex).

has_left_child_0 *Lemma, line 71.* [aux]

has_left_child_0 : top-level corollary of **has_left_child_fuel_0** – position 0 never has a left child.

has_left_child_fuel_monotone *Lemma, line 76.* [aux]

has_left_child_fuel_monotone : independence of fuel above the input size, the **has_left_child** analogue of **window_size_fuel_monotone**.

has_left_child_cons *Lemma, line 104.* [aux]

has_left_child_cons : structural unfolding of **has_left_child** on a **cons** by the **mm_pos** split, mirroring **window_size_cons**.

valid_mm *Fixpoint, line 152.* [aux]

valid_mm t : the tree t is a valid min-max tree, i.e. at every **Node l x r** the root sits at **mm_pos** of the in-order traversal. Specifies the trees in the image of **mmtree_of_seq_mm**.

valid_mm_build *Lemma, line 163.* [aux]

valid_mm_build : every output of **mmtree_of_seq_mm** satisfies **valid_mm**, the structural invariant used by **tree_structure**.

take_mm_eq *Lemma, line 217.* [aux]

take_mm_eq : at the `mm_pos` split, `take (size sl) s` recovers the left-subtree in-order `sl`.

drop_mm_eq *Lemma, line 222.* [aux]

drop_mm_eq : dual of `take_mm_eq` – `drop (size sl).+1 s` is the right-subtree in-order `sr`.

ws_bridge_left *Lemma, line 233.* [aux]

ws_bridge_left : on a valid node `s = sl ++ x :: sr`, when `i` is in the left subtree, `window_size i s` reduces to `window_size i sl`. Bridge between flat and structural recursion.

ws_bridge_root *Lemma, line 247.* [aux]

ws_bridge_root : at the root of the subtree, the M-window has size `size s - size sl` (`root + right subtree`).

ws_bridge_right *Lemma, line 260.* [aux]

ws_bridge_right : when `i` is in the right subtree, `window_size i s` reduces to `window_size (i - size sl - 1) sr` – the right-subtree descent step.

wa_bridge_left *Lemma, line 278.* [aux]

wa_bridge_left : the `window_at` analogue of `ws_bridge_left`.

wa_bridge_root *Lemma, line 292.* [aux]

wa_bridge_root : at the root, `window_at` is the suffix `drop (size sl) s` (`root + right-subtree labels`).

wa_bridge_right *Lemma, line 303.* [aux]

wa_bridge_right : the `window_at` analogue of `ws_bridge_right`.

hlc_bridge_left *Lemma, line 322.* [aux]

hlc_bridge_left : `has_left_child` structural bridge analogous to `ws_bridge_left`.

hlc_bridge_root *Lemma, line 336.* [aux]

hlc_bridge_root : at the root, `has_left_child` is true iff there is a nonempty left subtree.

hlc_bridge_right *Lemma, line 348.* [aux]

hlc_bridge_right : `has_left_child` structural bridge analogous to `ws_bridge_right`.

tree_props *Definition, line 525.* [aux]

tree_props `i w` is the conjunction of five structural properties at in-order position `i`: post-window extremum, two pre-window-vs-window inequalities, the descent flip, and the no-left-child extremum. Bundled to share a single structural induction.

pred_sub_add *Lemma, line 563.* [aux]

pred_sub_add : an arithmetic identity used to align in-order indices when descending into the right subtree.

tree_structure_via_tree *Lemma, line 584.* [aux]

tree_structure_via_tree : `tree_props` holds for every position `i` of the in-order traversal of any valid min-max tree. Proved by structural induction on `t`, dispatching on `i` vs the root index.

tree_structure *Lemma, line 1076.* [aux]

`tree_structure` : `tree_props i w` for every uniq `w`, obtained by re-routing through `mmtree_of_seq_mm` and `tree_structure_via_tree`.

post_window_extremum *Lemma, line 1090.* [aux]

`post_window_extremum` : the entry just past the window is either smaller than every window value or larger than every window value – the post-window position is a global extremum. First projection of `tree_props`.

pre_window_lt_max_when_min_head *Lemma, line 1119.* [aux]

`pre_window_lt_max_when_min_head` : at an LR vertex whose head is the window-min, the pre-window entry `w_{i-1}` is below the window-max. Discharges the `head = min` branch of `exactly_one_descent_LR`.

pre_window_gt_min_when_max_head *Lemma, line 1137.* [aux]

`pre_window_gt_min_when_max_head` : dual of `pre_window_lt_max_when_min_head` – at an LR vertex whose head is the window-max, `w_{i-1}` is above the window-min.

exactly_one_descent_LR *Lemma, line 1166.* [aux]

`exactly_one_descent_LR` : at an LR vertex with nontrivial window, exactly one of positions `i-1`, `i` is a descent of `w`. Used by the descent-effect theorems in `psi_descent_thms.v`.

pre_window_extremum_R *Lemma, line 1192.* [aux]

`pre_window_extremum_R` : at an R vertex (no left child), the entry `w_{i-1}` is itself a global extremum relative to the window. R-case counterpart of the LR pre-window lemmas.

qbin.v

qbin *Fixpoint, line 27.* [aux]

(no docstring)

qbin_n0 *Lemma, line 34.* [aux]

(no docstring)

qbin_OS *Lemma, line 37.* [aux]

(no docstring)

qbinS *Lemma, line 41.* [aux]

The defining q-Pascal recurrence.

qbin_small *Lemma, line 49.* [aux]

(no docstring)

qbin_nn *Lemma, line 55.* [aux]

(no docstring)

qbin_horner1 *Lemma, line 65.* [aux]

(no docstring)

qeul.v

q_eul_pol *Definition, line 41.* [Ch 6 §6.2]

`q_eul_pol n` is the q-Eulerian polynomial $A_n(q,t) = \sum_{\sigma \in S_{n+1}} q^{\text{maj } \sigma} t^{\text{des } \sigma}$, the bivariate joint generating function of `maj` and `des` over permutations of `n+1` letters. Stanley EC1 $\set{S}$ 1.4.

q_eul_pol_t1 *Theorem, line 50.* [Ch 6 §6.2]

Specialization $t = 1$ of $A_n(q, t)$ recovers the q-factorial: $A_n(q, 1) = [n+1]_q!$. Direct consequence of **maj_q_fact**.

eul_pol *Definition, line 68.* [Ch 6 §6.2]

eul_pol n is the classical Eulerian polynomial $\sum_{k \leq n} \text{eulerian}(n, k) * t^k$, with coefficients the Eulerian numbers counting permutations of $n+1$ letters by descent count. Stanley EC1 \S{}1.4.

sum_des_eq_eul_pol *Lemma, line 74.* [aux]

The descent generating function expressed as the Eulerian polynomial: $\sum_{\sigma \in S_{n+1}} X^{(\text{des } \sigma)} = \text{eul_pol } n$. Obtained by partitioning permutations according to their descent count.

q1_subst *Definition, line 96.* [Ch 6 §6.2]

q1_subst is the substitution $q := 1$ on a bivariate polynomial in $\{\text{poly } \{\text{poly int}\}\}$, implemented by evaluating each inner polynomial coefficient at 1.

q_eul_pol_q1 *Theorem, line 102.* [Ch 6 §6.2]

Specialization $q = 1$ of $A_n(q, t)$ recovers the classical Eulerian polynomial: $A_n(1, t) = \text{eul_pol } n$. Each $q^{(\text{maj } \sigma)}$ becomes 1, leaving the descent generating function.

qfact.v

q_int *Definition, line 28.* [Ch 6 §6.1]

q_int n is the q-integer $n_q = 1 + q + q^2 + \dots + q^{(n-1)}$, represented as the polynomial $\sum_{i < n} q^i$ in $\{\text{poly int}\}$. Stanley EC1 \S{}1.4.

q_fact *Definition, line 32.* [Ch 6 §6.1]

q_fact n is the q-factorial $n+1_q! = 1_q 2_q \dots n+1_q$, the product of **q_int** $k.+1$ for $k < n.+1$. Stanley EC1 \S{}1.4.

sum_rev_X *Lemma, line 41.* [aux]

Reindexing identity: summing $X^{(n-p)}$ over $p < n.+1$ equals **q_int** $n.+1$. Used to recognize the q-integer factor produced by expanding **exprD** over **val p** in the **inv_q_fact** inductive step.

is_inv_lift_pair *Lemma, line 70.* [aux]

Inversion correspondence on the "non-p rows": a pair $(\text{lift } p \ j1, \text{lift } p \ j2)$ is an inversion of **insert_max_perm** $t \ p$ iff $(j1, j2)$ is an inversion of t .

is_inv_lift_p *Lemma, line 85.* [aux]

Pairs $(\text{lift } p \ k, p)$ are never inversions of **insert_max_perm** $t \ p$: $\sigma_p = \text{ord_max}$ is the maximum value, so it cannot be exceeded by any earlier image $\sigma_p(\text{lift } p \ k)$.

sum_geq_p *Lemma, line 98.* [aux]

Counts the elements $k : I_{n.+1}$ with $p \leq \text{val } k$: there are exactly $n.+1 - p$ of them. Used to evaluate the p-row contribution in **inv_insert_max**.

inv_via_sum *Lemma, line 122.* [aux]

Reformulates **inv s** as the unconditional double sum $\sum_i \sum_j \text{is_inv } s \ i \ j$, unfolding the $\#\mid \dots \mid$ counting form.

inv_insert_max *Lemma, line 139.* [aux]

Headline arithmetic identity for the `insert_max_perm` bijection: `inv (insert_max_perm t p) = inv t + (n.+1 - val p)`. The new image `ord_max` at position `p` contributes exactly `n.+1 - val p` new inversions; the rest match `inv t` under the lift.

inv_q_fact *Theorem, line 204.* [Ch 6 §6.1]

Headline q-factorial identity for the inversion statistic (Stanley EC1 \S{}1.4, Cor 1.3.10): $\sum_{\sigma \in S_{n+1}} X^{\text{inv } \sigma} = [n+1]_q!$. Proved by induction on `n` using the `insert_max_perm` bijection and `inv_insert_max`.

maj_q_fact *Theorem, line 240.* [Ch 6 §6.1]

Headline q-factorial identity for the major-index statistic (Stanley EC1 \S{}1.4): $\sum_{\sigma \in S_{n+1}} X^{\text{maj } \sigma} = [n+1]_q!$. One-line transfer from `inv_q_fact` via the Foata bijection `foata_perm`, which sends `inv` to `maj`.

reflection.v

euler *Definition, line 28.* [aux]

(no docstring)

eulerA *Definition, line 32.* [aux]

(no docstring)

eulerA_0 *Lemma, line 39.* [aux]

(no docstring)

alt_desc_set_0 *Lemma, line 43.* [aux]

(no docstring)

euler_0 *Lemma, line 47.* [aux]

(no docstring)

eulerA_1 *Lemma, line 50.* [aux]

(no docstring)

alt_desc_set_1 *Lemma, line 55.* [aux]

(no docstring)

euler_1 *Lemma, line 62.* [aux]

(no docstring)

eulerA_2 *Lemma, line 65.* [aux]

(no docstring)

descent_set_insert_max_ord0 *Lemma, line 78.* [aux]

(no docstring)

descent_set_insert_max_ord_max *Lemma, line 101.* [aux]

(no docstring)

descent_set_insert_max_interior *Lemma, line 151.* [aux]

(no docstring)

widen_inj *Lemma, line 215.* [aux]

(no docstring)

leqj_n1 *Definition, line 223.* [aux]
(no docstring)

embed_left *Definition, line 226.* [aux]
(no docstring)

embed_left_inj *Lemma, line 228.* [aux]
(no docstring)

image_left *Definition, line 232.* [aux]
(no docstring)

t_embed_inj *Lemma, line 235.* [aux]
(no docstring)

card_image_left *Lemma, line 238.* [aux]
(no docstring)

mem_image_left *Lemma, line 244.* [aux]
(no docstring)

rank_left *Definition, line 269.* [aux]
(no docstring)

cast_to_j *Definition, line 274.* [aux]
(no docstring)

cast_to_j_inj *Lemma, line 277.* [aux]
(no docstring)

perm_left_fun *Definition, line 281.* [aux]
(no docstring)

perm_left_fun_inj *Lemma, line 285.* [aux]
(no docstring)

perm_left *Definition, line 294.* [aux]
(no docstring)

perm_leftE *Lemma, line 297.* [aux]
(no docstring)

enum_val_perm_left *Lemma, line 302.* [aux]
(no docstring)

perm_left_witness_indep *Lemma, line 317.* [aux]
(no docstring)

image_left_witness_pos *Lemma, line 331.* [aux]
(no docstring)

embed_right *Definition, line 347.* [aux]
(no docstring)

embed_right_inj *Lemma, line 350.* [aux]
(no docstring)

image_right *Definition, line 356.* [aux]
(no docstring)

t_embed_right_inj *Lemma, line 359.* [aux]
(no docstring)

card_image_right *Lemma, line 362.* [aux]
(no docstring)

mem_image_right *Lemma, line 367.* [aux]
(no docstring)

cast_to_subj *Definition, line 380.* [aux]
(no docstring)

cast_to_subj_inj *Lemma, line 383.* [aux]
(no docstring)

perm_right_fun *Definition, line 386.* [aux]
(no docstring)

perm_right_fun_inj *Lemma, line 390.* [aux]
(no docstring)

perm_right *Definition, line 399.* [aux]
(no docstring)

perm_rightE *Lemma, line 402.* [aux]
(no docstring)

enum_val_perm_right *Lemma, line 407.* [aux]
(no docstring)

perm_right_witness_indep *Lemma, line 418.* [aux]
(no docstring)

sorted_ltn_val_enum *Lemma, line 435.* [aux]
(no docstring)

enum_val_ltn *Lemma, line 448.* [aux]
(no docstring)

enum_rank_in_val_ltn *Lemma, line 477.* [aux]
(no docstring)

perm_left_lt_iff *Lemma, line 493.* [aux]
(no docstring)

perm_right_lt_iff *Lemma, line 506.* [aux]
(no docstring)

leqj_pred_n *Lemma, line 520.* [aux]
(no docstring)

embed_desc_left *Definition, line 546.* [aux]
(no docstring)

is_descent_perm_left *Lemma, line 548.* [aux]
(no docstring)

sub_succ *Lemma, line 573.* [aux]
(no docstring)

j_plus_lt_n *Lemma, line 581.* [aux]
(no docstring)

embed_desc_right *Definition, line 587.* [aux]
(no docstring)

is_descent_perm_right *Lemma, line 589.* [aux]
(no docstring)

image_left_right_disjoint *Lemma, line 614.* [aux]
(no docstring)

image_left_right_cover *Lemma, line 628.* [aux]
(no docstring)

image_left_card_plus_right *Lemma, line 640.* [aux]
(no docstring)

descent_left_of_t *Lemma, line 722.* [aux]
(no docstring)

descent_right_of_t *Lemma, line 739.* [aux]
(no docstring)

descent_left_part *Definition, line 770.* [aux]
(no docstring)

descent_boundary_part *Definition, line 773.* [aux]
(no docstring)

descent_right_part *Definition, line 776.* [aux]
(no docstring)

descent_set_decomp_partition *Lemma, line 779.* [aux]
(no docstring)

descent_left_boundary_disjoint *Lemma, line 788.* [aux]
(no docstring)

descent_left_right_disjoint *Lemma, line 796.* [aux]
(no docstring)

descent_boundary_right_disjoint *Lemma, line 804.* [aux]
(no docstring)

descent_left_part_image *Lemma, line 813.* [aux]
(no docstring)

descent_right_part_R *Definition, line 841.* [aux]
(no docstring)

descent_right_part_R_image *Lemma, line 844.* [aux]
(no docstring)

cardCL_eq *Lemma, line 891.* [aux]
(no docstring)

castL *Definition, line 896.* [aux]
(no docstring)

castR *Definition, line 897.* [aux]
(no docstring)

leqj *Definition, line 899.* [aux]
(no docstring)

split_pos *Definition, line 902.* [aux]
(no docstring)

assemble_fun *Definition, line 908.* [aux]
(no docstring)

split_pos_inj *Lemma, line 914.* [aux]
(no docstring)

assemble_fun_inj *Lemma, line 928.* [aux]
(no docstring)

assemble_perm *Definition, line 950.* [aux]
(no docstring)

assemble_permE *Lemma, line 952.* [aux]
(no docstring)

assemble_left *Lemma, line 969.* [aux]
(no docstring)

assemble_right *Lemma, line 987.* [aux]
(no docstring)

assemble_image_left *Lemma, line 1007.* [aux]
(no docstring)

assemble_image_right *Lemma, line 1024.* [aux]
(no docstring)

assemble_perm_left *Lemma, line 1053.* [aux]
(no docstring)

assemble_perm_right *Lemma, line 1082.* [aux]
(no docstring)

card_image_left_eq *Definition, line 1129.* [aux]
(no docstring)

assemble_decomp_inverse_left *Lemma, line 1135.* [aux]
(no docstring)

image_right_eq_compl *Lemma, line 1154.* [aux]
(no docstring)

assemble_decomp_inverse_right *Lemma, line 1178.* [aux]
(no docstring)

split_embed *Lemma, line 1206.* [aux]
(no docstring)

assemble_decomp_inverse *Lemma, line 1220.* [aux]
(no docstring)

assemble_decomp_unique *Lemma, line 1246.* [aux]
 (no docstring)

eulerA_S2 *Lemma, line 1262.* [aux]
 (no docstring)

two_eulerA_split *Lemma, line 1267.* [aux]
 (no docstring)

beta_eq_pair_sum *Lemma, line 1283.* [aux]
 (no docstring)

beta_eq_double_sum *Lemma, line 1294.* [aux]
 (no docstring)

beta_eq_triple_split *Lemma, line 1306.* [aux]
 (no docstring)

sum_partition_image_left *Lemma, line 1328.* [aux]
 (no docstring)

assemble_perm_pirr *Lemma, line 1340.* [aux]
 (no docstring)

assemble_decomp_inverse_gen *Lemma, line 1347.* [aux]
 (no docstring)

sub_gt0_n_j *Lemma, line 1359.* [aux]
 (no docstring)

fwdAss *Definition, line 1375.* [aux]
 (no docstring)

bwdAss *Definition, line 1378.* [aux]
 (no docstring)

bwd_fwd *Lemma, line 1381.* [aux]
 (no docstring)

fwd_bwd *Lemma, line 1387.* [aux]
 (no docstring)

sum_reindex_inner *Lemma, line 1396.* [aux]
 (no docstring)

imset_lift_eq *Lemma, line 1414.* [aux]
 (no docstring)

interior_descent_set_eq *Lemma, line 1438.* [aux]
 (no docstring)

alt_desc_set_is_alt *Lemma, line 1623.* [aux]
 (no docstring)

set_is_alt_indicator *Lemma, line 1633.* [aux]
 (no docstring)

alt_plus_nalt_as_set_is_alt_sum *Lemma, line 1664.* [aux]
 (no docstring)

sum_descent_set_indicator *Lemma, line 1680.* [aux]
(no docstring)

setU1_imset_lift_eq *Lemma, line 1688.* [aux]
(no docstring)

nat_of_andb *Lemma, line 1707.* [aux]
(no docstring)

sum_set_is_alt_ord0 *Lemma, line 1716.* [aux]
(no docstring)

sum_set_is_alt_ord_max *Lemma, line 1738.* [aux]
(no docstring)

andre_target *Definition, line 1770.* [aux]
(no docstring)

interior_set_is_alt *Lemma, line 1773.* [aux]
(no docstring)

andre_split_lt *Lemma, line 1801.* [aux]
(no docstring)

andre_s_ltn *Lemma, line 1806.* [aux]
(no docstring)

andre_target_lt *Lemma, line 1810.* [aux]
(no docstring)

andre_target_at *Lemma, line 1814.* [aux]
(no docstring)

andre_target_gt *Lemma, line 1817.* [aux]
(no docstring)

andre_union_eq_split *Lemma, line 1825.* [aux]
(no docstring)

beta_andre_left *Lemma, line 1872.* [aux]
(no docstring)

beta_andre_right *Lemma, line 1883.* [aux]
(no docstring)

andre_interior_count *Lemma, line 1893.* [aux]
(no docstring)

sum_set_is_alt_eq_andre_sum *Lemma, line 1947.* [aux]
(no docstring)

boundary_cancellation_alt *Lemma, line 1980.* [aux]
(no docstring)

euler_rec *Theorem, line 1988.* [aux]
(no docstring)

stanley_egf.v

rat_char0 *Lemma, line 35.* [aux]
rat has characteristic 0.

euler_egf *Definition, line 39.* [aux]
The exponential generating function of the Euler numbers, over rat.

euler_egf0 *Lemma, line 41.* [aux]
(no docstring)

euler_egf_ode *Lemma, line 46.* [aux]
(no docstring)

stanley_1_6_1 *Theorem, line 66.* [aux]
(no docstring)

stanley_1_6_1_coef *Corollary, line 73.* [aux]
Coefficient form: $E_n = n! * x^n (\sec x + \tan x)$.

stanley_ogf.v

pow_ogf *Definition, line 29.* [aux]
The OGF of the (n+1)-st powers: coefficient m is $(m+1)^{(n+1)}$.

stanley_1_4_div *Theorem, line 39.* [aux]
(no docstring)

stanley_1_4 *Theorem, line 67.* [aux]
(no docstring)

stanley_1_4_inv *Corollary, line 75.* [aux]
Display form with the explicit inverse of $1 - x$.

stirling_egf.v

natU_polyrat *Lemma, line 36.* [aux]
(no docstring)

factU_polyrat *Lemma, line 42.* [aux]
(no docstring)

stirling_c00 *Lemma, line 49.* [aux]
(no docstring)

stirling_cS0 *Lemma, line 61.* [aux]
Every permutation of a nonempty type has at least one cycle.

stirling_c_out *Lemma, line 70.* [aux]
A permutation of I_n has at most n cycles.

stirling_pol *Definition, line 82.* [aux]
(no docstring)

coef_stirling_pol *Lemma, line 85.* [aux]
(no docstring)

stirling_pol0 *Lemma, line 99.* [aux]
(no docstring)

stirling_pol_rec *Lemma, line 106.* [aux]
(no docstring)

stirling_egf *Definition, line 122.* [aux]
(no docstring)

stirling_egf0 *Lemma, line 125.* [aux]
(no docstring)

coef_fpsXfM *Lemma, line 129.* [aux]
Coefficient shift under multiplication by x.

fps_geom_deriv *Lemma, line 139.* [aux]
The geometric series solves $y' = y^2$.

stirling_egf_ode1 *Lemma, line 148.* [aux]
The formal ODE $(1-x) S' = t S$, from the row recurrence.

onesubX_pow_t *Definition, line 169.* [aux]
(no docstring)

onesubX_pow_t_ode1 *Lemma, line 172.* [aux]
(no docstring)

stirling_ode_convert *Lemma, line 190.* [aux]
Both equations convert to the standard linear form $y' = (t \text{ geom}) y$.

stirling_cycle_egf *Theorem, line 204.* [aux]
(no docstring)

stirling_fiber.v

insert_after_fun *Definition, line 34.* [aux]
`insert_after_fun j0 s` is the underlying function of `insert_after j0 s`: it splices `ord_max` into the `j0`-cycle of `s` right after `j0`. See `insert_after` for the perm wrapping.

insert_after_fun_inj *Lemma, line 44.* [aux]
`insert_after_fun j0 s` is injective on `'I_n.+1`. Used to package it as a `perm` in `insert_after`.

insert_after *Definition, line 82.* [aux]
`insert_after j0 s` : `{perm 'I_n.+1}` is the permutation that splices `ord_max` into the `j0`-cycle of `s` right after `j0`. An `s`-cycle `p1 -> ... -> j0 -> s j0 -> ...` becomes `p1 -> ... -> j0 -> ord_max -> s j0 -> ...`.

insert_afterE *Lemma, line 86.* [aux]
`insert_after` unfolds to `insert_after_fun` pointwise.

insert_after_ord_max *Lemma, line 92.* [aux]
Defining equation: `insert_after j0 s` sends `ord_max` to `lift ord_max (s j0)`.

insert_after_j0 *Lemma, line 98.* [aux]
Defining equation: `insert_after j0 s` sends `lift ord_max j0` to `ord_max` (the spliced-in element follows `j0` in its cycle).

insert_after_other *Lemma, line 104.* [aux]

Defining equation: on points `lift ord_max k` with $k \neq j_0$, `insert_after j0 s` acts like the `lift ord_max`-extension of `s`.

insert_after_ord_max_neq *Lemma, line 113.* [aux]

`insert_after j0 s` does not fix `ord_max`. Used to characterise the image of `insert_after` in the bijection of $\mathcal{S}^D$.

insert_after_decomp *Lemma, line 149.* [aux]

Decomposition: `insert_after j0 s` factors as the transposition `tperm ord_max (lift ord_max j0)` composed with `perm.lift_perm ord_max ord_max s`. Used in `cycle_count_insert_after` to apply `porbits_mul_tperm`.

cycle_count_insert_after *Lemma, line 169.* [aux]

Key invariant: `insert_after j0 s` preserves the cycle count of `s`. Splicing `ord_max` into the `j0`-cycle widens that cycle by one element without creating or destroying cycles.

insert_after_inj_pair *Lemma, line 201.* [aux]

`insert_after` is injective in the pair (j_0, s) : distinct (j_0, s) yield distinct permutations of $\mathcal{I}_{n+1}$.

insert_after_surj *Lemma, line 228.* [aux]

Surjectivity onto the non-fixing class: every $\sigma : \{\text{perm } \mathcal{I}_{n+1}\}$ with $\sigma \text{ ord_max} \neq \text{ord_max}$ arises as `insert_after j0 s` for some `j0` and `s`. Constructed by conjugating σ with a transposition to land in the fixed-`ord_max` class, then applying `drop_perm`.

card_neq_ord_max_fiber *Lemma, line 266.* [aux]

Count of permutations σ of $\mathcal{I}_{n+1}$ with $\sigma \text{ ord_max} \neq \text{ord_max}$ and exactly $k+1$ cycles equals $n * \text{stirling_c } n \ k.+1$. This is the $j \neq \text{ord_max}$ half of the Stirling fiber bijection.

card_eq_ord_max_fiber *Lemma, line 309.* [aux]

Count of permutations σ of $\mathcal{I}_{n+1}$ fixing `ord_max` with exactly $k+1$ cycles equals $\text{stirling_c } n \ k$. Bijection via `perm.lift_perm ord_max ord_max` and `drop_perm`. This is the $j = \text{ord_max}$ half of the Stirling fiber bijection.

stirling_c_rec *Lemma, line 364.* [Ch 2 §2.2]

Stirling cycle-number recurrence: $c(n+1, k+1) = n * c(n, k+1) + c(n, k)$ (Stanley EC1 §1.3.2). Final assembly from `card_neq_ord_max_fiber` and `card_eq_ord_max_fiber`.

worpitzky.v

worpitzky_bin_step *Lemma, line 33.* [aux]

(no docstring)

worpitzky *Theorem, line 56.* [aux]

(no docstring)

coef_eul_pol *Lemma, line 92.* [aux]

(no docstring)

Appendix B

Glossary of mathcomp primitives

A quick-reference companion to Chapter 1. Entries are ordered roughly by first appearance in the chapter; each gives the mathcomp source file, a one-line gloss, and an example.

`'I_n (fintype.v)` The finite type of natural numbers strictly less than n . Inhabitant: a pair $(i, \text{proof of } i < n)$. Cardinality n .

`ord0 : 'I_n.+1 (fintype.v)` The value 0, with its proof of $0 < n + 1$.

`ord_max : 'I_n.+1 (fintype.v)` The value n , the largest element.

`val i : nat (eqtype.v)` Strips an ordinal back to its underlying natural number.

`widen_ord (h : m \leq n) i (fintype.v)` Casts $i : 'I_m$ to $'I_n$ keeping the same numerical value. Pure cast.

`lift (p : 'I_n.+1) j (fintype.v)` Casts $j : 'I_n$ to $'I_n.+1$ “skipping” the value p . With $p = \text{ord0}$, this is $j \mapsto j + 1$.

`unlift p i : option 'I_n (fintype.v)` Partial inverse of `lift p`. Returns `None` when $i = p$, else `Some j` with $i = \text{lift } p \ j$.

`rev_ord i (fintype.v)` The involution $i \mapsto n - 1 - i$ on $'I_n$.

`{set T} (finset.v)` The type of finite subsets of T (a finite type), with extensional equality.

`[set i : T | P i] (finset.v)` Set comprehension: $\{i : T \mid P \ i\}$.

`[set f i | i in S] (finset.v)` Image of set S under function f .

`#|S| (fintype.v)` Cardinality of a finite set or predicate.

`i \in S (finset.v)` Membership test, returning `bool`.

`~ : S (finset.v)` Complement of S in its ambient finite type.

`{perm T} (perm.v)` Type of bijections $T \rightarrow T$ for T a finite type. Inhabitant: a function plus its proof of injectivity.

`1%g (fingroup.v)` Group identity. For `{perm T}`, the identity permutation.

`(s * t)%g (fingroup.v)` Group multiplication. For perms: $(s * t) \ i = t \ (s \ i)$; see lemma `permM`.

`perm (h : injective f) (perm.v)` Constructor: turn an injective function into a `{perm T}`.

`\sum_(i : T) f i (bigop.v)` Sum of $f\ i$ over every $i : T$.

`\sum_(i < n) f i (bigop.v)` Same, indexed by $i : 'I_n$.

`\sum_(i \in S) f i (bigop.v)` Sum over the elements of an explicit set S .

`\sum_(i | P i) f i (bigop.v)` Sum over i satisfying boolean filter P . The vertical bar is a filter, not disjunction.

`partition_big f xpredT (bigop.v)` Rewrite a sum as a nested sum partitioned by the value of f .

`n.+1, n.+2, n.-1 (ssrnat.v)` Successor, double-successor, predecessor on `nat`. Read as $n + 1$, $n + 2$, $n - 1$.

`~~ b` Boolean negation.

`a == b (eqtype.v)` Decidable equality on an `eqType`, returning `bool`. Distinct from propositional `=`; bridged by lemma `eqP`.

Section X. Variable n. ... End X. (Rocq) Sections introduce parameters that are auto-generalised on `End`. Inside the section, `n` reads as a fixed natural; outside, every definition is implicitly parameterised.

Appendix C

Source map

A complete file index is in `docs/READING_GUIDE.md` in the [mathcomp-eulerian](#) repository. Files covered (or planned) by this document, in Stanley reading order:

`cycles.v`, `stirling_fiber.v` Stanley §1.3.1–2 (Chapter 2).

`inversions.v` Stanley §1.3.3 (Chapter 3).

`foata.v` Stanley §1.3.4 (Chapter 4).

`descent.v`, `eulerian.v`, `beta.v` Stanley §1.4 (Chapter 5).

`qfact.v`, `qeul.v` Stanley §1.4 q -analogues (Chapter 6).

`altsub.v` Stanley §1.6.2 (Chapter 7).

`beta_omega.v`, `beta_swap.v`, `perm_seq_bridge.v` Stanley §1.6.3 (Chapter 8); the `psi_*.v` files are project-internal scaffolding for the cd-index machinery underlying `omega_proper_beta_lt`.

`reflection.v` Stanley §1.6.4 (Chapter 9); André’s reflection method and the recurrence `euler_rec`, fully proved.

`fps/*.v`, `stanley_egf.v` formal power series sub-library (namespace `mathcomp_fps`) and Stanley §1.6.4 Proposition 1.6.1, $\sum E_n x^n / n! = \sec x + \tan x$ (Chapter 9); this layer uses the classical axioms of `mathcomp-classical`.